

# K-Bound: When Is Label-Free Adaptation Knowable?

Full Technical Report on Directional Action Under Distribution Shift

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## Abstract

An unlabeled adaptation procedure can produce an update without establishing that deploying it will help. We study when the available evidence supports ADAPT, FREEZE, or ABSTAIN for a fixed model pair. For binary 0/1 loss and positive disagreement mass, K-Bound characterizes the exact compatible-benefit interval under a declared bound  $\beta$  on a hidden target calibration residual. In the full label-kernel class, a strict direction is uniformly justified exactly when  $|M| > \beta$ , where  $M$  is the observable disagreement margin. A complementary evidence-fibre theorem gives the smallest residual bound that any uniformly valid audit of the same information must cover. Additional processing cannot remove ambiguity left by that information. Knowability-Guided Adaptation (KGA) addresses a separate empirical target: the measured score difference between a candidate and its frozen baseline on an evaluation cell. Labeled development and disjoint residual-calibration cells fit its benefit predictor and error interval; the action uses no scored-cell outcome. On the opened, dependent six-family CIFAR-10-C diagnostic, Tent and EATA have lower point regret than both fixed policies. Tent makes 2 false ADAPT decisions among 1,091 ADAPT actions and 1 false FREEZE among 337 FREEZE actions; EATA makes neither error among 1,207 ADAPT and 105 FREEZE actions. SAR instead favors always-adapt. On CCT-20, all 45 cells retain the frozen model, including one false FREEZE; five-checkpoint DomainNet development evaluation yields universal abstention. Thus these natural-shift studies support conservative retention, not useful selective routing. The theory distinguishes structural ambiguity from prediction uncertainty; the experiments do not establish coverage or population protection on unseen shifts.

**Report scope.** This book-level report expands the maintained paper with proof anatomy, current protocol dossiers, formal-coverage boundaries, and reproducibility guidance. Accuracy and regret are reported as scoped utility consequences; the headline is whether the available information supports ADAPT, FREEZE, or explicit ABSTAIN.

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## Part I

# Core Manuscript

## 1 Introduction

Test-time adaptation changes a model using unlabeled deployment data. Entropy minimization, filtering, regularization, and related methods can recover accuracy under shift. The same update can also make predictions worse under unfavorable shifts (Wang et al., 2021; Niu et al., 2022, 2023; Wang et al., 2022). The central deployment question is therefore directional action correctness: an adapter supplies a candidate, not a reason to deploy it.

Our primary research question is: *When can evidence available without new target labels justify replacing a frozen predictor with a fixed adapted candidate?* Let  $f_0$  be the frozen

predictor and  $f_a$  the candidate. Their population benefit is

$$\Delta = R_T(f_0) - R_T(f_a). \quad (1)$$

A positive value favors the update; a negative value favors the frozen model. We use three decisions. ADAPT asserts positive benefit, FREEZE asserts negative benefit, and ABSTAIN asserts neither. Both non-adapt decisions serve  $f_0$ , but only FREEZE makes a directional claim.

K-Bound treats this as partial identification. In a binary model under 0/1 loss, we fix the input law, the predictors, and every label-free observable. We then vary the target label kernel within a declared class. The compatible benefits give an exact frontier for uniform strict decisions. The class is indexed by a bound on a hidden target calibration residual. That bound is an assumption, not a quantity supplied by an unlabeled batch. A second theorem shows that an audit using the same evidence must cover the residual range that the evidence leaves unresolved.

Knowability-Guided Adaptation (KGA) is the empirical companion. It predicts a model pair’s measured benefit on an evaluation cell. It acts only when a residual-calibrated interval excludes zero. KGA is label-free at the scored decision, not during development and calibration: labeled development cells train the predictor, and disjoint residual-calibration cells calibrate its errors. The specified cross-fit excludes each scored cell’s outcome from its own prediction, radius, and decision. This computational separation does not make an already opened retrospective panel prospective; outcome access is reported separately for each study. K-Bound is the population partial-identification framework; KGA is the empirical cell-benefit interval gate. They share three-action decision logic, but not a target or guarantee: K-Bound reasons about population benefit  $\Delta$  through  $(M, \gamma, \beta)$ , whereas KGA predicts measured cell benefit  $\Delta^{\text{cell}}$  through  $(\hat{\Delta}, \varepsilon)$ .

## 1.1 Three Scientific Contributions

1. **Exact population frontier.** We construct target label laws with identical label-free evidence, derive their feasible benefit interval, and characterize its strict positive, strict negative, and unresolved regions, including the zero-benefit boundary.
2. **Evidence-fibre audit limit.** We identify the least residual bound that the same observable law can support uniformly, explaining why historical labels help only through a stated link to the deployment domain.
3. **Empirical paired-benefit gate.** We give a cell-outcome-disjoint three-way cross-fit for the controlled grids and evaluate its ADAPT/FREEZE/ABSTAIN behavior on constructed shifts and its retention behavior on natural shifts. We report action exposure and false-action denominators before secondary accuracy and regret consequences. The standard coverage-to-action implication states exactly what follows if the interval covers its declared target.

The opened CIFAR-10-C diagnostic first asks whether each committed direction was correct and reports the corresponding action exposure. Tent makes two false ADAPT actions and one false FREEZE; EATA makes neither; SAR makes no FREEZE decisions. The secondary utility result is candidate-specific: Tent and EATA have lower point regret than

KGA empirical decision flow

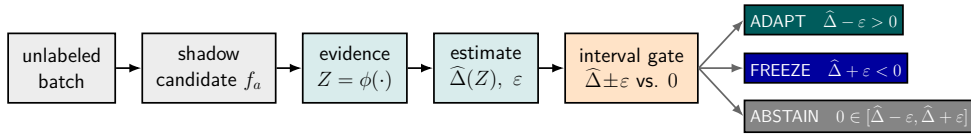


Figure 1: KGA’s empirical decision flow for one evaluation cell. A shadow adapter creates  $f_a$  from frozen  $f_0$ . Labeled development cells fit the benefit predictor, and disjoint residual-calibration cells determine its error radius. The scored cell contributes label-free evidence only. An interval above zero selects ADAPT, one below zero selects FREEZE, and an interval crossing zero selects ABSTAIN. The figure defines the action rule; it does not establish interval coverage.

either fixed policy, whereas SAR favors always-adapt. CCT-20 instead preserves frozen-model utility in all 45 cells with no ADAPT exposure. This is conservative frozen-model retention, not selective routing on an unseen natural shift. The appendices retain adverse, incomplete, and withheld studies so that this boundary does not depend on removing unfavorable evidence.

Newer painting-development evaluations retain the same limitation: the five-checkpoint cohort and separate reference-source candidates yielded universal abstention, with adverse queue and normalization-state diagnostics preserved as development evidence.

Sections 3–5 define the population target, derive the strict-commitment frontier, and establish the evidence-fibre audit floor. Section 6 introduces the separate empirical gate. Sections 7–8 report the controlled CIFAR diagnostic and CCT-20 retention result. Section 9 states the assumptions, missing evidence, and deployment limits.

## 2 Related Work

### 2.1 Candidate Adaptation and Update Selection

Tent, EATA, SAR, CoTTA, MEMO, SHOT, and test-time training construct candidates through entropy objectives, filtering, regularization, temporal stabilization, augmentation, or auxiliary tasks (Wang et al., 2021; Niu et al., 2022, 2023; Wang et al., 2022; Zhang et al., 2022; Liang et al., 2020; Sun et al., 2020). Other methods decide where or when to update. TTSA selects modality-specific adapters (Chen et al., 2025). GALA selects layers and filters samples by gradient alignment (Sahoo et al., 2025). Informed Adaptation learns topological features associated with useful updates (Mutlu et al., 2026). KGA starts later in this pipeline: it compares one realized candidate with its frozen baseline. Learned routing and rollback are established ideas, not our novelty claim.

### 2.2 Label-Free Performance and Strategy Estimation

ATC, agreement-on-the-line, AETTA, and prediction-matrix statistics estimate performance from unlabeled outputs and suitable development information (Garg et al., 2022; Baek et al., 2022; Miller et al., 2021; Lee et al., 2024; Deng et al., 2023; Xie et al., 2024). The closest

empirical comparator is [Kim et al. \(2024\)](#), whose TTA/ALine method uses agreement-based estimators after adaptation, supports hyperparameter selection, and compares adaptation strategies without labeled OOD data.

KGA directly fits the paired score difference for one frozen/candidate pair and converts its prediction interval into a three-way action. Separate adapted- and frozen-model accuracy estimates can also be differenced; a synchronized current-policy comparison with official TTA/ALine and AETTA implementations remains missing.

## 2.3 Monitoring and Adaptation Controllers

POEM protects adaptation through entropy matching and online betting ([Bar et al., 2024](#)). [Schirmer et al. \(2025\)](#) monitor an adapting model’s running risk against a source-risk reference under a proxy-informativeness condition; a supervised precursor uses a labeled test stream ([Podkopaev and Ramdas, 2022](#)). Drift-to-Action Controllers use delayed labels and sequential evidence to gate adaptation, abstention, rollback, and retraining ([Lamaakal et al., 2026](#)). These methods act during an evolving stream. KGA instead decides whether one proposed candidate should be committed before it replaces the frozen state; it makes no anytime-valid risk claim.

## 2.4 Partial Identification and Robust Decisions

An identified set contains the values compatible with the observations and assumptions. Decision theory under partial identification studies actions when that set leaves payoffs ambiguous. [Christensen et al. \(2026\)](#), for example, combine worst-case payoff uncertainty with statistical learning of identified parameters. K-Bound asks the narrower question of whether one strict benefit sign is uniform over the set. It neither solves the general optimal-decision problem nor claims interval thresholding as a new idea.

Indistinguishable problems are already central to domain-adaptation impossibility ([Ben-David et al., 2010a,b](#)). Our contribution is a disagreement-conditional label-kernel construction, the resulting clipped benefit set, and its residual-audit floor. The proof does not introduce label-free impossibility. Risk-estimation assumptions based on conditional independence ([Steinhardt and Liang, 2016](#)), label shift ([Lipton et al., 2018](#); [Garg et al., 2020](#)), or disagreement discrepancy ([Rosenfeld and Garg, 2023](#)) restrict the compatible laws. Likewise, MMD-credal risk intervals impose RKHS and covariate-shift structure ([Ariq, 2026](#)). Each restriction can shrink the identified set. A converse theorem for a richer class does not contradict those results.

## 2.5 Marginal and Conditional Coverage

Selective classification and reject-option learning abstain on individual predictions ([Geifman and El-Yaniv, 2017](#); [Chow, 1970](#); [Cortes et al., 2016](#)). KGA instead abstains on an update. Its coverage-to-action proposition uses standard interval logic ([Tibshirani et al., 2019](#); [Barber et al., 2023](#); [Bates et al., 2021](#); [Angelopoulos et al., 2025](#)). The statistical burden is to justify coverage for the intended unit, especially after selection or under shift.

Table 1: Neighboring approaches differ in inferential target, decision timing, and guarantee. The comparison is conceptual; the archived empirical ports are not an official synchronized benchmark.

| Approach                         | Target                                          | Decision timing             | Guarantee relevant here                       |
|----------------------------------|-------------------------------------------------|-----------------------------|-----------------------------------------------|
| TTA/ALine;<br>AETTA              | adapted-model performance                       | after candidate evaluation  | performance estimate                          |
| POEM; risk monitors              | running adaptation risk                         | during or after adaptation  | sequential monitoring under named assumptions |
| Partial-identification decisions | payoff set across compatible laws               | decision under ambiguity    | robust-decision context                       |
| K-Bound / KGA                    | population benefit set / predicted cell benefit | before candidate commitment | strict frontier / coverage-conditional action |

Jin and Ren (2025)’s JOMI procedure targets coverage conditional on selection, assuming exchangeability and a compatible selection rule. Gibbs et al. (2025) obtain conditional guarantees over specified shift classes through a different calibration construction. KGA’s ordinary residual interval implements neither method. Its marginal false-adapt implication must therefore remain separate from error conditional on ADAPT and from coverage within each target environment. Learn Then Test addresses the separate problem of configuration selection and multiplicity (Angelopoulos et al., 2025).

### 3 Setup, Notation, and Claim Levels

This section fixes the probability target, evaluation unit, model pair, and gate before defining actions or error events. Table 2 separates the population and empirical symbols used throughout.

#### 3.1 Model Pair, Evidence, and Evaluation Unit

Let  $(\Omega, \mathcal{F}, P_T)$  support a target draw  $(X, Y)$ , and let  $P_S$  denote the source law. For loss  $\ell$ , define

$$R_T(f) = \mathbb{E}_{P_T}[\ell(f(X), Y)].$$

We fix a realized pair of measurable predictors for the population comparison. The frozen predictor is  $f_0$ ; an adapter  $\mathcal{A}$  uses an unlabeled batch  $X_{1:m}$  to construct the fixed candidate  $f_a = \mathcal{A}(f_0, X_{1:m})$ . Population risk then integrates a fresh target draw while holding this pair fixed.

The gate may read label-free evidence

$$Z = \phi(X_{1:m}, f_0, f_a),$$

such as predictions, logits, and update statistics. It may not read the outcome of the cell it is about to score.

Table 2: Population and empirical quantities.  $\beta$  bounds a population residual by assumption;  $\varepsilon$  is an empirical prediction-error radius. They are not interchangeable.

| Quantity                    | Level      | Meaning                                                                |
|-----------------------------|------------|------------------------------------------------------------------------|
| $f_0, f_a$                  | both       | frozen predictor and realized shadow candidate                         |
| $\Delta$                    | population | expected risk reduction $R_T(f_0) - R_T(f_a)$                          |
| $D, \mu_T(D)$               | population | disagreement region and its target probability                         |
| $M$                         | population | label-free mean correctness-score margin on $D$                        |
| $\gamma, \beta$             | population | hidden calibration residual and its externally declared absolute bound |
| $\Delta^{\text{cell}}$      | empirical  | observed score difference on one predeclared evaluation unit           |
| $\hat{\Delta}, \varepsilon$ | empirical  | predicted cell benefit and residual-based interval radius              |
| $\text{FA}_u$               | empirical  | probability of ADAPT and a nonpositive cell benefit                    |
| $\text{FA}_c$               | empirical  | probability of nonpositive cell benefit conditional on ADAPT           |
| $\text{FF}_u$               | empirical  | probability of FREEZE and a nonnegative cell benefit                   |
| $\text{FF}_c$               | empirical  | probability of nonnegative cell benefit conditional on FREEZE          |
| Commitment rate             | empirical  | fraction of ADAPT or FREEZE decisions; not interval coverage           |

**Scope of the matched-world observation experiment.** The population evidence-law and randomized-decision statements use the same fixed observation experiment in each compatible world and the same fixed decision kernel. The observation law factors through the fixed input marginal; finite iid input sampling is one proved realization. Other dependent sampling designs require their own common-law premise. Randomized decisions require a common joint law of observation and random seed; an independent common seed is one sufficient construction, whereas equal seed marginals alone do not establish that joint law. Equality of one-input marginals does not imply equality for arbitrary world-dependent batch couplings. These are the formal model’s premises, not conclusions about the reported data. Holding the model pair fixed in this experiment does not justify conditioning on a pair adapted on that same batch. We do not establish that the transductive benchmark satisfies this observation model: outcome blindness alone supplies neither that conditioning argument nor iid evaluation.

An *evaluation cell*  $j$  is a predeclared collection of records  $\mathcal{E}_j$  on which both models are scored. For a fixed higher-is-better score  $S$ , its measured benefit is

$$\Delta_j^{\text{cell}} = S(f_a; \mathcal{E}_j) - S(f_0; \mathcal{E}_j).$$

This is not the population benefit  $\Delta$ . For accuracy it is an average of paired correctness differences. For macro-F1 it is a difference between two cell-level scores, not an average of per-example losses. We therefore report metric scales separately.

## 3.2 Actions and Their Operational Meaning

**Definition 3.1** (Decision semantics). The gate  $g$  maps its permitted evidence and frozen calibration artifacts to  $\{\text{ADAPT}, \text{FREEZE}, \text{ABSTAIN}\}$ . ADAPT commits  $f_a$  when an interval supports positive benefit. FREEZE rejects  $f_a$  when an interval supports negative benefit. ABSTAIN retains  $f_0$  when neither strict direction is supported or the assessment is unavailable.

Freeze is a supported rejection; abstention is an unresolved assessment. An operator may attach monitoring, label acquisition, or review to an abstention. The experiments here evaluate only the frozen fallback and uncertainty record, not those additional interventions. If evidence is missing, retaining  $f_0$  must be logged as ABSTAIN when the interface returns an unresolved assessment. An integrity failure may instead raise an exception; the caller must preserve the frozen service and record the failure, not a certified FREEZE. Algorithm 1 describes the decision semantics after this integrity boundary has been handled.

The observed-cell oracle chooses the model with the larger score. Regret is the policy’s score shortfall from that oracle, averaged over the named cells with the stated weights. Always-freeze regret is  $\max(\Delta_j^{\text{cell}}, 0)$ ; always-adapt regret is  $\max(-\Delta_j^{\text{cell}}, 0)$ . KGA serves the candidate only after ADAPT. Lower regret is better. A zero-benefit tie has zero regret for either model but supports neither strict sign.

### 3.3 Commitment, Interval Coverage, and Error

Let  $J$  be a random evaluation unit drawn from a predeclared deployment-unit distribution  $Q$ . The common probability space includes fitting and calibration randomness, candidate construction,  $J$ , and the records in  $\mathcal{E}_J$ . Write  $\Delta^{\text{cell}} = \Delta_J^{\text{cell}}$  and  $g = g_J$ . The probabilities below are prospective marginal quantities under that joint law. When we instead summarize a finite recorded grid, we use the equal-weight empirical measure over its cells and call the result descriptive. It does not estimate the prospective law without an additional sampling design.

With this law and the gate fixed, we distinguish three measurements:

$$\text{Commit} = \mathbb{P}(g \in \{\text{ADAPT}, \text{FREEZE}\}), \quad (2)$$

$$\text{Cov}_{\text{cell}} = \mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon), \quad (3)$$

$$\text{FA}_u = \mathbb{P}(g = \text{ADAPT}, \Delta^{\text{cell}} \leq 0). \quad (4)$$

Conditional false adaptation is  $\text{FA}_c = \mathbb{P}(\Delta^{\text{cell}} \leq 0 \mid g = \text{ADAPT})$ , defined only when  $\mathbb{P}(g = \text{ADAPT}) > 0$ . The operational proposition below bounds only the marginal error, and only when the stated interval coverage premise holds. It does not bound  $\text{FA}_c$ . A gate that never adapts has zero false adaptations mechanically, so every error report includes action exposure. A false FREEZE is defined analogously by  $g = \text{FREEZE}$  and  $\Delta^{\text{cell}} \geq 0$ . Its conditional rate is undefined when no FREEZE action occurs.

The population theorem uses binary labels and 0/1 loss. The empirical gate can target a predeclared scalar score difference, including a multiclass metric, but that interface does not extend the binary theorem. The reviewer guide in Appendix L records the multiclass identity. **Six-symbol guide.**  $M$  is the observable population-side disagreement margin;  $\gamma$  is the hidden target calibration residual; and  $\beta$  is an externally declared bound on  $|\gamma|$ . These quantities bracket population benefit  $\Delta$ . At the empirical level,  $\Delta^{\text{cell}}$  is the measured cell benefit,  $\hat{\Delta}$  is its prediction, and  $\varepsilon$  is the calibrated prediction-error radius. Thus  $(M, \gamma, \beta)$  supports population reasoning, whereas  $(\hat{\Delta}, \varepsilon)$  brackets  $\Delta^{\text{cell}}$ ; the two uncertainty objects are not interchangeable.



Table 3: Validity obligations for the paper’s three claim levels. Each row states its target, required premise, supported conclusion, and excluded inference.

| Claim level             | Required premise                                                                           | Supported conclusion                                 | Not implied                                     |
|-------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------|-------------------------------------------------|
| Population frontier     | binary 0/1 loss, fixed augmented evidence, declared residual class; richness for necessity | exact strict-sign frontier within that class         | learning $\beta$ from unlabeled data            |
| Operational proposition | marginal interval coverage for a named scalar target and unit                              | marginal false commitment for that target            | conditional, group, or anytime control          |
| Empirical evaluation    | fixed candidates, splits, metric, and action rule                                          | measured cell regret, inclusion, and action exposure | population transfer or prospective confirmation |

## 4 Population Partial-Identification Frontier

The decision depends on one unknown sign: the sign of  $\Delta$ . Target labels hide that sign.  $M$  supplies the visible direction,  $\gamma$  records what labels still hide, and  $\beta$  limits how large that hidden term may be. Together they yield the exact three-action frontier.

### 4.1 Disagreement-Region Model

The two predictors can differ in risk only where they disagree. We therefore isolate that region before introducing the score and its hidden residual.

**Assumption 4.1** (Deployment setup). Let  $Y \in \{0, 1\}$  and use 0/1 loss. Fix measurable binary predictors  $f_0$  and  $f_a$ . Their disagreement region is  $D = \{x : f_0(x) \neq f_a(x)\}$ ; assume  $\mu_T(D) > 0$ .

Let  $s : \mathcal{X} \rightarrow [0, 1]$  be a fixed measurable candidate-correctness score fitted from source information. The target law has the measurable correctness kernel  $\eta_a(x) = \mathbb{P}_T(f_a(X) = Y \mid X = x)$ . Conditional on  $D$ , define

$$\begin{aligned} M &= \mathbb{E}[s(X) \mid D] - \frac{1}{2}, \\ \gamma &= \mathbb{E}[\eta_a(X) - s(X) \mid D]. \end{aligned} \tag{5}$$

The class radius  $\beta \geq 0$  is supplied externally. It defines  $\mathcal{C}_\beta = \{P_T : |\gamma(P_T)| \leq \beta\}$ . Because  $s \in [0, 1]$ , the feasible margin is  $M \in [-\frac{1}{2}, \frac{1}{2}]$ . The identities below do not assume that  $s$  is source-calibrated.

*Remark 4.2* (Residual identity). By definition,  $M + \gamma = \mathbb{E}[\eta_a(X) \mid D] - \frac{1}{2}$ . This is an algebraic decomposition, not evidence that calibration transfers. We call  $\gamma$  the target disagreement-conditional calibration residual; conditioning can make it nonzero even without distribution shift. The reviewer guide in Appendix L gives the full counterexample and calibration qualification.

**Definition 4.3** (Risk alignment). Evidence is risk-aligned when evidence-identical laws cannot have opposite nonzero benefit signs. This is weaker than uniform strict-sign identification: a fibre may still contain zero and positive benefit. We use the augmented evidence  $\tilde{Z} = (Z, M)$ .

**Definition 4.4** (Strict directional soundness). At a fixed augmented-evidence law, ADAPT is sound when  $\Delta(P) > 0$  for every compatible  $P \in \mathcal{C}_\beta$ , and FREEZE is sound when  $\Delta(P) < 0$  throughout the fibre. A pointwise maximal rule commits exactly when one strict action is sound and abstains otherwise;  $\Delta = 0$  supports neither strict claim.

## 4.2 Label-Kernel Freedom on the Disagreement Region

The central obstacle is now visible. Labels can change on  $D$  while the input distribution and all model outputs stay fixed. The next lemma isolates that freedom.

**Lemma 4.5** (Label-kernel freedom on disagreement). *Fix  $(\mu_T, P_S, f_0, f_a, s)$ , and therefore fix  $D$  and  $M$ . Also fix the conditional label law outside  $D$ . For every measurable  $\eta : D \rightarrow [0, 1]$ , there is a target law with input marginal  $\mu_T$  and  $\mathbb{P}(f_a(X) = Y \mid X = x) = \eta(x)$  on  $D$ . Across these laws, every label-free observable formed from the input batch, fixed models, and model outputs under the declared common observation experiment has the same distribution. Meanwhile,*

$$\gamma = \mathbb{E}[\eta(X) \mid D] - \frac{1}{2} - M.$$

*Proof.* On  $D$ , one binary predictor outputs 0 and the other outputs 1. Assign conditional probability  $\eta(x)$  to  $Y = f_a(x)$  and  $1 - \eta(x)$  to  $Y = f_0(x)$ . Retain the fixed label kernel outside  $D$ . This construction changes neither the input marginal nor any model output. The declared observation law factors through that fixed marginal, so its label-free evidence laws are preserved. Equation (5) gives the displayed expression for  $\gamma$ .  $\square$

Consequently, every statistic formed from target inputs and fixed model outputs under this observation experiment has unchanged law, although the models' relative correctness can change.

## 4.3 Benefit Reduction and Matched-Evidence Impossibility

The first result connects the decision target to the three population symbols. Because the models agree off  $D$  and binary correctness is complementary on  $D$ , benefit depends only on  $M + \gamma$ .

**Lemma 4.6** (Disagreement-region reduction). *Under Assumption 4.1, with  $\bar{a} := \mathbb{P}_T(f_a=Y \mid D)$ ,*

$$\begin{aligned} \Delta &= 2\mu_T(D)\left(\bar{a} - \frac{1}{2}\right), \\ \text{sign } \Delta &= \text{sign}(\bar{a} - \frac{1}{2}) = \text{sign}(M + \gamma). \end{aligned}$$

*The identity  $\text{sign } \Delta = \text{sign}(M + \gamma)$  holds for every target law; it does not assume  $|\gamma| \leq \beta$ .*

*Proof.* Outside  $D$ , the two predictors agree and incur the same 0/1 loss. On  $D$ , exactly one is correct. The adapted and frozen accuracies conditional on  $D$  are therefore  $\bar{a}$  and  $1 - \bar{a}$ . Hence  $\Delta = 2\mu_T(D)(\bar{a} - \frac{1}{2})$ . The definitions of  $M$  and  $\gamma$  give

$$\begin{aligned} M + \gamma &= \left(\mathbb{E}_{\mu_T|D}[s] - \frac{1}{2}\right) + \mathbb{E}_{\mu_T|D}[\eta_a - s] \\ &= \mathbb{E}_{\mu_T|D}[\eta_a] - \frac{1}{2} = \bar{a} - \frac{1}{2}. \end{aligned}$$

Because  $\mu_T(D) > 0$ , the three quantities have the same sign.  $\square$

The reduction tells us which hidden quantity controls the sign. The harder question is identification. Can two target worlds agree on every permitted observation while assigning opposite signs to  $\Delta$ ? Inside the declared band, the answer is yes.

**Theorem 4.7** (Interior matched-evidence impossibility). *Fix  $\beta > 0$ . For any feasible margin  $M \in [-\frac{1}{2}, \frac{1}{2}]$  with  $|M| < \beta$ , there exist  $P_T^+, P_T^- \in \mathcal{C}_\beta$  with the following properties:*

$$\begin{aligned} \text{Law}(\tilde{Z} \mid P_T^+) &= \text{Law}(\tilde{Z} \mid P_T^-), \\ M(P_T^+) &= M(P_T^-) = M, \\ \text{sign } \Delta_+ &= -\text{sign } \Delta_- \neq 0, \end{aligned}$$

where  $\tilde{Z} = (Z, M)$ . If  $\beta = 0$ , then  $\mathcal{C}_0$  forces  $\gamma = 0$  and  $\text{sign } \Delta = \text{sign } M$ ; matched-evidence ambiguity therefore requires  $\beta > 0$ .

*Proof.* Choose  $0 < \delta < \min\{\beta - |M|, \frac{1}{2}\}$ . For  $\theta \in \{+1, -1\}$ , apply Lemma 4.5 with the constant correctness kernel

$$\eta_\theta(x) = \frac{1}{2} + \theta\delta, \quad x \in D,$$

and use the same fixed label kernel outside  $D$ . Denote the resulting target law by  $P_T^\theta$ . The input marginal, predictors, score, and source law are fixed, so  $M$  is the same under  $P_T^+$  and  $P_T^-$ . Lemma 4.5 also gives the same law of  $Z$  in the two worlds. Consequently,  $\text{Law}(\tilde{Z} \mid P_T^+) = \text{Law}(\tilde{Z} \mid P_T^-)$  for  $\tilde{Z} = (Z, M)$ .

It remains to verify class membership and the benefit signs. Under  $P_T^\theta$ ,  $\bar{a}_\theta = \frac{1}{2} + \theta\delta$  and

$$\gamma_\theta = \theta\delta - M, \quad |\gamma_\theta| \leq \delta + |M| < \beta.$$

Thus both laws belong to  $\mathcal{C}_\beta$ . Lemma 4.6 yields  $\Delta_\theta = 2\mu_T(D)\theta\delta$ . The two benefits are nonzero and have opposite signs.  $\square$

The evidence laws coincide exactly—their total-variation distance is zero—so this is exact non-identifiability, not a rate-dependent testing bound. Randomization cannot resolve two worlds that induce exactly the same evidence law. Controlling both directional errors therefore requires abstention.

**Corollary 4.8** (Matched-evidence abstention lower bound). *Fix  $\alpha \in (0, \frac{1}{2})$  and the shared evidence law from Theorem 4.7. Let  $g$  be any possibly randomized label-free rule. If, in every admissible world,*

$$\begin{aligned} \mathbb{P}[g = \text{ADAPT}, \Delta \leq 0] &\leq \alpha, \\ \mathbb{P}[g = \text{FREEZE}, \Delta \geq 0] &\leq \alpha, \end{aligned}$$

then  $\mathbb{P}[g = \text{ABSTAIN}] \geq 1 - 2\alpha$ .

*Proof.* The rule has the same action probabilities in both worlds. Its ADAPT probability is at most  $\alpha$  in the negative world. Its FREEZE probability is at most  $\alpha$  in the positive world. The three action probabilities sum to one.  $\square$

## 4.4 Boundary Case

Inside the band, both strict signs may be feasible. At the boundary, zero benefit is feasible. These constructions differ, but both prevent a uniform strict direction.

**Proposition 4.9** (Boundary case and closed-band abstention). *Fix  $\alpha \in (0, \frac{1}{2})$ . Under Definition 4.4, abstention is the pointwise maximal sound action throughout  $|M| \leq \beta$ . Any label-free rule that controls both marginal directional errors at level  $\alpha$  must abstain there with probability at least  $1 - 2\alpha$ . Opposite nonzero benefit signs are asserted only when  $|M| < \beta$ .*

*Proof.* For  $|M| < \beta$ , apply Theorem 4.7 and Corollary 4.8.

Now suppose  $|M| = \beta > 0$ . The declared class contains a zero-benefit world with  $\gamma = -M$  and a strict-benefit world with  $\gamma = 0$ . These worlds have the same augmented-evidence law. In the zero-benefit world, both directional error events occur whenever the rule commits. Dual error control therefore gives  $\mathbb{P}[\text{ADAPT}] \leq \alpha$  and  $\mathbb{P}[\text{FREEZE}] \leq \alpha$ . It follows that  $\mathbb{P}[\text{ABSTAIN}] \geq 1 - 2\alpha$ , and neither strict action is uniformly sound.

Finally, if  $M = \beta = 0$ , then  $\mathcal{C}_0$  forces  $\gamma = 0$  and  $\Delta \equiv 0$ . The same conclusion follows directly.  $\square$

This convention is stricter than task regret. When  $\Delta = 0$ , adapt and freeze are risk-equivalent, but neither certifies a strict sign.

## 4.5 Exact Strict-Commitment Frontier

We can now answer the population question completely. The theorem separates four jobs: reduction, sufficiency outside the band, necessity inside the band, and maximality of the resulting rule.

**Theorem 4.10** (Exact strict-commitment frontier). *Under Assumption 4.1, fix  $(\mu_T, P_S, f_0, f_a, s)$  and let  $\beta \geq 0$ . Necessity and maximality hold for  $\mathcal{C}_\beta$ , or for a subclass closed under the constructions in Theorem 4.7 and Proposition 4.9; otherwise only sufficiency follows without an additional richness assumption.*

- (i) **Reduction.** *For every  $P_T$ ,  $\text{sign } \Delta = \text{sign}(M + \gamma)$  (Lemma 4.6).*
- (ii) **Sufficiency.** *If  $|M| > \beta$  and  $P_T \in \mathcal{C}_\beta$ , then  $\text{sign } \Delta = \text{sign } M$ .*
- (iii) **Necessity.** *If  $|M| < \beta$ , two laws in  $\mathcal{C}_\beta$  have the same augmented-evidence law and opposite nonzero benefit signs. If  $|M| = \beta > 0$ , the admissible residual  $\gamma = -\text{sign}(M)\beta$  gives  $\Delta = 0$ . If  $M = \beta = 0$ , then  $\Delta \equiv 0$  throughout  $\mathcal{C}_0$ . Thus  $|M| \leq \beta$  supports no uniform strict direction.*
- (iv) **Strict-commitment iff.** *At the fixed augmented-evidence law, a strict action is uniformly directionally sound if and only if  $|M| > \beta$ . On the closed band, abstention is the pointwise maximal sound action in the sense of Definition 4.4.*

*The population rule  $M > \beta \Rightarrow \text{ADAPT}$ ,  $M < -\beta \Rightarrow \text{FREEZE}$ ,  $|M| \leq \beta \Rightarrow \text{ABSTAIN}$  is the pointwise maximal sound rule on  $\mathcal{C}_\beta$ .*

*Proof.* Part (i) is Lemma 4.6. For (ii), suppose  $M > \beta$  and  $|\gamma| \leq \beta$ . Then  $M + \gamma > 0$ , so Lemma 4.6 gives  $\Delta > 0$ . The negative case is symmetric.

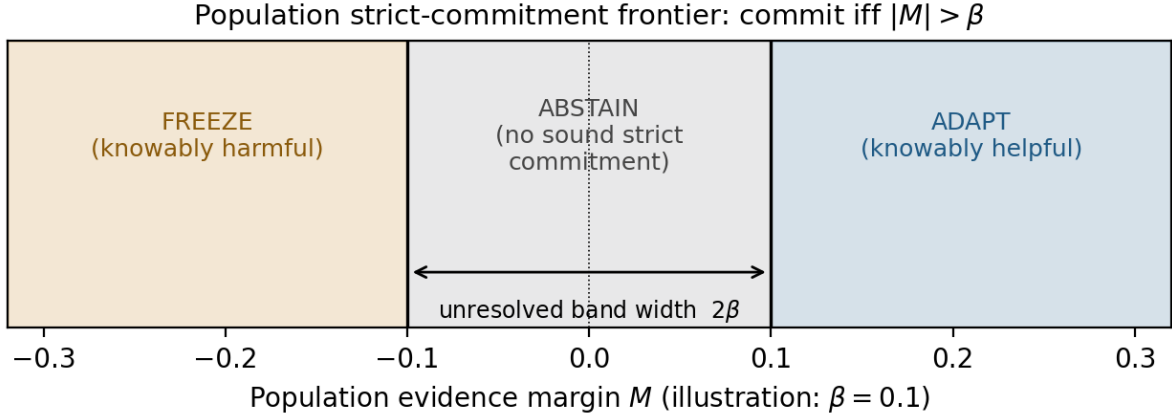


Figure 2: Population strict-commitment frontier for a fixed binary model pair under 0/1 loss. ADAPT is uniformly supported when  $M > \beta$ , FREEZE when  $M < -\beta$ , and neither strict direction is supported on  $|M| \leq \beta$ . This population result is not KGA’s numerical cell-level rule.

For (iii), Theorem 4.7 supplies opposite nonzero signs when  $|M| < \beta$ . At  $|M| = \beta > 0$ , the boundary law with  $\gamma = -\text{sign}(M)\beta$  has zero benefit. When  $M = \beta = 0$ , every law in  $\mathcal{C}_0$  has zero benefit. These cases preclude a uniform strict direction throughout the closed band.

Part (iv) combines (ii), (iii), and Proposition 4.9. For a nonempty compatible subclass, sufficiency gives the unique sound strict direction outside the band. This individual directional uniqueness requires a realized member: on an empty subclass, both universal directional statements are vacuous. The full compatible class is nonempty. The impossibility and boundary results rule out both strict directions on the band.  $\square$

Outside the band, the residual cannot reverse  $M$ ; on the band, the same evidence admits a failed strict claim, so abstention is part of the identified answer. Let  $d = \mu_T(D) > 0$ . The full class identifies

$$\Delta(\mathcal{C}_\beta) = 2d \left[ \max\left\{-\frac{1}{2}, M - \beta\right\}, \min\left\{\frac{1}{2}, M + \beta\right\} \right].$$

The bracket equals  $[-\frac{1}{2}, \frac{1}{2}] \cap [M - \beta, M + \beta]$ , and constant correctness kernels attain every point. A strict direction exists exactly when this interval excludes zero.

| Region            | Meaning                        | Action  |
|-------------------|--------------------------------|---------|
| $M > \beta$       | strictly positive benefit      | ADAPT   |
| $M < -\beta$      | strictly negative benefit      | FREEZE  |
| $ M  < \beta$     | both strict signs are feasible | ABSTAIN |
| $ M  = \beta > 0$ | zero benefit is feasible       | ABSTAIN |
| $M = \beta = 0$   | benefit is identically zero    | ABSTAIN |

## 4.6 Interpretation and Limits of the Frontier

The theorem is exact only for the declared class. Choosing  $\beta = 0$  assumes a zero target disagreement-conditional residual. It is not a conservative default. Without a credible upper bound on  $|\gamma|$ , the frontier cannot support a robust strict action. More unlabeled samples do not close this gap because the missing information lies in the target label kernel.

Empirical abstention does not prove structural unknowability. A finite-sample interval may be wide because calibration data are scarce, prediction is poor, or transfer fails. This distinction leads to the next question: can the required population residual bound be learned from the same label-free evidence?

## 5 What Label-Free Evidence Cannot Learn

The population frontier becomes usable only after  $\beta$  has been declared. Can the same label-free evidence learn a smaller valid budget? An audit must cover every residual value left unresolved on its evidence fibre.

### 5.1 Audit Setup and Evidence Fibres

Let  $W$  collect everything a deployment-time audit may read: the unlabeled target batch, fixed models, score function, and derived label-free statistics. It may also contain a source or calibration sample whose law does not depend on the target label kernel. A randomized procedure returns a measurable real-valued audit  $\hat{\beta} = g(W, U) \geq 0$ , where the seed  $U$  has the same distribution in every world and is independent of  $W$  in every world.

For a declared class  $\mathcal{C}$  and an observable-law value  $z$ , define the evidence fibre

$$\mathcal{C}(z) = \{P \in \mathcal{C} : \text{Law}_P(W) = z\}$$

and its unresolved residual radius

$$\Gamma_z(\mathcal{C}) = \sup_{P \in \mathcal{C}(z)} |\gamma(P)|. \quad (6)$$

The fibre contains target laws that the audit cannot distinguish. Its radius is the smallest uniform bound on  $|\gamma|$  that this observable law can support.

### 5.2 Exact Fibre-Wise Audit Floor

The answer is a floor, not an estimator. Any audit that is valid throughout one evidence fibre must cover the largest residual that remains possible in that fibre.

**Theorem 5.1** (Exact label-free audit floor). *Fix a nonempty fibre  $\mathcal{C}(z)$  with bounded attainable residuals  $\{|\gamma(P)| : P \in \mathcal{C}(z)\}$  and  $\delta \in [0, 1)$ . Let  $\mathbb{P}_z$  denote the common joint law of  $(W, U)$  on this fibre. Suppose the measurable nonnegative audit  $\hat{\beta}$  satisfies  $\mathbb{P}_P(|\gamma(P)| \leq \hat{\beta}) \geq 1 - \delta$  for every  $P \in \mathcal{C}(z)$ . Then*

$$\mathbb{P}_z(\hat{\beta} \geq \Gamma_z(\mathcal{C})) \geq 1 - \delta.$$

The constant audit  $\hat{\beta} \equiv \Gamma_z(\mathcal{C})$  is valid with  $\delta = 0$ . Consequently, on the displayed high-probability event, an audited frontier can commit only where the frontier with radius  $\Gamma_z(\mathcal{C})$  already commits.

*Proof.* The common joint law of  $(W, U)$  induces the same law for  $\hat{\beta}$  in every world. Choose  $P_n \in \mathcal{C}(z)$  so that  $|\gamma(P_n)| \uparrow \Gamma_z(\mathcal{C})$ . Under the common law, each event  $\{\hat{\beta} \geq |\gamma(P_n)|\}$  has probability at least  $1 - \delta$ . These events decrease to  $\{\hat{\beta} \geq \Gamma_z(\mathcal{C})\}$ , so continuity from above proves the bound. The constant audit is valid by the definition of the supremum. Finally,  $\hat{\beta} \geq \Gamma_z$  and  $|M| > \hat{\beta}$  imply  $|M| > \Gamma_z$ .  $\square$

The theorem does not prohibit useful budgets. An audit using the same observable law must cover every residual left possible on that fibre. A smaller bound requires labels or structure that genuinely refines the compatible set;  $\Gamma_z(\mathcal{C})$  is an oracle fibre benchmark, not a general finite-batch estimator.

**Declared budget and abstention radius.** For the full declared class  $\mathcal{C}_\beta = \{P : |\gamma(P)| \leq \beta\}$  with  $0 \leq \beta \leq \frac{1}{2}$ , feasible  $M \in [-\frac{1}{2}, \frac{1}{2}]$ , and positive disagreement mass, the label-kernel construction in Lemma 4.5 gives

$$\Gamma_z(\mathcal{C}_\beta) = \beta. \quad (7)$$

Thus the least valid fibre-wide residual bound equals the population abstention radius,  $\beta$ . This connection does not turn the declared  $\beta$  into an estimate.

**When calibration labels do not shrink the target fibre.** A labeled calibration domain does not reduce the target fibre unless the declared class links its label kernel to deployment. Lemma 4.5 can otherwise vary deployment labels while leaving the calibration sample unchanged. Labels therefore help through representativeness or structural coupling, not merely through their existence.

**What additional structure can buy.** Extra information can shrink the fibre, but it changes the problem. The available routes are representative deployment labels, exchangeable historical deployment episodes, a structural cross-domain coupling, or a domain-specific physical bound. Each gain comes from labels or assumptions, not the current unlabeled batch.

### 5.3 Implications for KGA

KGA does not estimate  $M$ ,  $\gamma$ , or  $\beta$ , and does not apply  $|M| > \beta$  numerically. It predicts  $\Delta^{\text{cell}}$  and uses an empirical radius  $\varepsilon$ . Transferring cell-level coverage to population benefit requires an additional sampling-error argument that the reported experiments do not provide. The next section therefore changes the question from population identification to deployment-time prediction.

## 6 KGA: An Empirical Benefit-Interval Gate

For the current evaluation unit, can KGA support which side of zero the benefit lies on? It predicts cell benefit from label-free evidence and attaches a residual-calibrated radius.

### 6.1 Shared Decision Logic, Different Targets

**Scope of KGA.** KGA predicts measured evaluation-cell benefit  $\Delta^{\text{cell}}$ , not population benefit  $\Delta$ . Its radius  $\varepsilon$  describes empirical prediction error under a named calibration protocol. It neither estimates the population residual bound  $\beta$  nor establishes population-risk coverage.

Both layers commit only when their respective uncertainty set supports one strict sign, but they use different targets:

$$\begin{aligned}\Delta &= 2\mu_T(D)(M + \gamma), \\ \Delta^{\text{cell}} &= S(f_a; \mathcal{E}) - S(f_0; \mathcal{E}), \quad \hat{\Delta} = h_\theta(Z).\end{aligned}\tag{8}$$

Development cells fit  $h_\theta$ , and disjoint residual-calibration cells supply residuals. A scored cell contributes  $Z$ , but not its outcome. The empirical coverage premise is

$$\mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon) \geq 1 - \alpha.\tag{9}$$

Proposition 6.1 applies only when this premise is justified.

### 6.2 The Coverage-to-Action Proposition

**Proposition 6.1** (Coverage-to-action implication). *Fix the scalar benefit target and its evaluation unit before calibration. Let  $B$  and  $\hat{B}$  be finite real random variables, and let  $\varepsilon \in [0, \infty]$  be a measurable random radius on a common probability space containing all calibration, benefit-model fitting, candidate-construction, and evaluation-unit randomness. The prediction  $\hat{B}$  and radius  $\varepsilon$  must be available without reading the new unit's evaluation labels. For  $\alpha \in (0, 1)$ , suppose the interval satisfies marginal coverage*

$$\mathbb{P}[|\hat{B} - B| \leq \varepsilon] \geq 1 - \alpha.$$

Define

$$g = \begin{cases} \text{ADAPT} & \hat{B} - \varepsilon > 0, \\ \text{FREEZE} & \hat{B} + \varepsilon < 0, \\ \text{ABSTAIN} & \text{otherwise.} \end{cases}$$

Then, marginally over that common probability space,

$$\begin{aligned}\mathbb{P}[g = \text{ADAPT}, B \leq 0] &\leq \alpha, \\ \mathbb{P}[g = \text{FREEZE}, B \geq 0] &\leq \alpha.\end{aligned}$$



*Proof.* On the event  $\{|\hat{B} - B| \leq \varepsilon\}$ , the condition  $\hat{B} - \varepsilon > 0$  implies  $B > 0$ , so a false adapt requires the complement of this event, which has probability at most  $\alpha$ . The freeze case is symmetric.  $\square$

For the empirical panels,  $B = \Delta^{\text{cell}}$  and  $\hat{B} = \hat{\Delta}$ . A population-risk application instead sets  $B = \Delta$  and requires coverage for that quantity; cell-outcome coverage alone does not provide it.

*Remark 6.2* (Scope of the implication). Proposition 6.1 controls marginal error for the declared target. It does not control error conditional on commitment, group-specific coverage, candidate-selection multiplicity, or repeated-time behavior. The coverage premise needs separate justification. When it is unsupported, a declared fallback may retain  $f_0$  and record ABSTAIN, but that response does not prove  $f_0$  safe or optimal and is not a certified FREEZE decision. For this certificate, a population sign claim requires interval coverage for population benefit itself, or a justified cell-to-population transfer bound. Once that coverage premise holds, the event-containment argument applies directly; it does not need an additional risk-alignment assumption.

### 6.3 Cell-to-Population Transfer

Cell coverage alone does not imply population coverage. The following proposition states the additional sampling premise needed to change the target while preserving the three-way decision semantics. It is a conditional extension, not a replacement of KGA’s cell-level calibration. The evaluation unit, target-environment population, model-pair construction protocol, score functional, and units must be declared before calibration.

**Proposition 6.3** (Cell-to-population transfer). *Let  $\hat{\Delta}$ ,  $\Delta^{\text{cell}}$ , and  $\Delta$  be real-valued measurable quantities on a common probability space, with cell and population benefit defined for the same model pair and score functional. Let  $\varepsilon, b \geq 0$  be finite measurable radii. Suppose that  $\alpha_{\text{cell}}, \delta \in [0, 1)$ ,  $\alpha_{\text{cell}} + \delta < 1$ , and*

$$\begin{aligned} \mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon) &\geq 1 - \alpha_{\text{cell}}, \\ \mathbb{P}(|\Delta^{\text{cell}} - \Delta| \leq b) &\geq 1 - \delta. \end{aligned} \tag{10}$$

*Then, without requiring independence of these two events,*

$$\mathbb{P}(|\hat{\Delta} - \Delta| \leq \varepsilon + b) \geq 1 - \alpha_{\text{cell}} - \delta. \tag{11}$$

*If  $\hat{\Delta}$ ,  $\varepsilon$ , and  $b$  are available before the new evaluation labels, define  $L_{\text{pop}} = \hat{\Delta} - (\varepsilon + b)$  and  $U_{\text{pop}} = \hat{\Delta} + (\varepsilon + b)$ . The rule*

$$g_{\text{pop}} = \begin{cases} \text{ADAPT}, & L_{\text{pop}} > 0, \\ \text{FREEZE}, & U_{\text{pop}} < 0, \\ \text{ABSTAIN}, & \text{otherwise} \end{cases} \tag{12}$$

*satisfies*

$$\begin{aligned} \mathcal{E}_{\text{pop}} &= \{g_{\text{pop}} = \text{ADAPT}, \Delta \leq 0\} \\ &\quad \cup \{g_{\text{pop}} = \text{FREEZE}, \Delta \geq 0\}, \\ \mathbb{P}(\mathcal{E}_{\text{pop}}) &\leq \alpha_{\text{cell}} + \delta. \end{aligned} \tag{13}$$

*Proof.* On the intersection of the two premise events, the triangle inequality gives  $|\hat{\Delta} - \Delta| \leq |\hat{\Delta} - \Delta^{\text{cell}}| + |\Delta^{\text{cell}} - \Delta| \leq \varepsilon + b$ . The complement of that intersection has probability at most  $\alpha_{\text{cell}} + \delta$  by the union bound. On the intersection,  $L_{\text{pop}} > 0$  implies  $\Delta > 0$  and  $U_{\text{pop}} < 0$  implies  $\Delta < 0$ , so either false strict direction requires failure of at least one premise event.  $\square$

The bound is marginal over the declared evaluation unit and its protocol randomness, not conditional on commitment, groupwise, or anytime-valid. The radii need separately justified coverage premises; computing their sum does not establish those premises. No such sampling-error bound is established for the reported panels, whose gates continue to use  $\varepsilon$ , not  $\varepsilon + b$ . An unavailable premise or non-finite radius therefore supports no population certificate: the operational fallback is ABSTAIN while retaining  $f_0$ , not a certified FREEZE. The sampling radius  $b$  is distinct from the population residual bound  $\beta$ . Appendix A records the sampling and score-functional restrictions.

## 6.4 Evidence, Calibration, and the Action Rule

Each candidate-specific model uses label-free summaries of confidence, class balance, model disagreement, distribution shift, and update magnitude. Natural tracks retain protocol-specific models and schemas. The evidence inventory in Appendix I records their exact specifications.

Given residuals  $r_i = |\hat{\Delta}_i - \Delta_i^{\text{cell}}|$ , ordered as  $r_{(1)} \leq \dots \leq r_{(n)}$ , the order-statistic rule is

$$k = \lceil (n+1)(1-\alpha) \rceil, \quad \varepsilon = \begin{cases} r_{(k)}, & k \leq n, \\ +\infty, & k > n. \end{cases} \quad (14)$$

The rank rule has split-conformal coverage when calibration and next-unit scores are jointly exchangeable under the full fitting and candidate-construction protocol (Tibshirani et al., 2019; Barber et al., 2023); here  $n$  counts cells, not images. Disjoint fit, calibration, and check labels prevent the specified reuse; this does not establish exchangeability. The painting-development inclusion rates below are descriptive checks on opened cells, not validation of group-conditional, repeated-use, or population-risk coverage.

A deterministic five-fold cross-fit is keyed by stable cell IDs. For each score fold, its complement is divided into disjoint estimator-fit and residual-calibration subsets. Fit outcomes determine the predictor, and the exact-rank rule on calibration residuals determines the radius. No scored outcome can affect its own prediction, radius, or action; the whole-pipeline invariance test verifies this property. This separation blocks direct outcome flow but does not create independent environments or establish exchangeability. Natural tracks use separate partitions only where their archived protocols record them.

For radius 0.01, predictions 0.04,  $-0.04$ , and 0.005 produce ADAPT, FREEZE, and ABSTAIN, respectively. These values illustrate the rule; they are not experimental results.

Runtime includes shadow adaptation, two-model inference, evidence extraction, and storage; comparable end-to-end latency measurements are unavailable.

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**Algorithm 1** KGA decision for one proposed candidate

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**Require:** Frozen  $f_0$ , adapter  $\mathcal{A}$ , unlabeled  $X$ , frozen  $h_\theta$  and calibration artifacts

- 1: Create  $f_a \leftarrow \mathcal{A}(f_0, X)$  in isolated shadow state
  - 2: Extract  $Z = \phi(X, f_0, f_a)$  and validate the assessment inputs
  - 3: **if** the assessment is unavailable **then**
  - 4:     retain  $f_0$ , log the reason, and return ABSTAIN
  - 5: **end if**
  - 6: Obtain  $\varepsilon$  from the calibrated residual rule in Eq. (14)
  - 7:  $\hat{\Delta} \leftarrow h_\theta(Z)$
  - 8:  $L \leftarrow \hat{\Delta} - \varepsilon$ ;  $U \leftarrow \hat{\Delta} + \varepsilon$
  - 9: **if**  $L > 0$  **then**
  - 10:     commit  $f_a$  and return ADAPT
  - 11: **else if**  $U < 0$  **then**
  - 12:     discard  $f_a$  and return FREEZE
  - 13: **else**
  - 14:     retain  $f_0$ , log uncertainty, and return ABSTAIN
  - 15: **end if**
- 

Table 4: Evidence status and admissible claim. “Primary” means central to the paper, not prospectively confirmatory; unfavorable and incomplete studies remain in scope.

| Evidence                   | Design and status                                                                                                          | Question it can answer                                                    |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| CIFAR-10-C                 | opened, dependent, constructed; cell-outcome-disjoint cross-fit                                                            | retrospective candidate-specific routing                                  |
| CCT-20                     | cell-outcome-unopened, internally sealed execution; aggregate target metadata previously inspected                         | conservative retention under the locked endpoint                          |
| So2Sat-LCZ42               | development stop before target access                                                                                      | candidate failed eligibility                                              |
| DomainNet painting         | opened development; five checkpoints on shared cells and separate reference-source candidates; fixed follow-up diagnostics | candidate feasibility, abstention, and mechanism limits; not confirmation |
| Auxiliary replays          | opened or one-sided diagnostics                                                                                            | transfer limits and supporting behavior                                   |
| Invalid or partial records | PACS partial; iWildCam metric invalid; historical ports superseded; Office-Home serialization invalid                      | audit history only                                                        |

## 7 Primary Experimental Protocols

### 7.1 Two Primary Questions and the Evidence Hierarchy

The CIFAR-10-C diagnostic asks whether KGA distinguishes helpful from harmful updates on an opened constructed grid. CCT-20 asks whether a frozen gate avoids measured candidate harm across camera locations. The first exercises directional branches retrospectively; the second evaluates governed retention and may pass without ADAPT exposure.

The opened, dependent, constructed CIFAR-10-C diagnostic (Hendrycks and Dietterich, 2019) uses six corruption families at severities 1 and 5. It crosses three batch sizes (200, 16,

and 8), three compositions (IID, imbalanced, and single-class), two update schedules (10 steps at  $10^{-3}$  and 50 at  $2.5 \times 10^{-3}$ ), and two repeats: 432 cells per RNG seed and 2,160 per candidate over seeds 0–4.

Tent, EATA, and SAR are local ports applied to one ResNet-18 checkpoint; the five seeds repeat streams and adapter randomness, not checkpoint training. Point estimates weight cells equally; family sensitivity first averages within each family and then weights the six means equally.

The recorded candidates are local, protocol-matched composite operating points rather than untouched reproductions of published defaults. They combine evaluation-batch BatchNorm statistics with normalization-affine updates; Appendix D records the contract and unexecuted BatchNorm-only control.

The three-way cross-fit removes each scored fold’s outcomes from fitting and residual calibration, but does not make the shared checkpoint, corpus, or cells independent. This is a retrospective diagnostic, not prospective confirmation or an untouched deployment trial.

## 7.2 CCT-20 Design and Locked Utility-Preservation Endpoint

The CCT-20 study (Beery et al., 2018) crosses 5 independent checkpoints with 9 target locations, giving 45 cells and 6950/16325 probe/evaluation images (23275 total). Set-membership top-1 accuracy credits any archived category, not an official single-label target. Official Tent makes one non-episodic Adam update per batch at  $10^{-3}$ ; pre-update logits score it, state carries within a cell, and resets between cells.

Individual outcomes remained unopened until one-shot scoring, but aggregate target-label metadata had previously been inspected; CCT-20 was therefore cell-outcome-unopened and internally sealed, not globally label-unopened or publicly preregistered. Ridge fitting used 10 rows from two camera locations and calibration used 45 rows across nine others; transfer to target locations is assumed.

For fixed policy  $b$ ,  $G_b$  is mean baseline regret minus mean KGA regret. The limited, predeclared utility check uses nominal pointwise 95% intervals and passes when their lower endpoints satisfy

$$L_{\text{adapt}} > 0, \quad L_{\text{freeze}} > -0.005. \quad (15)$$

The second threshold is frozen-policy noninferiority; this check can pass when KGA always serves  $f_0$  and is not a population guarantee.

The separate strong-success rule requires complete data for all 45 cells, nominal 97.5% Bonferroni intervals above zero against both fixed policies, both exact nine-location tests surviving Holm at 0.05, ADAPT and FREEZE rates at least 0.10, and total commitment at least 0.30. It therefore requires interval-supported improvement over both fixed policies; any failure precludes strong success.

The CCT utility table 6 uses pointwise 95% intervals; the separate strong-success audit 27 uses 97.5% Bonferroni intervals. They are not alternate versions of the same interval.

**Scoring and inference.** Actions are fixed before outcomes are joined, but candidate TTA is transductive because it reads unlabeled evaluation inputs. CIFAR sensitivity resamples the six corruption-family means conditional on the archived checkpoint. CCT-20 uses a paired

Table 5: Opened, dependent CIFAR-10-C diagnostic over 2,160 equally weighted cells per candidate. Lower regret is better. A/F/U gives action exposure; directional errors use the corresponding action denominator. Tent’s all-cell errors are 2/2,160 for false ADAPT and 1/2,160 for false FREEZE. All cells reuse one checkpoint, so this is retrospective rather than independent environment validation. Canonical panel SHA-256: `35d4c165843de1ece3cb35fb4ac50dbfcbfa33b646754e2b6d133fbdfa78c6e`.

| Candidate | $n$  | KGA    | Adapt  | Freeze | A/F/U        | False A/A | False F/F | Commitment |
|-----------|------|--------|--------|--------|--------------|-----------|-----------|------------|
| Tent      | 2160 | 0.0019 | 0.0080 | 0.1239 | 1091/337/732 | 2/1091    | 1/337     | 0.6611     |
| EATA      | 2160 | 0.0016 | 0.0033 | 0.1313 | 1207/105/848 | 0/1207    | 0/105     | 0.6074     |
| SAR       | 2160 | 0.0018 | 0.0003 | 0.1405 | 1414/0/746   | 0/1414    | –         | 0.6546     |

product bootstrap over checkpoint rows and location columns; its separate exact sign-flip reference calculation uses nine location means and requires a stronger joint sign-symmetry condition than exchangeability alone. All reported levels are nominal. The inference audit in Appendix F gives the metrics, sign convention, full assumptions, and retrospective status (Hemerik and Goeman, 2018).

## 8 Primary Results

### 8.1 CIFAR-10-C: Directional Actions and Candidate-Specific Utility

Table 5 first exposes the ADAPT/FREEZE/ABSTAIN record under one current rule. Tent makes two false ADAPT decisions and one false FREEZE; EATA makes neither; SAR makes no FREEZE decision, so that error direction is unexercised. These counts require their action and all-cell denominators below. Utility is a secondary measured consequence: Tent and EATA have lower point regret than both fixed policies, whereas always-adapt has lower regret than KGA for SAR. The panel therefore supports only candidate-specific, retrospective action and utility results.

Tent regret is 0.0019, compared with 0.0080 for always-adapt and 0.1239 for always-freeze. EATA’s values are 0.0016, 0.0033, and 0.1313. Their ADAPT/FREEZE/ABSTAIN counts are 1,091/337/732 and 1,207/105/848, respectively. Tent makes two false ADAPT decisions; EATA makes none. SAR makes 1,414/0/746 decisions. Its KGA regret is 0.0018, compared with 0.0003 for always-adapt and 0.1405 for always-freeze. Abstention can prevent harmful updates while still forgoing helpful ones.

The nominal six-family intervals distinguish Tent from EATA. Gaps are baseline regret minus KGA regret, so positive values favor KGA. Tent’s gap versus always-adapt is 0.0061, with nominal family-bootstrap interval  $[0.0021, 0.0095]$ ; its interval versus always-freeze is also strictly positive. EATA’s interval versus always-adapt crosses zero,  $[-0.0013, 0.0044]$ , although its interval versus always-freeze is positive. Tent therefore has the favorable nominal two-baseline pattern; EATA does not.

The retrospective reference tests do not make either pattern confirmatory. Across the six candidate-by-baseline contrasts in the retrospective comparison, Tent’s Holm values are

0.140625 against always-adapt and 0.09375 against always-freeze. EATA’s are 0.3125 and 0.09375, respectively. The replay, family unit, tests, and multiplicity analysis were selected after the grid was open, and all six families share one checkpoint. Table 23 reports every interval and reference value. Nominal interval exclusion and retrospective Holm adjustment answer different questions; neither establishes prospective coverage.

## 8.2 Retrospective Interval Diagnostics

Retrospective inclusion is 1,976/2,160 for Tent, 1,950/2,160 for EATA, and 1,949/2,160 for SAR. These values describe opened cross-fit cells; they are not independent validation of coverage on new environments. The inference appendix F reports interval widths and family-level diagnostics. Tent’s two false ADAPT actions correspond to marginal error 2/2160 and conditional error 2/1091. It also makes one false FREEZE among 337 FREEZE decisions. EATA makes no false ADAPT and no false FREEZE. SAR makes no false ADAPT and no FREEZE decisions, so its conditional false-freeze rate is undefined. Exposure and denominators are essential: a zero count alone is not evidence of safety.

## 8.3 CCT-20: Conservative Retention, Not Selective Routing

All 45 checkpoint–location cells completed. KGA made 0/44/1 ADAPT/FREEZE/ABSTAIN decisions. All 45 served predictions used the frozen model, but the sole helpful cell received FREEZE, yielding one false FREEZE among 44 FREEZE actions (1/45 overall). Tent helped in 1 cell, tied in 0, and harmed in 44; 43 harmful cells received FREEZE, and the remaining harmful cell received ABSTAIN. The study therefore passes only the locked utility-preservation endpoint and does not demonstrate selective routing or strong success.

Table 6: CCT-20 action record and locked utility-preservation endpoint over 45 checkpoint–location cells. Positive policy contrasts favor KGA. The 95% intervals belong to the utility endpoint; Appendix G reports the separate 97.5% Bonferroni strong-success audit.

| Item             | Result                                                    |
|------------------|-----------------------------------------------------------|
| Actions          | 0 ADAPT / 44 FREEZE / 1 ABSTAIN                           |
| Measured effects | 1 helpful / 0 tied / 44 harmful                           |
| False FREEZE     | 1/44 conditional; 1/45 overall                            |
| Locked endpoint  | Pass: utility preservation only                           |
| Strong success   | Fail: no ADAPT exposure or improvement over always-freeze |

| Comparator    | Mean contrast | Nominal 95% CI   | Required lower bound |
|---------------|---------------|------------------|----------------------|
| Always adapt  | 0.1819        | [0.0937, 0.2914] | $L > 0.0000$         |
| Always freeze | 0.0000        | [0.0000, 0.0000] | $L > -0.0050$        |

KGA matched always-freeze and avoided the measured degradation from always-adapt. The unchanged locked rule therefore returns pass on the stored cells (generated flag: yes). The archived verdict token remains SAFE UTILITY ONLY, but the supported description is utility preservation only. This is a scoped endpoint, not a finite-sample population-safety guarantee.

The complete strong-success audit is more informative. Completeness passed. Against always-adapt, the nominal 97.5% Bonferroni interval is  $[0.08501071, 0.31066374]$  and the Holm-adjusted location sign-flip value is 0.00390625; both pass their locked thresholds. Against always-freeze, the contrast and interval are zero and the Holm value is 1; neither supports strict improvement. FREEZE exposure and total commitment exceed 0.10 and 0.30, but ADAPT exposure is zero and fails its 0.10 threshold. The strong-success criterion therefore fails. This is conservative frozen-model retention, not learned selective routing. The false-adapt count (0/45) is mechanically zero because ADAPT exposure is zero.

Every development cell was harmful for Tent, so the fit saw no helpful example. Only two camera locations supplied fitting data, and calibration-to-target transfer is unverified. The complete location ledger and primary inference appear in the CCT-20 ledger in Appendix G.1. We infer no selective-routing success from this result.

## 8.4 Development Stops and Adverse Diagnostics

So2Sat-LCZ42 (Zhu et al., 2020) screened Tent and SAR in a development-only, multi-city, multi-checkpoint feasibility study. Tent produced both helpful and harmful development cases, but neither candidate passed every predeclared eligibility check. The study therefore stopped before gate calibration. No gate-calibration rows or target inputs, pixels, or labels were accessed, so there is no target natural-shift score. The gate-fit outcomes are now open; retuning them cannot create fresh confirmation.

**DomainNet painting development.** Five separately trained checkpoints were evaluated on the same 24 check cells per checkpoint. KGA abstained on all 120 seed-cell records; candidate effects were helpful in 12, harmful in 15, and tied in 93. Equal-seed regret was 0.110677 percentage points for both KGA and always-freeze, versus 0.123547 for always-adapt. These repetitions do not create 120 independent environments. Using a separate reference-source ResNet-50 checkpoint, a one-pass local entropy candidate restored the source BatchNorm running buffers before inference. Its 24-cell check panel had 7 helpful, 14 harmful, and 3 tied cells. A separate AdaContrast-derived single-process recipe had 1 helpful and 23 harmful cells. Both gates abstained on all 24 cells. The two recipes produced different frozen prediction totals and are not a matched adapter comparison. These are opened-development negatives, not selective-routing or unseen-shift coverage evidence. Fixed queue and normalization-state diagnostics did not rescue the evaluated candidates; their scope and separate numerical summaries appear in the DomainNet development appendix B.11.

The auxiliary studies expose further limits. ImageNet-R and PACS favor always-adapt, while ImageNet-C depends strongly on candidate and operating point. Office-Home and RxRx1 mainly retain the frozen model; the opened Camelyon17 OOD panel adapts throughout. The auxiliary-results appendix B reports valid diagnostic rows on separate metric scales. It also explains why iWildCam, the historical POEM/AETTA ports, and the focused Office-Home route cannot support current headline comparisons.

**ImageNet-C SAR operating point.** The ImageNet-C SAR result uses a local port at learning rate  $4 \times 10^{-3}$ , which is  $16\times$  the published  $2.5 \times 10^{-4}$  setting. The port adapts

the recorded BatchNorm affine parameters, including the upper ResNet block that the official default excludes. It also combines target-batch BatchNorm-statistics replacement with gradient updates. This is an aggressive local diagnostic, not an official paper-default configuration. We ran no matched control at SAR’s published operating point, so the result concerns this port and cannot be read as evidence against SAR as published.

**ImageNet-R backbone pattern.** KGA has higher regret than always-adapt on 8 of 10 backbones. On `efficientnet_b0`, KGA improves over always-adapt but ties always-freeze at zero regret. On `swin_t`, it also improves over always-adapt but ties always-freeze at 0.00473958 regret. Thus KGA has lower regret than both fixed policies on 0 of 10 backbones. Four of the eight adapt-favoring backbones have a 0% harmful-update base rate, so routing cannot improve on always-adapt there. This opened, descriptive pattern is consistent with routing being most useful when harmful updates occur. It is not one deployable multi-backbone policy or confidence-supported natural-shift evidence.

## 9 Limitations and Broader Impact

The measured action record does not demonstrate universal safety. The theory is exact relative to its declared class, whereas the empirical intervals are protocol-scoped and require separate coverage assumptions. The recorded experiments measure directional errors and utility only for the fixed candidates, metrics, and evaluation units studied. Neither layer establishes unconditional deployment safety.

### 9.1 Formal Assumptions and Empirical Calibration Scope

The population frontier is exact within its declared rich binary class. A credible  $\beta$  must come from domain knowledge, target labels, or a stated transfer assumption. A narrower class may permit stronger decisions. An unsupported bound cannot justify a strict population claim. The substantive object is the class-dependent feasible set, not the inequality by itself.

**The development-based surrogate failed.** We tested estimating  $\beta$  from source-like development cells rather than declaring it. The estimate is 1.4–50 $\times$  too small on real evaluation cells (24–73% of cells fall outside the declared class  $\mathcal{C}_\beta$ ; commit error 0.4–16.6%), and returns zero commitments on all 405 ImageNet-C cells in 5 of 10 configurations. We therefore withdraw the estimation route and declare  $\beta$  from domain knowledge instead.

This test used an operational development proxy; it could not observe the theorem’s target disagreement-conditional residual directly. The repository retains the reported result text but not the named raw beta-sweep artifact. The negative result is therefore historically documented but not presently regenerable from the canonical bundle. KGA neither estimates nor numerically uses  $\beta$ .

For KGA, empirical interval coverage needs separate support at the intended decision unit. A larger  $\varepsilon$  reduces commitment but may increase regret when  $f_0$  is worse. A smaller  $\varepsilon$  increases exposure and the cost of calibration failure. Changing the adapter, checkpoint,



feature schema, or update settings changes the prediction problem. Each change requires recalibration or a justified transfer argument.

**Repeated deployment needs a separate guarantee.** A predeclared policy does not by itself create exchangeability or time-uniform safety. Reusing calibration data, replacing the frozen baseline after an accepted update, or selecting later candidates from earlier outcomes changes the joint process. Adaptive conformal inference studies online coverage frequency with feedback (Gibbs and Candes, 2021); it does not automatically give time-uniform control of KGA’s false commitments. Selection-conditional and group-conditional guarantees require additional methods, as Section 2 explains.

## 9.2 Evidence Needed for Prospective Confirmation

The controlled results reuse one checkpoint, one corpus, and related corruption cells. Their cell-outcome-disjoint split fixes a leakage defect; it does not remove dependence. The interval inclusion table is retrospective, not an independent calibration test. The natural/development record remains retention-heavy or adverse: CCT-20 has no ADAPT exposure, So2Sat stops during development, and the painting-development gates abstain throughout. Five-checkpoint repetition documents development behavior but supplies neither independent unseen environments nor a prospective routing success. Empirical abstention may reflect weak prediction, limited data, a wide  $\varepsilon$ , or transfer failure. It does not establish population non-identifiability.

The search history is substantial. One superseded census reported 1,387 determinations and 326 positives (23.5%). A broader archived scan reported 1,427 determinations and 345 positives (24.2%). Neither census froze a reproducible family definition, file list, and repository state. They are overlapping historical search summaries, not two predeclared families and not the multiplicity family for the current panel. The controlled results remain explicitly post hoc.

Table 7 states the four missing tests and the claim each would inform. They cannot be supplied by reclassifying the existing panel.

Table 7: Evidence needed for prospective confirmation. Each row names a missing test rather than reinterpreting a completed diagnostic.

| Missing test                                          | Question it would answer                                                                            | Present boundary                                                                                                                                                                                                   |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Synchronized official comparators                     | Does the current policy add value beyond official TTA/ALine, AETTA, POEM, and simple benefit gates? | Only historical local-port diagnostics exist.                                                                                                                                                                      |
| New natural shift with helpful and harmful candidates | Can a frozen policy exercise nonzero ADAPT and FREEZE exposure prospectively?                       | Current natural tracks are one-sided, retention-heavy, or stopped.                                                                                                                                                 |
| Untouched-group coverage evaluation                   | Do interval inclusion and directional errors transfer beyond opened cross-fit cells?                | No held-out group-coverage result is available.                                                                                                                                                                    |
| Independent checkpoints and environments              | Does the inference extend beyond one checkpoint and related constructed conditions?                 | Five separately trained checkpoints have development results on shared painting cells; independent-environment prospective inference remains unavailable. The primary CIFAR grid reuses one checkpoint and corpus. |

The confirmation protocol in Appendix K specifies a future protocol.

### 9.3 Operational Risks and Deployment Scope

Rejecting an update can prevent adaptation harm. A frozen fallback can prevent adaptation-induced harm while still preserving a model that is inaccurate, biased, or unsuitable for the application. Treating abstention as a safety certificate would be especially risky. A deployment should retain the assessment reason and route unresolved cases to domain-appropriate review or evidence acquisition. Our experiments do not validate clinical, safety-critical, or autonomous operation. We claim neither a held-out physical camera deployment nor a common end-to-end latency result.

The maintained software distinguishes an unavailable assessment from a supported negative interval. Its tests and proof artifacts document a scoped validity, commitment-control, and rollback layer. They do not certify every data pipeline or production system. The appendices record reproduction instructions, formal scope, excluded historical extensions, and release checks.

## 10 Conclusion

K-Bound separates missing target-label information from estimable uncertainty. For a fixed binary model pair under 0/1 loss, a declared residual class induces a feasible set of population benefits. That set supports a strict direction exactly outside the closed band  $|M| \leq \beta$ . The evidence-fibre audit theorem explains why additional functions of the same label-free evidence cannot remove residual ambiguity that remains on the fibre.

KGA addresses a different target. It predicts measured evaluation-cell benefit and uses a separately calibrated interval to ADAPT, FREEZE, or ABSTAIN. Its empirical criterion is directional action correctness, with exposure and false-action denominators reported separately. The utility preservation endpoint is reported separately, not as proof of universal safety. The current CIFAR-10-C diagnostic exercises both directional actions retrospectively, whereas CCT-20 demonstrates conservative all-frozen retention rather than selective routing. These results support evidence-aware commitment under recorded protocols, not unconditional safety or population transfer. The newer opened development record likewise shows abstention and unrecovered candidate harm.

The practical lesson is simple: commit a proposed update only when the relevant target, evidence, and assumptions support a direction; otherwise preserve the fallback and record the unresolved uncertainty.

## Part II

# A Long-Form Guide to the Population Frontier

## 11 Keeping the Estimands Apart

The compact presentation develops two related decision problems. The first is a population partial-identification problem. It asks whether the risk difference between a fixed frozen model and a fixed candidate has one sign in every target law allowed by a declared class. The second is an empirical prediction problem. It asks whether a fitted interval for the measured benefit on a named evaluation unit lies strictly on one side of zero. The two problems share an action vocabulary, but they do not share an estimand or an uncertainty radius. This chapter slows down the notation and proofs so that no empirical interval is mistaken for a population identified set.

Throughout the population analysis, the model pair is held fixed. If an adapter used an unlabeled batch to construct  $f_a$ , that construction has already happened. Population risk is then taken over a fresh target draw with both  $f_0$  and  $f_a$  fixed. The binary result for zero-one loss also holds the target input marginal, the source information, the score  $s$ , and every label-free model output fixed. Only the target label kernel is allowed to vary inside the declared class. These choices are part of the theorem, not implementation details that may be changed silently.

### 11.1 A symbol atlas by inferential level

Let

$$d := \mu_T(D), \quad D := \{x : f_0(x) \neq f_a(x)\}. \quad (16)$$

The maintained population theorem assumes  $d > 0$ . When  $d = 0$ , the models agree almost surely and their zero-one risks are equal; the interesting directional problem disappears. On  $D$ ,

Table 8: The same model pair appears at two inferential levels. The population class and the cell-prediction interval answer different questions.

| Symbol                 | Level       | Role                                                         |
|------------------------|-------------|--------------------------------------------------------------|
| $d$                    | population  | target probability of the fixed disagreement region $D$      |
| $M$                    | population  | label-free mean score margin on $D$                          |
| $\gamma$               | population  | hidden target disagreement-conditional score residual        |
| $\beta$                | population  | externally declared upper bound on $ \gamma $                |
| $\Delta$               | population  | expected zero-one risk reduction for the fixed model pair    |
| $\Delta^{\text{cell}}$ | empirical   | observed score difference on one declared evaluation unit    |
| $\hat{\Delta}$         | empirical   | label-free-at-scoring prediction of that cell difference     |
| $\varepsilon$          | empirical   | calibrated prediction-error radius for the named cell target |
| $B$                    | operational | generic scalar target in the coverage-to-action proposition  |

let  $s(x) \in [0, 1]$  be the fixed candidate-correctness score and  $\eta_a(x) = \mathbb{P}_T(f_a(X) = Y \mid X = x)$ . The four central population symbols are

$$M := \mathbb{E}[s(X) \mid D] - \frac{1}{2}, \quad (17)$$

$$\gamma := \mathbb{E}[\eta_a(X) - s(X) \mid D], \quad (18)$$

$$\beta \geq 0 \quad \text{with the declared restriction } |\gamma| \leq \beta, \quad (19)$$

$$\Delta := R_T(f_0) - R_T(f_a). \quad (20)$$

Here  $M$  is a margin determined by the fixed score and target input law. The residual  $\gamma$  is a property of the unobserved target label kernel after conditioning on disagreement. The number  $\beta$  is an external class parameter. Finally,  $\Delta$  is the population risk reduction. In particular,  $\beta$  is not an estimator,  $\gamma$  is not automatically a source-to-target drift quantity, and  $M$  is not the measured accuracy difference on a finite evaluation cell.

The empirical companion uses a separate collection of symbols. For a predeclared cell  $\mathcal{E}_j$  and a higher-is-better score  $S$ ,

$$\Delta_j^{\text{cell}} := S(f_a; \mathcal{E}_j) - S(f_0; \mathcal{E}_j). \quad (21)$$

A fitted model produces  $\hat{\Delta}_j$ , and a calibration procedure supplies a nonnegative radius  $\varepsilon_j$ . The interval is  $[\hat{\Delta}_j - \varepsilon_j, \hat{\Delta}_j + \varepsilon_j]$ . These objects describe a prediction error for the chosen cell-level score. They do not estimate  $(M, \gamma, \beta)$ , and  $\varepsilon$  is not a finite-sample version of  $\beta$ .

The generic symbol  $B$  is useful because the interval implication is agnostic about the scientific meaning of its scalar target. Setting  $B = \Delta^{\text{cell}}$  gives a cell-level statement. Setting  $B = \Delta$  would give a population statement only if coverage had actually been justified for  $\Delta$ . Substituting one symbol for the other in prose cannot create that justification.

## 11.2 Why the conditioning event matters

The residual is conditional on  $D$ . Even if a score has a useful calibration property under some source distribution, that fact need not survive either target transfer or conditioning on

the event that the two models disagree. Disagreement can select a highly atypical region of the input space. A score that behaves well on average may have a different residual there. This is why the maintained text calls  $\gamma$  a calibration residual rather than treating it as a synonym for distribution shift.

The distinction is visible from the identity

$$M + \gamma = \mathbb{E}[\eta_a(X) \mid D] - \frac{1}{2}. \quad (22)$$

The score cancels. Equation (22) is always true under the definitions; it does not say that either summand is known. The label-free part  $M$  is fixed by the augmented evidence law used in the theorem. The hidden part  $\gamma$  records exactly how far the average candidate correctness on  $D$  can depart from that score margin.

Treating  $M$  as fixed also deserves care. In the population theorem,  $M$  is part of the augmented observable. If a practical system estimates  $M$  from a finite batch, then estimation error creates another uncertainty term. The exact frontier does not make that term vanish. A valid practical analysis must account for that estimation error, for example by enlarging the class or proving a separate sampling statement. Conditioning is an alternative only under an explicitly justified conditional observation and model law; conditioning on a model pair adapted on the same batch is not justified by the fixed-pair theorem. None of those steps is obtained merely by reusing the symbol  $M$  for a sample average.

### 11.3 A disciplined order for reading a claim

Every directional statement can be checked in four passes. First identify the estimand: population risk, a cell score, or another scalar. Second identify the information set: label-free target outputs, labeled historical cells, target labels, or a frozen calibration artifact. Third identify the uncertainty object: a declared class radius, a prediction-error radius, or a sampling-error bound. Fourth identify the probability law and denominator: all evaluation units, only committed units, one time point, or an entire sequence. A statement that omits one of these items is not yet specific enough to compare with either the population theorem or the empirical proposition.

This order prevents several common substitutions. A wide  $\varepsilon$  does not demonstrate that  $\beta$  is large. A narrow empirical interval does not show that the population label kernel lies in a narrow class. A low number of observed false actions does not establish interval coverage. And an interval-coverage premise for a random cell does not imply conditional error control among the cells on which the gate acts. The remainder of this part derives these boundaries rather than leaving them as warnings.

## 12 Proof Anatomy of the Exact Population Frontier

The frontier follows from three ingredients: cancellation outside disagreement, binary complementarity on disagreement, and an exact description of the residual values allowed by the declared class. Each ingredient has a distinct role. Separating them shows both why the formula is simple and why its assumptions are restrictive.

## 12.1 Step one: cancel the agreement region

Under zero-one loss, define

$$C_a := \mathbf{1}\{f_a(X) = Y\}, \quad C_0 := \mathbf{1}\{f_0(X) = Y\}. \quad (23)$$

Because  $\mathbf{1}\{f(X) \neq Y\} = 1 - \mathbf{1}\{f(X) = Y\}$ , the pointwise loss difference is

$$\mathbf{1}\{f_0(X) \neq Y\} - \mathbf{1}\{f_a(X) \neq Y\} = C_a - C_0. \quad (24)$$

On  $D^c$ , the two predictions are equal, so  $C_a = C_0$ . The integrand is zero there. Therefore

$$\Delta = \mathbb{E}[C_a - C_0] \quad (25)$$

$$= d \mathbb{E}[C_a - C_0 \mid D]. \quad (26)$$

This cancellation uses only equality of the predictions on  $D^c$ . It is not a calibration argument and does not use the class bound.

## 12.2 Step two: use binary complementarity

On  $D$ , the two binary predictions differ, so one predicts zero and the other predicts one. Since the label is also binary, exactly one prediction is correct. Thus

$$C_0 = 1 - C_a \quad \text{on } D. \quad (27)$$

Substituting into Equation (26) gives

$$\Delta = d \mathbb{E}[2C_a - 1 \mid D] \quad (28)$$

$$= 2d \left( \mathbb{P}_T(f_a(X) = Y \mid D) - \frac{1}{2} \right). \quad (29)$$

The tower property identifies the conditional correctness probability with  $\mathbb{E}[\eta_a(X) \mid D]$ . Equation (22) then yields

$$\boxed{\Delta = 2d(M + \gamma)}. \quad (30)$$

This is the maintained reduction written with  $d = \mu_T(D)$ . Because  $d > 0$ , multiplying by  $2d$  does not change a sign. Hence  $\text{sign}(\Delta) = \text{sign}(M + \gamma)$  for every target law in the setup, whether or not that law satisfies a particular residual budget.

Binary complementarity is the step that fails in a direct multiclass transcription. With three or more labels, two different predictions can both be wrong. Cancellation outside  $D$  remains valid, but  $C_0 = 1 - C_a$  need not hold on  $D$ . The open-extension discussion later in this part begins from that exact point rather than carrying the binary factor of two into a setting where it is unjustified.

## 12.3 Step three: translate the class into a compatible set

Let

$$q := \mathbb{E}[\eta_a(X) \mid D] - \frac{1}{2} = M + \gamma. \quad (31)$$

The conditional correctness probability lies in  $[0, 1]$ , so  $q \in [-\frac{1}{2}, \frac{1}{2}]$ . The declared restriction  $|\gamma| \leq \beta$  gives  $q \in [M - \beta, M + \beta]$ . Both restrictions must hold. The compatible normalized-benefit set is therefore

$$\mathcal{Q}(M, \beta) = [-\frac{1}{2}, \frac{1}{2}] \cap [M - \beta, M + \beta] = [\max\{-\frac{1}{2}, M - \beta\}, \min\{\frac{1}{2}, M + \beta\}]. \quad (32)$$

Scaling by the positive number  $2d$  gives the compatible population-benefit set

$$\mathcal{I}_\Delta(M, \beta, d) = 2d \mathcal{Q}(M, \beta). \quad (33)$$

The clipping by  $[-\frac{1}{2}, \frac{1}{2}]$  enforces the elementary feasibility of a correctness probability. It is not an optional numerical safeguard. Omitting it would describe residuals whose implied average correctness is below zero or above one.

Every point in Equation (32) is attainable in the full declared class. To see this, choose any  $q$  in that interval and set the candidate-correctness kernel on  $D$  to the constant

$$\eta_q(x) := \frac{1}{2} + q. \quad (34)$$

It is a valid probability because  $q \in [-\frac{1}{2}, \frac{1}{2}]$ . Its residual is  $\gamma_q = q - M$ , whose absolute value is at most  $\beta$  because  $q \in [M - \beta, M + \beta]$ . Holding the input law, models, score, and label kernel on  $D^c$  fixed then produces a target law in the class with benefit  $2dq$ . Thus the interval is an identified set, not merely an outer bound, for the rich class used by the necessity result.

## 12.4 Step four: locate the strict-sign regions

A uniformly positive benefit requires the lower endpoint of  $\mathcal{Q}(M, \beta)$  to be strictly positive. Because  $-\frac{1}{2}$  is not positive, this happens exactly when  $M - \beta > 0$ , or  $M > \beta$ . Similarly, a uniformly negative benefit requires the upper endpoint to be strictly negative. Since  $\frac{1}{2}$  is not negative, this occurs exactly when  $M + \beta < 0$ , or  $M < -\beta$ . The maximal strict population rule is consequently

$$M > \beta \Rightarrow \text{ADAPT}, \quad M < -\beta \Rightarrow \text{FREEZE}, \quad |M| \leq \beta \Rightarrow \text{ABSTAIN}. \quad (35)$$

The inequalities are strict because the action words make strict directional assertions. A lower endpoint equal to zero does not justify the claim that every compatible benefit is positive. An upper endpoint equal to zero does not justify the claim that every compatible benefit is negative.

For  $|M| < \beta$ , the interval contains values on both sides of zero. One may select a small  $r > 0$  satisfying  $r < \min\{\beta - |M|, \frac{1}{2}\}$  and use the two constant kernels  $\eta_+(x) = \frac{1}{2} + r$  and  $\eta_-(x) = \frac{1}{2} - r$ . Their residuals are  $r - M$  and  $-r - M$ ; both have absolute value below  $\beta$ . Under the fixed observation experiment in Section 13.2, they preserve the label-free evidence law and yield benefits  $2dr$  and  $-2dr$ . The ambiguity is exact: no additional sample from the same label-free evidence law can distinguish these two target worlds.

For  $|M| = \beta > 0$ , opposite nonzero signs need not both be available, but zero is. Taking  $\gamma = -M$  gives  $q = 0$  and hence  $\Delta = 0$ . The class also contains a strict-benefit law on the side indicated by  $M$ . Because one compatible law has zero benefit, the strict statement still fails uniformly. Finally, when  $M = \beta = 0$ , the class forces  $\gamma = 0$  and the benefit is identically zero. This last case is not uncertainty about the sign: the value is identified, but neither strict directional assertion is true.

## 12.5 Sufficiency, necessity, and class richness

The outside-band direction needs only the bound  $|\gamma| \leq \beta$ . If  $M > \beta$ , then  $M + \gamma \geq M - \beta > 0$  for every admissible law; the negative case is symmetric. This is a sufficiency statement and remains valid for any narrower subclass obeying the same bound.

The inside-band conclusion is different. To show that no strict direction is uniformly supported, the class must contain the evidence-matched kernels used above, including the zero-benefit boundary law. A narrower scientific model might exclude them. For example, a verified structural relation could tie the target label kernel to an observed quantity and leave only one possible sign. The maintained theorem therefore states its necessity and maximality for the full rich class. It does not say that every imaginable restricted problem has the same identified set.

This distinction is central to using impossibility results responsibly. An impossibility theorem does not prohibit action after stronger information is supplied. It identifies the information that the present evidence and class fail to provide. Escaping the result requires changing an assumption, observing a new channel, or narrowing the target class in a way that can be defended for the deployment at hand.

## 13 Evidence Fibres and the Exact Audit Floor

The frontier assumes a declared residual budget. A natural next question is whether the same label-free information used by the gate can discover a smaller valid budget. Evidence fibres make the obstacle precise. They group together target laws that induce exactly the same distribution for everything the audit may observe. An audit can transform its evidence, randomize, and compute arbitrarily complicated statistics, but it cannot tell apart two laws in the same fibre.

### 13.1 The observable object and its fibres

Let  $W$  contain the audit's full information set. It may include the unlabeled target batch, predictions from both models, logits, features derived from those outputs, the fixed score, and source or development data whose law is unaffected by the deployment label kernel. Let an audit also use an independent random seed  $U$  and return the measurable real-valued output

$$\hat{\beta} = g(W, U) \in [0, \infty). \quad (36)$$

Including  $U$  makes the argument cover randomized procedures. It gives the audit no additional information about which compatible target law generated the observable data.

For a declared class  $\mathcal{C}$ , two laws are evidence-equivalent when they induce the same law of  $W$ . If  $z$  denotes one such observable law, define

$$\mathcal{C}(z) := \{P \in \mathcal{C} : \text{Law}_P(W) = z\}, \quad (37)$$

$$\Gamma_z(\mathcal{C}) := \sup_{P \in \mathcal{C}(z)} |\gamma(P)|. \quad (38)$$

The first object is an evidence fibre. The second is the largest absolute residual still possible within it. Notice that  $z$  indexes a probability law, not merely one realized value of a summary



statistic. Equal realized summaries would be a weaker relation; the exact impossibility uses equal observable distributions.

The radius is conditional on both the declared class and the chosen evidence. Enlarging the class can enlarge a fibre. Adding a genuinely informative observable can split a fibre into smaller ones. Merely computing a new deterministic feature of  $W$  cannot do so, because equal laws of  $W$  imply equal laws of every measurable function of  $W$ .

## 13.2 How the label-kernel construction preserves evidence

Fix the target input marginal, source information, predictors, score, and the label kernel outside  $D$ . On  $D$ , choose any measurable  $\eta : D \rightarrow [0, 1]$  and define the conditional label law by assigning probability  $\eta(x)$  to  $Y = f_a(x)$  and probability  $1 - \eta(x)$  to  $Y = f_0(x)$ . This is a complete binary conditional distribution because the two predictions are distinct on  $D$ . Use one fixed marginal-based observation experiment for all target laws, including a fixed joint law for any source information; finite iid target sampling is one realization.

Changing  $\eta$  in this construction changes target labels but changes neither the distribution of  $X$  nor either model's output at a given  $X$ . Under this common observation experiment, changing the target label kernel preserves the law of all recorded label-free evidence. At the same time, the residual becomes

$$\gamma(\eta) = \mathbb{E}[\eta(X) \mid D] - \frac{1}{2} - M. \quad (39)$$

Thus one evidence law can coexist with a range of target residuals. The construction isolates the missing information: correctness on the target disagreement region.

This is stronger than saying that a finite sample is noisy. The two worlds can induce identical laws for samples of every size drawn from the permitted label-free channel. More data from that same channel can estimate its distribution more accurately, but it cannot reveal which target label kernel was paired with it. The obstacle is identification, not a slow convergence rate.

## 13.3 Deriving the audit floor line by line

Suppose the audit is required to cover the true absolute residual throughout one nonempty fibre whose residual set is bounded (as it is for the stated binary score):

$$\mathbb{P}_P\{\hat{\beta} \geq |\gamma(P)|\} \geq 1 - \delta \quad \text{for every } P \in \mathcal{C}(z), \quad (40)$$

where  $0 \leq \delta < 1$ . Every law in the fibre gives  $(W, U)$  the same distribution. Therefore  $\hat{\beta} = g(W, U)$  has one common distribution, which we denote by  $\mathbb{P}_z$ .

If the supremum in  $\Gamma_z(\mathcal{C})$  is attained at a law  $P^*$ , substituting  $P^*$  into Equation (40) immediately gives

$$\mathbb{P}_z\{\hat{\beta} \geq \Gamma_z(\mathcal{C})\} \geq 1 - \delta. \quad (41)$$

The supremum need not be attained, so the general proof uses approximation. Select laws  $P_n$  in the fibre such that  $a_n := |\gamma(P_n)|$  increases to  $\Gamma_z(\mathcal{C})$ . The events  $A_n := \{\hat{\beta} \geq a_n\}$  form

a decreasing sequence. Uniform validity gives  $\mathbb{P}_z(A_n) \geq 1 - \delta$  for every  $n$ . By continuity of probability from above,

$$\mathbb{P}_z\{\hat{\beta} \geq \Gamma_z(\mathcal{C})\} = \mathbb{P}_z\left(\bigcap_{n=1}^{\infty} A_n\right) \quad (42)$$

$$= \lim_{n \rightarrow \infty} \mathbb{P}_z(A_n) \geq 1 - \delta. \quad (43)$$

This is the exact fibre-wise floor. A valid audit must return at least the unresolved fibre radius with the same high probability at which it promises to cover every compatible residual. The constant procedure  $\hat{\beta} \equiv \Gamma_z(\mathcal{C})$  is valid on that fixed fibre, so the floor is sharp as an oracle benchmark. Calling it an oracle benchmark matters: the statement does not provide an algorithm that learns the fibre index or computes its radius from a finite sample.

For the full class  $\mathcal{C}_\beta$  and the maintained feasibility range  $0 \leq \beta \leq \frac{1}{2}$ , the constant-kernel construction attains residuals of magnitude  $\beta$  while preserving the evidence law. Hence

$$\Gamma_z(\mathcal{C}_\beta) = \beta. \quad (44)$$

In this setting the label-free audit cannot uniformly certify a radius below the one already declared. If the allowed budget is larger than the maintained range, probability feasibility can clip the attainable residuals, and Equation (44) must not be repeated without recalculating the fibre. The general audit-floor statement remains Equation (43), with  $\Gamma_z(\mathcal{C})$  rather than an assumed numerical substitute.

### 13.4 Why labels from an unrelated domain do not shrink the fibre

Now add a labeled source or calibration sample  $L_S$  to the observable object, so  $W = (W_0, L_S)$ . If the declared class places no restriction linking the law of  $L_S$  to the target label kernel on  $D$ , hold the full joint observation law of  $(W_0, L_S)$  fixed while applying the target kernel construction. These worlds then have the same recorded evidence law and can still have different values of  $\gamma$ . The labeled sample may estimate source correctness extremely well; it does not identify target correctness on a region whose label law is allowed to change independently.

The point is not that historical labels are intrinsically useless. They become informative after a relation is stated and defended. Examples include a sampling design that makes them representative of the intended target units, a transport model that bounds the change in a conditional label law, or a stability premise that is itself externally validated. In each case, the gain comes from the link, not from the word “labeled.” Without such a link, adding  $L_S$  to  $W$  leaves the decisive target fibre intact.

This observation also clarifies development-based residual surrogates. A proxy computed from source-like cells can be useful for engineering exploration. It does not become a uniform bound on a deployment residual unless the class restricts how those two quantities differ. The audit theorem asks whether the claimed bound covers every target law in the fibre. Empirical correlation on development domains is not, by itself, that class restriction.

## 13.5 Information refinement is set inclusion

Suppose a new observable  $H$  is added. At a joint observable law  $(z, h)$ , the refined fibre obeys

$$\mathcal{C}(z, h) \subseteq \mathcal{C}(z), \quad \Gamma_{z,h}(\mathcal{C}) \leq \Gamma_z(\mathcal{C}). \quad (45)$$

The inequality is simply monotonicity of a supremum under set inclusion. It gives a clean language for escape routes. New information helps only to the extent that it excludes target laws with large residuals. If  $H$  carries representative target labels or a validated structural measurement, the inclusion may be strict.

If  $H$  is a deterministic function of  $W$ , then for every compatible joint observable law  $(z, h)$ , the refined fibre equals the original fibre:

$$\mathcal{C}(z, h) = \mathcal{C}(z). \quad (46)$$

Indeed, the law of  $(W, H)$  is already determined by the law  $z$  of  $W$ . A deterministic recoding can make the evidence easier to use, but it cannot exclude a target law that was compatible with the original evidence distribution.

Equation (45) is deliberately one-sided for genuinely new evidence. It does not say that any particular label-shift statistic, rank condition, or symmetry assumption produces a symmetric interval or an exact if-and-only-if frontier. Those conclusions require a separate model-specific proof. Likewise, a closure property under swapping labels does not automatically make arbitrary regional benefit signs feasible. The maintained population theorem obtains exactness from its explicit binary kernel construction, not from an abstract slogan about symmetry.

## 14 Strict Actions, Closed Bands, and Coverage

The three action names carry logical content. They are not merely labels for which model is served. An adapt decision asserts that the candidate has strictly positive benefit. A FREEZE decision asserts that the candidate has strictly negative benefit. An abstain decision makes neither assertion. In the implementation described in the main text, both FREEZE and abstain retain  $f_0$ , but their audit meanings remain different: one is a supported rejection and the other is a record of unresolved evidence.

### 14.1 Why the boundary is closed

The population compatible set supports adapt only if every member is greater than zero, and it supports FREEZE only if every member is less than zero. The word “every” explains the closed abstention band. At  $M = \beta > 0$ , the smallest compatible normalized benefit is zero. At  $M = -\beta < 0$ , the largest is zero. At  $M = \beta = 0$ , zero is the only compatible benefit. In all three cases a strict statement is false for at least one admissible law.

This convention is intentionally stronger than choosing a model with zero regret. When  $\Delta = 0$ , either model has the same population risk, so serving either one incurs no loss relative to the pairwise oracle. Yet neither “the candidate is better” nor “the frozen model is better” is true. The frontier concerns warranted directional assertions, not only realized task loss.

The same logic applies to an empirical interval. Let

$$L := \hat{B} - \varepsilon, \quad U := \hat{B} + \varepsilon. \quad (47)$$

The operational rule uses  $L > 0$  for `ADAPT` and  $U < 0$  for `FREEZE`. If  $L = 0$  or  $U = 0$ , the interval touches the boundary and the rule abstains. Replacing either strict comparison by a weak one would change the claim: it would permit a directional action even though the interval contains a value at which the corresponding strict assertion fails.

Unavailable evidence is another boundary case. If required features, a candidate, or a frozen calibration artifact are unavailable, there is no interval on which to base a negative claim. Retaining  $f_0$  may be the appropriate service behavior, but its audit record is `abstain`, together with the reason. Reclassifying missing assessment as `FREEZE` would make the apparent negative action count larger without supplying negative evidence.

## 14.2 What exact matched evidence forces a randomized rule to do

Consider two compatible population laws with exactly the same evidence distribution and opposite nonzero benefit signs. A possibly randomized label-free rule has the same action probabilities in both worlds. If its marginal false-adapt probability may not exceed  $\alpha$  in any world, then its probability of `ADAPT` is at most  $\alpha$  in the negative-benefit world. Because the action distribution is shared, the same upper bound applies in the positive-benefit world. Symmetrically, control of false `FREEZE` limits its `FREEZE` probability to at most  $\alpha$ . The remaining mass must satisfy

$$\mathbb{P}\{g = \text{ABSTAIN}\} \geq 1 - 2\alpha. \quad (48)$$

Randomization therefore cannot manufacture information. It can distribute a permitted error budget across unsupported actions, but a rule that controls both directions must leave most mass on abstention when the evidence law is shared by opposite worlds. This is a probabilistic error-control statement. It is distinct from the deterministic pointwise-maximal rule, which abstains everywhere that no strict action is uniformly sound.

## 14.3 Coverage-to-action as a set containment

The operational proposition has an even shorter proof because it begins from a coverage premise. Let

$$E := \{|\hat{B} - B| \leq \varepsilon\}. \quad (49)$$

On  $E$ , an `ADAPT` decision implies  $B \geq \hat{B} - \varepsilon > 0$ . Therefore

$$\{g = \text{ADAPT}, B \leq 0\} \subseteq E^c. \quad (50)$$

Likewise, on  $E$ , a `FREEZE` decision implies  $B \leq \hat{B} + \varepsilon < 0$ , and hence

$$\{g = \text{FREEZE}, B \geq 0\} \subseteq E^c. \quad (51)$$

If  $\mathbb{P}(E) \geq 1 - \alpha$ , each marginal false-direction probability is at most  $\alpha$ . No distributional approximation appears in these two inclusions. All statistical difficulty has been placed in the premise that the interval covers the predeclared target under the intended joint law.

The proposition does not prove its own premise. An order statistic, a regression algorithm, or the name of a calibration method does not by itself establish coverage for a changed candidate, environment, or evaluation unit. One must specify the unit distribution, all fitting and candidate construction randomness, and the assumptions under which calibration units and a future unit are related. The action implication is exact conditional on that work being valid; it cannot replace that work.

## 14.4 Marginal error, conditional error, and exposure

Equation (50) bounds  $\mathbb{P}(g = \text{ADAPT}, B \leq 0)$ , whose denominator is the entire probability space. It does not directly bound

$$\mathbb{P}(B \leq 0 \mid g = \text{ADAPT}), \quad (52)$$

whose denominator is only the adapt actions. When  $\mathbb{P}(g = \text{ADAPT}) > 0$ , the marginal statement gives the elementary ratio bound

$$\mathbb{P}(B \leq 0 \mid g = \text{ADAPT}) \leq \min \left\{ 1, \frac{\alpha}{\mathbb{P}(g = \text{ADAPT})} \right\}. \quad (53)$$

This can be uninformative when action exposure is small. Obtaining a useful fixed level conditional on selection requires a different guarantee. If the rule never adapts, its marginal false-adapt probability is zero mechanically, while the conditional rate is undefined. Thus every safety summary must report exposure alongside error.

The same denominator discipline applies to finite recorded panels. A count divided by all cells answers a marginal descriptive question under equal cell weighting. A count divided by actions of one type answers a conditional descriptive question on the exposed subset. Neither ratio is an independent estimate for a new environment unless the sampling and dependence structure support that interpretation.

## 14.5 Choosing the target $B$

For the empirical gate,  $B = \Delta^{\text{cell}}$  yields a statement about the named cell-level score. This target may be an accuracy difference, a macro-averaged score difference, or another predeclared scalar. The population theorem instead concerns  $B = \Delta$ , the expected zero-one risk reduction for a fresh target draw and a fixed binary model pair. Coverage of the first does not entail coverage of the second.

A bridge can be built only by adding an explicit relation. For example, suppose one event gives  $|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon$  and another gives  $|\Delta^{\text{cell}} - \Delta| \leq b$ . On their intersection, the triangle inequality gives

$$|\hat{\Delta} - \Delta| \leq \varepsilon + b. \quad (54)$$

A probability statement then needs valid coverage levels for both events and an appropriate union bound or joint argument. Without the second event, the extra radius  $b$ , and their assumptions, the empirical interval remains a cell-benefit interval.

Selection and time introduce further distinctions. Marginal coverage for each of several candidates does not automatically cover the candidate selected after inspecting their intervals.

Pointwise coverage at each time does not control the probability of any error over an indefinite sequence. Reusing outcomes to tune the predictor or radius changes the probability space. These are not defects in the elementary containment proof; they are different statistical targets that need stronger premises.

## 15 What Must Change to Escape the Impossibility

An escape route must exclude some of the evidence-matched label kernels. That can happen through a new observation or through a narrower class, but the new restriction must be stated at the same target level as the desired conclusion. A useful way to audit a proposed route is to ask three questions: What target laws does it remove from the fibre? Why is that removal credible in the deployment domain? What finite-sample uncertainty remains after the structural restriction is accepted?

### 15.1 Representative target labels

Labels sampled from the intended target-unit distribution directly reveal target correctness. In the binary setup, labeled observations on  $D$  can estimate  $\mathbb{E}[\eta_a(X) \mid D]$  and hence  $M + \gamma$ . Their usefulness depends on disagreement exposure: if  $D$  is rare, a general target sample can contain few informative labels. A sampling design that targets disagreements may be more efficient, but then its weighting and selection mechanism must be accounted for. In either design, labels replace an identification gap with a sampling problem; they do not make uncertainty disappear.

Historical deployment labels can play the same role when the past and future units are connected by a defensible sampling or stability model. The resulting statement is episode- or environment- specific. If conditions can change after calibration, that change requires its own bound or monitoring assumption. The evidence fibre shrinks because the class couples past and future target kernels, not because past labels possess universal transfer value.

### 15.2 Structural coupling across domains

A model may restrict how the target label kernel differs from a source or calibration kernel. A uniform bound on an appropriate conditional discrepancy, a verified invariance, or a mechanistic constraint can imply a residual budget. Such a route is scientifically meaningful when the constraint has a domain interpretation and can be stress-tested independently. Simply naming the observed change “covariate shift” does not establish the needed equality of conditional label laws.

Label-shift models offer another possible restriction: fixed class-conditional feature laws plus a changing class prior can make some target quantities recoverable through an evidence operator. At the abstract level, adding informative restrictions cannot enlarge a fibre, as Equation (45) shows. Stronger conclusions—including a symmetric identified interval, an exact sign frontier, or an if-and-only-if condition—depend on the particular operator, feasibility constraints, and invertibility assumptions. They remain open here and are not consequences of monotonicity alone.

### 15.3 Additional views and proxy assumptions

Multiple predictions, augmentations, sensors, or independently trained models can supply a richer observable  $H$ . They help only under a relation between their agreement pattern and correctness. Conditional-independence or error-structure assumptions can sometimes provide such a relation, but agreement alone is not truth. A rank pattern may be a useful diagnostic without being a two-sided certificate of latent error. In particular, a small observable residual does not upper-bound an unobservable model deviation when the proved inequality goes in the opposite direction.

A domain expert may instead declare a physically meaningful upper bound on the residual. This is not a statistical estimate, but it can be the right object when the mechanism supplies knowledge that the data channel cannot. The decision is then conditional on that declared class. A good report should vary the bound in a sensitivity analysis and explain what evidence would falsify the assumption, while avoiding the claim that the unlabeled batch verified it.

### 15.4 An assumption ledger for any proposed escape

Before promoting an escape route to a decision rule, its ledger should name: the target population; the fixed or adaptively constructed model pair; the exact information available at action time; the new structural restriction; the parameter or residual it bounds; the unit over which any coverage statement is marginal; and the behavior when a premise cannot be checked. It should also separate identification from estimation. The first asks what would be known with the evidence distribution known exactly. The second asks how accurately a finite sample recovers the identified object. A rate calculation cannot repair non-identification, while an identification argument does not supply a finite-sample rate.

## 16 Open Extensions Beyond the Maintained Binary Pair

The following directions are research programs, not additional results claimed by this report. Each starts from an identity that remains valid and then states the missing ingredient needed for a frontier or certificate. This format prevents an attractive analogy from being presented as a proved extension.

### 16.1 Multiclass prediction

For multiclass zero-one loss, cancellation outside  $D$  still gives

$$\Delta = d(p_a - p_0), \quad p_a := \mathbb{P}_T(f_a(X) = Y \mid D), \quad p_0 := \mathbb{P}_T(f_0(X) = Y \mid D). \quad (55)$$

The difference from the binary proof is exact: two distinct multiclass predictions may both be wrong, so  $p_0$  need not equal  $1 - p_a$ . A useful generalization would model the conditional correctness advantage

$$A(x) := \mathbb{P}_T(f_a(X) = Y \mid X = x) - \mathbb{P}_T(f_0(X) = Y \mid X = x) \quad (56)$$

and compare it on  $D$  with a label-free proxy  $u(x)$  that is integrable under  $P_X(\cdot | D)$ . Defining a proxy mean and a residual yields an algebraic decomposition of  $\Delta$ , and an externally justified residual bound gives an immediate outside-band sufficiency statement. An exact necessity result would require characterizing which advantage functions can arise from one multiclass label kernel while all model outputs remain fixed. The feasibility geometry is not the binary interval, so the maintained binary theorem does not settle it.

## 16.2 Regression and general losses

For a general loss with both model losses integrable under the target joint law, define the random loss advantage

$$G(X, Y) := \ell(f_0(X), Y) - \ell(f_a(X), Y), \quad \Delta = \mathbb{E}[G(X, Y)]. \quad (57)$$

For a label-free proxy  $t(X)$ , assume  $\mathbb{E}[|G(X, Y)|] < \infty$  and  $\mathbb{E}[|t(X)|] < \infty$  under the same target joint law. These integrability conditions justify the identity

$$\Delta = \mathbb{E}[t(X)] + \mathbb{E}[G(X, Y) - t(X)]. \quad (58)$$

This decomposition suggests a margin-plus-residual analysis. A declared absolute bound on the second term is sufficient for a strict sign outside the corresponding band. Exactness is harder. It depends on the loss, the allowed response distributions, integrability, output bounds, and which label kernels preserve the evidence. Unbounded losses also require tail conditions before a finite residual budget is meaningful. Previously suggested regression brackets and constants are not imported here; a valid extension needs a fresh statement and proof under explicit assumptions.

## 16.3 More than one candidate

With candidates  $f_1, \dots, f_K$ , define each benefit relative to the same frozen reference,  $B_k = R_T(f_0) - R_T(f_k)$  or its cell-level analogue. Marginally valid intervals for the  $K$  predeclared targets do not automatically remain valid after selecting the candidate with the most favorable predicted benefit. A sound design could reserve independent information for selection and evaluation, construct a simultaneous event for all candidates, or use a procedure proved valid conditional on the selection rule. Each choice changes exposure and power.

The population identification problem also changes. Pairwise model agreements constrain a shared label kernel, and the feasible vector  $(B_1, \dots, B_K)$  need not be a product of pairwise intervals. Low-rank agreement patterns can suggest structure, but a one-sided fit diagnostic is not an upper certificate for violation of that structure. Nor does pairwise agreement by itself establish conditional independence of correctness indicators. A complete theory would characterize the joint feasible benefit set, state the sign anchor that orients it, and prove how an observable test controls departures from the assumed model. Those steps remain open.

## 16.4 Sequential and adaptive deployment

In a sequence, let  $\mathcal{F}_t$  contain all information available before action  $g_t$ . The candidate at time  $t$  may depend on earlier inputs and actions, and an accepted update may become the



next baseline. The target  $B_t$ , its predictor, and its calibration distribution can therefore all change with the history. A marginal guarantee at each fixed time does not imply

$$\mathbb{P}\{\text{some false directional action occurs over time}\} \leq \alpha. \quad (59)$$

Controlling this event requires a sequential design, such as a valid allocation of error across times, a time-uniform confidence sequence, or another supermartingale-based construction, together with assumptions robust to the feedback created by adaptation. Recalibrating on outcomes selected because an earlier gate was uncertain can also induce selection bias.

The population fibre becomes history-dependent as well. New labels can shrink it, while model updates and distribution change can redefine the disagreement region and residual. A sequential extension should specify when the declared class is refreshed, what remains fixed between actions, and whether safety is required per action, over a finite horizon, or uniformly in time. The present theory and ordinary cell-level interval do not provide that guarantee.

## 16.5 A research checklist for extensions

An extension is ready to join the maintained theorem stack only after six items are explicit:

1. the estimand and action semantics;
2. the observable evidence and its probability law;
3. the admissible target class and feasibility constraints;
4. an exact or conservative compatible-value set with a proved construction or bound;
5. the statistical premise needed to estimate or cover the intended target; and
6. the denominator, selection rule, and time horizon of every error statement.

Numerical checks may reveal counterexamples and test algebra, but they do not substitute for these items. Conversely, a population proof does not certify a software pipeline or empirical transfer. The value of the K-Bound decomposition is that it makes this division of labor visible: structural assumptions determine what can be known, statistical calibration governs finite-sample uncertainty, and action semantics determine which claims are made when either layer is inconclusive.

## Part III

# Current Evidence and Protocol Atlas

## 17 How to Read the Empirical Record

The empirical record answers a narrower question than the population theory. The theory asks which benefit signs are uniform over a declared class of target laws. The experiments instead record what a fixed gate did on named evaluation cells after a particular candidate, evidence map, benefit predictor, and calibration rule had been frozen. Keeping those two questions separate is not a stylistic precaution. It determines which denominator is meaningful, which sources may support a sentence, and what an observed error can falsify.

This atlas therefore uses an evidence-role taxonomy. The roles are not cumulative stages, and a record does not “move upward” merely by accumulating artifacts. A strong checksum cannot compensate for a weak study design, and a prospectively governed design cannot compensate for an invalid metric. Conversely, an unfavorable or stopped study remains informative when its status and stopping rule are stated accurately. The objective is an auditable account of directional action, error exposure, and utility consequences, not a tournament table of adaptation methods.

Table 9: Evidence-role taxonomy used in this report. The rows are non-cumulative logical roles, not a ranking of datasets or stages on a promotion ladder. The last column states the largest conclusion that a record in the named role can support.

| Role                    | Minimum obligation                                                                 | Permitted conclusion                                          |
|-------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Population theorem      | assumptions, quantifiers, and declared class stated explicitly                     | uniform sign claim only inside that class                     |
| Operational implication | coverage or testing premise for one named target and unit                          | marginal wrong-direction event control for that target        |
| Software invariant      | executable dependency test over the complete decision path                         | absence of the tested information-flow defect                 |
| Current diagnostic      | valid metric, fixed reconstruction rule, current authority, and stated dependence  | measured action and utility behavior on recorded cells        |
| Governed target study   | pre-outcome lock, complete receipts, one-pass scoring, and declared inference unit | result under the locked endpoint, including a negative result |
| Historical provenance   | immutable old artifact whose protocol is no longer current                         | what was once run; never current comparative support          |
| Missing evidence        | protocol or comparator not completed under the required contract                   | an explicit research gap, not an implied result               |

## 17.1 The object that receives an action

An evaluation cell is the smallest unit to which the recorded gate assigns one action. A cell can be a corruption condition, a checkpoint–location pair, or another protocol-defined collection of examples. For a higher-is-better score  $S$ , the measured outcome for cell  $i$  is

$$B_i = \Delta_i^{\text{cell}} = S(f_a; \mathcal{E}_i) - S(f_0; \mathcal{E}_i). \quad (60)$$

This is an observed score difference after labels are joined offline. It is not automatically an estimate of the population benefit  $\Delta$ , and it need not be an average of per-example losses. For example, a difference in cell-level macro-F1 is a nonlinear functional of a confusion table. No sample-size argument converts such a difference into the population quantity merely because the action itself did not read the labels.

The prediction  $\hat{\Delta}_i$  and its empirical radius  $\varepsilon_i$  are separate objects. The gate commits to ADAPT when  $\hat{\Delta}_i - \varepsilon_i > 0$ , commits to FREEZE when  $\hat{\Delta}_i + \varepsilon_i < 0$ , and otherwise records

ABSTAIN while serving the frozen model. FREEZE and ABSTAIN therefore have the same served predictor but different meanings. FREEZE is a directional rejection of the candidate. ABSTAIN records unresolved or unavailable evidence. A system that silently converts every missing assessment into FREEZE would manufacture apparent negative-direction confidence that the evidence never supplied.

## 17.2 Four denominators that must not be conflated

Let  $n$  be the number of equally weighted recorded cells,  $A$  the number of ADAPT actions,  $F$  the number of FREEZE actions, and  $U = n - A - F$  the number of abstentions. Let  $A^-$  count ADAPT actions on cells with  $B_i \leq 0$ , and let  $F^+$  count FREEZE actions on cells with  $B_i \geq 0$ . The basic descriptive quantities are

$$\text{exposure}_A = A/n, \quad \text{exposure}_F = F/n, \quad \text{commitment} = (A + F)/n, \quad (61)$$

$$\widehat{\text{FA}}_u = A^-/n, \quad \widehat{\text{FA}}_c = A^-/A, \quad \widehat{\text{FF}}_u = F^+/n, \quad \widehat{\text{FF}}_c = F^+/F. \quad (62)$$

The conditional ratios are undefined when the corresponding action count is zero. The all-cell and action-conditional denominators answer different operational questions. The first asks how often a wrong directional commitment appears in the stream of all decisions. The second asks how often a particular commitment is wrong among the times that branch is exercised. Neither is the population event involving the unobserved population benefit, and neither is an interval-coverage rate.

A zero numerator is uninterpretable without exposure. Zero false adaptations among many ADAPT actions describes a different operating record from zero false adaptations produced by a gate that never adapts. Likewise, an all-frozen policy may preserve utility when the candidate is harmful, but it has supplied no evidence about its ability to recognize a helpful candidate. For that reason every result discussion below begins with  $A/F/U$  and the relevant error numerator before turning to accuracy or regret.

Regret is a secondary utility measure. On each cell the observed oracle serves the model with the higher recorded score. The policy regret is its shortfall from that oracle, averaged using the protocol's declared weights. Always-adapt and always-freeze are useful reference policies because they expose the cost of the two unconditional choices. A lower point regret for the gate is a candidate- and panel-specific utility fact. It is not a definition of safety and cannot establish a uniform advantage over future candidates or target laws.

## 18 Canonical Authority and Reconstruction Chain

The current controlled and auxiliary panel is bound to `experiments/kbound/results/reconciled_panels_v1/canonical_panel_results.json`. Its SHA-256 is 35d4c165843de1ece3cb35fffb4ac50dbfcbfa33b646754e2b6d133fbdfa78c6e. The corresponding input inventory is

`experiments/kbound/results/reconciled_panels_v1/source_manifest.json`, whose SHA-256 is 03b1d2b1e9e5ed1cf835126f871ed83eb8497a1ac81727b3955af0d69eb89742. These two hashes have different jobs. The first identifies the derived panel consumed by the

manuscript. The second identifies the inventory of compact sources from which that panel was reconciled. Equality of one digest does not imply equality of the other.

The storage manifest records the canonical panel as a tracked derived release authority and gives its exact reconstruction command. It also records that large datasets, checkpoints, and raw prediction arrays are external rather than silently treating their absence from Git as proof that they never existed. Where a historical run failed to serialize dataset, population, checkpoint, or code identity, the gap remains a gap. A post-hoc tree hash may document current local bytes, but it does not retroactively identify the inputs used by the historical run.

Table 10: Authority names and roles for quantitative claims in this atlas. Exact repository-relative paths follow the table. The status column prevents one artifact from being used for a role it was not designed to fill.

| Authority                               | Role                                                  | Status boundary                                                          |
|-----------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------|
| Canonical reconciled panel              | current cell-level panel and reconstructed decisions  | current point-estimate authority; study design remains protocol-specific |
| Canonical source manifest               | compact-input inventory and source digests            | provenance binding, not a statistical guarantee                          |
| Current-policy family sensitivity       | six-family CIFAR sensitivity                          | retrospective analysis, separate from the canonical point panel          |
| CCT-20 release manifest and receipt     | CCT-20 target actions, effects, and inference         | separate governed study with verdict<br>SAFE_UTILITY_ONLY                |
| So2Sat candidate selection and receipts | So2Sat development selection and stopping decision    | development-stage authority; no target result exists                     |
| Claim ledger                            | allowed wording, forbidden wording, and support links | hand-maintained claim/status authority, not a numerical generator        |
| Storage manifest                        | storage class, expected path, size, and digest        | evidence inventory; existence alone does not validate a claim            |

#### Exact repository-relative authority paths.

- **Canonical reconciled panel.** experiments/kbound/results/reconciled\_panels\_v1/canonical\_panel\_results.json.
- **Canonical source manifest.** experiments/kbound/results/reconciled\_panels\_v1/source\_manifest.json.
- **Current-policy family sensitivity.** experiments/kbound/results/reconciled\_panels\_v1/current\_policy\_cluster\_inference.json.
- **CCT-20 release manifest and receipt.** docs/research/kbound/paper/generated/cct20\_release\_manifest.json and docs/research/kbound/paper/generated/cct20\_release\_manifest.json.receipt.json.
- **So2Sat candidate selection and receipt.** experiments/kbound/results/so2sat\_lcz42\_prospective\_v1/development\_mps\_

- `bn_fix_v1/so2sat_candidate_selection.json` and  
`experiments/kbound/results/so2sat_lcz42_prospective_v1/development_mps_  
bn_fix_v1/so2sat_candidate_selection.json.receipt.json`.
- **Claim ledger.** `docs/research/kbound/claim_ledger.json`.
- **Storage manifest.** `docs/research/kbound/STORAGE_MANIFEST.json`.

## 18.1 Why a generated table is not a new experiment

The presentation tables under `paper/generated/` are consumers of the authorities above. They round values and arrange columns for LaTeX. They do not create observations, change action rules, or promote a diagnostic. In this report, a literal number not expanded through a macro is transcribed only from one of those current generated surfaces or from exact maintained text whose artifact is named. The presentation layer must be regenerated when an authority changes; it must not be edited to force agreement with prose.

This distinction is especially important for later studies. CCT-20 and the So2Sat development screen occurred after the canonical 106-source panel was assembled. Their separate, receipt-linked authorities are not merged into that JSON. Treating absence from the canonical panel as absence of the study would be wrong. Silently copying their rows into the canonical panel would also be wrong, because it would erase the independent protocol and release boundary. The claim ledger links each study to the authority that actually governs it.

## 18.2 A reproducible claim is a tuple, not a sentence

For audit purposes, an empirical claim should be read as the tuple

$$Q = (\text{candidate, checkpoint, cell unit, metric, gate, calibration, weights, status, authority}). \quad (63)$$

Changing any component creates a different empirical question. A new adapter learning rate changes the candidate. Replacing accuracy by macro-F1 changes the outcome. Pooling cells instead of weighting families equally changes the estimand. Reusing the same radius after changing the checkpoint changes the calibration identity. Calling an opened grid prospective changes the status without changing the data, which is precisely why it is not allowed.

The tuple also clarifies what a checksum can establish. A digest can show that the bytes named by the authority have not changed. It cannot show that the metric was appropriate, that candidate construction was independent of the test inputs, that residual scores are exchangeable, or that a corruption family represents a future deployment environment. Those are semantic or statistical premises and require their own evidence.

## 19 Controlled CIFAR-10-C Dossier

The CIFAR-10-C study is the largest current record of both directional branches. Its value comes from dense controlled variation and explicit outcome-disjoint reconstruction. Its limitation is equally important: the grid is fully opened, its cells are related constructions from one corpus, and all candidates reuse one source checkpoint. The correct description is

Table 11: Construction of one candidate–seed CIFAR-10-C grid. The product gives 432 cells; five RNG seeds give 2,160 cells per candidate. The table describes the protocol, not independent sample sizes.

| Factor            | Levels | Recorded values                                             |
|-------------------|--------|-------------------------------------------------------------|
| Corruption family | 6      | Gaussian noise, defocus blur, fog, contrast, pixelate, JPEG |
| Severity          | 2      | 1, 5                                                        |
| Batch size        | 3      | 200, 16, 8                                                  |
| Composition       | 3      | IID, imbalanced, single-class                               |
| Update schedule   | 2      | 10 at $10^{-3}$ ; 50 at $2.5 \times 10^{-3}$                |
| Repeat            | 2      | two recorded repeats                                        |

therefore an *opened, dependent, constructed six-family diagnostic*. It is neither prospective confirmation nor a sample of independent deployments.

## 19.1 Factorial construction and weighting

For each candidate and RNG seed, the grid crosses six corruption families—Gaussian noise, defocus blur, fog, contrast, pixelate, and JPEG compression—with severities 1 and 5. Each of those 12 corruption–severity states is crossed with three batch sizes (200, 16, and 8), three compositions (IID, imbalanced, and single-class), two update schedules (10 steps at learning rate  $10^{-3}$  and 50 steps at  $2.5 \times 10^{-3}$ ), and two repeats. The product is

$$6 \times 2 \times 3 \times 3 \times 2 \times 2 = 432 \quad \text{cells per RNG seed.} \quad (64)$$

Seeds 0 through 4 give 2,160 cells for each candidate. These are five stream/run seeds conditional on one archived ResNet-18 checkpoint. They repeat stream construction and adapter randomness; they are not five independently trained models.

Primary point summaries assign equal weight to all 2,160 cells. The family sensitivity has a different, explicitly hierarchical aggregation: it first averages seeds and condition cells within each corruption family, then assigns equal weight to the six family means. The two summaries answer different questions. A cell-weighted average emphasizes every factorial condition equally. A family-weighted analysis treats the six named corruption families as its six resampling units. Neither weighting scheme makes the families independent, and neither turns the opened grid into a future-domain sample.

## 19.2 Local ports and the composite operating point

Tent, EATA, and SAR in this grid are local protocol-matched ports, not untouched reproductions of each paper’s default implementation. Candidate preview runs in training mode. Target-batch statistics replace stored BatchNorm running statistics, while gradient updates modify the selected normalization affine parameters, including LayerNorm parameters where present. Scoring retains the train-mode BatchNorm behavior in evaluation chunks of 512 images. Consequently a recorded cell benefit combines at least two mechanisms: replacement

of running statistics by evaluation-batch statistics, and parameter updates from the adapter objective.

This composite is the candidate that was actually measured. It would be inaccurate to attribute the full cell delta to gradients alone, and it would be equally inaccurate to call the delta a BatchNorm-only effect. A BatchNorm-statistics-only control exists in code but was not executed on this grid. Candidate names identify the local ports; they do not erase the operating-point details. Any comparison to a published method must therefore carry the local-port qualifier.

The ImageNet-C SAR port makes the consequence concrete. Its learning rate is  $4 \times 10^{-3}$ , compared with the published  $2.5 \times 10^{-4}$  setting, a factor of 16. The port also adapts the recorded BatchNorm affine parameters in an upper ResNet block excluded by the published default, and it combines target-batch statistics with gradient updates. No matched control was run at the published SAR operating point. The ImageNet-C record therefore describes this aggressive local candidate, not SAR as published.

### 19.3 Three-way cross-fitting as an information-flow contract

The current reconstruction does not reuse the scored outcome in either of the two artifacts that determine its action. Stable sample identifiers are sorted by a SHA-256-derived order and assigned round-robin to five score folds. For a given score fold  $S_k$ , the complement is reordered with a fold-specific salt. Approximately 30 percent of that complement, subject to the exact-rank minimum, is assigned to residual calibration; the remainder is assigned to estimator fitting. Writing  $T_k$  for the fit indices and  $C_k$  for the calibration indices gives

$$S_k \cap T_k = S_k \cap C_k = T_k \cap C_k = \emptyset. \quad (65)$$

The benefit regressor is fitted on  $(Z_j, B_j)$  for  $j \in T_k$ . Absolute residuals on  $C_k$  produce the exact-rank radius. Only then are prediction, radius, and action computed for  $i \in S_k$ .

This design repairs a subtle leak in the older two-stage replay. Excluding  $B_i$  from the model that predicts cell  $i$  is not sufficient if  $B_i$  can alter another cell’s fitted prediction, enter that cell’s residual, and then flow back into the radius used for cell  $i$ . The separate calibration role blocks that indirect path. The maintained implementation records hashes of the score, fit, and calibration IDs for each fold together with hashes of normalized inputs and live code dependencies.

The contract is tested at the whole-pipeline level. The test changes one scored  $B_i$ , reruns all fits and calibrations, and requires that the same cell’s  $\hat{\Delta}_i$ ,  $\varepsilon_i$ , and action are unchanged. The test lives at `tests/test_controlled_grid_crossfit.py`; it also demonstrates that the retired radius path changes under this perturbation. This is stronger than inspecting a function signature because it follows the complete implemented dependency path.

The invariant proves only what it tests: scored-cell outcome disjointness. It does not prove that different cells are independent or exchangeable. Score folds reuse overlapping complements. Cells share the source checkpoint, corpus, corruption generator, and factorial design. Exact-rank coverage for a future unit would need a defensible calibration/deployment relation beyond this software invariant. The current inclusion rates are consequently descriptive audits of opened cells, not verified coverage promises for a new corruption family.

Table 12: Current CIFAR-10-C action and error exposure. Every row has 2,160 equally weighted opened cells. Conditional error is undefined when its action denominator is zero. Counts are transcribed from the generated primary and interval-diagnostic authorities.

| Candidate | A    | F   | U   | false A | false-A denominators | false-F exposure              |
|-----------|------|-----|-----|---------|----------------------|-------------------------------|
| Tent      | 1091 | 337 | 732 | 2       | 2/2160; 2/1091       | 1/2160; 1/337                 |
| EATA      | 1207 | 105 | 848 | 0       | 0/2160; 0/1207       | 0/2160; 0/105                 |
| SAR       | 1414 | 0   | 746 | 0       | 0/2160; 0/1414       | 0/2160; conditional undefined |

## 19.4 Action exposure and directional errors

The current generated primary table and interval-diagnostic authority record the following  $A/F/U$  triplets: Tent 1,091/337/732, EATA 1,207/105/848, and SAR 1,414/0/746. Each triplet has denominator 2,160 and sums to that denominator. Thus all three candidates exercise ADAPT, but only Tent and EATA exercise FREEZE. SAR’s false-freeze conditional rate is undefined because its FREEZE denominator is zero.

Tent records two false ADAPT actions. Its all-cell frequency is 2/2160, reported after rounding as 0.0009, and its action-conditional frequency is 2/1091, reported as 0.0018. Tent also records one false FREEZE: 1/2160 on the all-cell denominator and 1/337 on the FREEZE-action denominator. EATA records no false ADAPT among 1,207 ADAPT actions and no false FREEZE among 105 FREEZE actions. SAR records no false ADAPT among 1,414 ADAPT actions. These are observed measured-cell signs on an opened diagnostic, not population error probabilities.

For SAR, the false-FREEZE all-cell numerator is also zero, written 0/2160. Its conditional false-FREEZE rate is undefined because the policy issued no FREEZE action; it must not be printed as zero.

Observed interval inclusion supplies another denominator. The nominal setting is 90 percent. The current intervals include 1,976 of 2,160 Tent outcomes, 1,950 of 2,160 EATA outcomes, and 1,949 of 2,160 SAR outcomes. Inclusion is not commitment: an interval can include the measured outcome while crossing zero, and an interval can exclude the outcome while still triggering an action. The near-nominal pooled values are not independent validation because the grid is opened, dependent, and cross-fitted through overlapping complements.

The point-regret consequences are candidate-specific. Tent’s KGA, always-adapt, and always-freeze regrets are 0.0019, 0.0080, and 0.1239. EATA’s are 0.0016, 0.0033, and 0.1313. For SAR the generated table reports 0.0018, 0.0003, and 0.1405, respectively. Thus the current point rule has lower regret than each fixed policy for Tent and EATA, while always-adapt has lower regret for SAR. These values describe three separately calibrated candidates. Averaging them into one headline would create a policy that was never deployed.

## 19.5 Six-family sensitivity and the multiplicity boundary

For the family analysis, define a paired gap as fixed-policy regret minus KGA regret, so positive values favor KGA. Tent’s gap against always-adapt is 0.00606 with ordinary, unadjusted nominal 95 percent family-bootstrap interval [0.00209, 0.00954]. Its gap against always-freeze is 0.12202 with interval [0.06282, 0.17757]. EATA’s adapt-side interval is [−0.00135, 0.00438]



and therefore crosses zero, although its freeze-side interval is positive. SAR’s adapt-side gap is negative. This is why Tent and EATA must not be described with the same interval sentence: Tent has two favorable nominal intervals, whereas EATA does not.

The ordinary intervals and the retrospective reference tests are different inferential objects. The retrospective family contains all six candidate-by-baseline contrasts; the exact-rank replay, corruption-family bootstrap, sign-flip calculation, and Holm analysis were chosen after the grid was open. Under Holm adjustment over all six sign-flip values, Tent’s adjusted values are 0.140625 against always-adapt and 0.09375 against always-freeze. EATA’s are 0.3125 and 0.09375. No candidate clears both comparisons at family-wise level 0.05.

A within-Tent two-comparison adjustment gives smaller post-hoc values, but it is not the declared six-comparison family and cannot replace it. Likewise, a positive unadjusted interval endpoint is not a multiplicity-adjusted confidence statement. The family count is small, the clusters share a checkpoint, and coordinatewise sign-flip invariance is an unverified reference assumption. The proper conclusion is a favorable nominal Tent pattern accompanied by a failed retrospective Holm gate, not confirmatory or cluster-robust evidence.

## 20 Natural and Architecture-Panel Dossiers

The natural and semantic-shift records test different failure modes from the constructed CIFAR grid. They do not form one pooled benchmark. Some predominantly exercise retention, some expose a candidate-specific adverse pattern, and one stops before the target stage. Their scientific value lies in showing where the gate’s action branches were and were not exercised.

### 20.1 CCT-20: a governed all-frozen outcome

CCT-20 crosses 5 independently trained source checkpoints with 9 target camera locations, giving 45 checkpoint–location cells. The release macros record 6950 probe images and 16325 evaluation images, 23275 in total. Its primary metric is set-membership top-1 accuracy: a prediction is correct if it matches any distinct archived category for the image. It is not the official single-label leaderboard metric.

The Tent candidate uses the official implementation with one non-episodic Adam update per incoming batch at learning rate  $10^{-3}$ . Each batch is scored using pre-update logits, adaptation state continues within one checkpoint–location cell, and state resets between cells. This was a cell-outcome-unopened, internally sealed execution: individual cell outcomes remained unopened until one-shot scoring. Aggregate target-label metadata had previously been inspected, so the study was not globally label-unopened or publicly preregistered.

Two endpoints were locked. The limited safe-utility endpoint uses nominal pointwise 95 percent paired product-bootstrap lower endpoints. With  $L_a$  denoting the gap against always-adapt and  $L_f$  the gap against always-freeze, it requires

$$L_a > 0, \quad L_f > -0.005. \tag{66}$$

The generated margin -0.005 is on the accuracy-proportion scale. The second inequality is noninferiority to the frozen policy and can be satisfied by a controller that always serves  $f_0$ . The endpoint name does not turn this check into a population-safety theorem.

The strong-success rule is more demanding and is the appropriate test of bidirectional selective routing. It requires all 45 cells; nominal 97.5 percent Bonferroni intervals strictly above zero against both fixed policies; both exact one-sided tests over nine location means to survive Holm adjustment at 0.05; ADAPT rate at least 0.10; FREEZE rate at least 0.10; and total commitment at least 0.30. Failure of any component blocks that claim. These thresholds were not added after seeing the result.

The nominal pointwise 95 percent intervals in the CCT safe-utility table belong to the locked safe-utility endpoint. The nominal 97.5 percent Bonferroni intervals in the strong-success audit belong to a separate, stricter two-comparison criterion; they are not alternate versions of the same interval.

The realized action record was 0 ADAPT, 44 FREEZE, and 1 ABSTAIN. Because both FREEZE and ABSTAIN serve  $f_0$ , every evaluated prediction came from the frozen model. Tent helped in 1 cell, tied in 0, and harmed in 44 cells. The gate issued FREEZE on the one helpful cell and on 43 harmful cells; on the remaining harmful cell, the ABSTAIN action served the frozen model. The observed false-FREEZE frequency is  $1/45$  over all cells and  $1/44$  conditional on a FREEZE action. The false-ADAPT frequency is  $0/45$ , while the conditional false-ADAPT rate is undefined because the ADAPT count is zero. The zero false-ADAPT numerator is therefore mechanical for that branch.

Against always-adapt, the baseline-minus-KGA point contrast is 0.18190227 with nominal 97.5 percent interval  $[0.08501071, 0.31066374]$ , exact location sign-flip value 0.00195312, and Holm value 0.00390625. Against always-freeze, the contrast and interval endpoints are 0,  $[0, 0]$ , with Holm value 1. The stored decision therefore matches always-freeze and avoids the measured loss from always-adapt. It passes the limited safe-utility endpoint but fails strict improvement over freeze and fails the ADAPT-exposure threshold. Its release verdict remains SAFE UTILITY ONLY: all-frozen fail-closed behavior, not selective-routing success.

The inferential assumptions remain visible. The two-way bootstrap resamples checkpoint rows and location columns; the exact test enumerates sign flips of nine location means. Enumeration removes Monte Carlo error, but not the need for coordinatewise sign-flip invariance. The ridge estimator was fitted using 10 rows from only two camera locations and calibrated on 45 rows aggregated to nine cis-test locations. The target locations are trans-test. All 55 development cells were harmful for Tent, so the estimator saw no helpful development example. These facts explain why the result is evidence of conservative retention rather than evidence that a helpful/harmful boundary was learned across natural locations.

## 20.2 ImageNet-R: architecture variation is not environment replication

The current ImageNet-R authority contains four registered run seeds, ten backbone-specific candidates, and 12 conditions per backbone and seed, for 480 cells. The architecture-panel aggregate is descriptive because each backbone is a separately calibrated candidate. It is not one router applied to ten independent target domains.

Before utility is considered, the aggregate action record is  $A/F/U = 179/34/267$ : 179 ADAPT, 34 FREEZE, and 267 ABSTAIN actions. There are six false ADAPT actions, giving  $6/480$  on the all-cell denominator and  $6/179$  conditional on ADAPT. These are descriptive

sums across separately calibrated backbone candidates, not error estimates for one deployable policy.

The canonical aggregate gives KGA regret 0.0137, always-adapt regret 0.0064, and always-freeze regret 0.0325. KGA has higher regret than always-adapt on eight of the ten backbones. Four of those eight have a 0 percent harmful-update base rate, leaving no candidate harm for a router to avoid. The two remaining backbone patterns are conservative ties. On `efficientnet_b0`, KGA has lower regret than always-adapt but exactly matches always-freeze at zero regret. On `swin_t`, it again has lower regret than always-adapt but matches always-freeze at 0.00473958. Both rows make no ADAPT decisions. Thus KGA has lower regret than each fixed comparator on zero of ten backbones.

This breakdown is adverse but useful. It shows that an aggregate gap can hide two structurally different limits: when adaptation is uniformly helpful, abstention or freezing forgoes utility; when adaptation is harmful, a conservative controller may merely recover the frozen policy. The panel does not support a universal ranking, but it directly constrains any claim that the current gate transfers across architectures.

### 20.3 ImageNet-C: candidate identity and an aggressive SAR port

The current ImageNet-C authority has 27 conditions per run seed and five run seeds, or 135 cells per candidate. The candidate rows differ sharply. The SAR row has 28/9/98 ADAPT/FREEZE/ABSTAIN actions and records 0/135 marginal false-adapt events, or 0/28 conditional on ADAPT. Its point regrets are 0.0227 for KGA, 0.0529 for always-adapt, and 0.0319 for always-freeze. Descriptive run-seed intervals have positive lower endpoints against the two fixed policies, but the run seeds do not record independent checkpoint identities. Those intervals cannot support an independent-checkpoint or confidence-robust promotion.

The operating-point disclosure is inseparable from the numerical row. As noted above, SAR uses the  $4 \times 10^{-3}$  local learning rate rather than the published  $2.5 \times 10^{-4}$  setting, includes a broader set of BatchNorm affine parameters, and combines evaluation-batch statistics with updates. No published-setting control exists. The point record is therefore a diagnosis of that candidate configuration only. It is not evidence for a statement about SAR under its published default.

### 20.4 So2Sat-LCZ42: stopping is the result

The locked So2Sat-LCZ42 v1 study evaluated two candidates, Tent and SAR, at a development-only feasibility stage. Its current generated surface records nine development cities, five independent source checkpoints, and 45 development cells per candidate, hence 90 opened development outcomes. Tent showed both helpful and harmful cities, but neither candidate passed all nine predeclared feasibility checks. The locked decision was to stop before gate calibration.

The access boundary is more important than a target score because no target score exists. Gate-calibration rows read before selection were zero. Target-input count, target-pixel reads, and target-label reads were all zero. The target runner was not executed. The v1 protocol also named one action per city while the dormant runner would have produced city-checkpoint

actions. That mismatch did not affect an action because the study stopped before any target action was created.

The development outcomes are now open. Retuning on them cannot make the same panel fresh. A v2 design resolves the action unit, but it remains a no-go without its own pre-calibration seal and target authorization. The defensible conclusion from v1 is a negative development-stage eligibility result with no target access. It is not a natural-shift routing result, and the absence of a target table must not be filled with exploratory handoff notes.

## 21 Negative Results That Define the Present Boundary

Negative results are not ancillary in a decision paper. They identify which tempting shortcuts do not support the desired action claim. Three records are particularly important: the failure of a development proxy for the population budget, the mismatch between exploratory search counts and a frozen multiplicity family, and the historical status of local comparator ports.

### 21.1 The development surrogate for $\beta$ failed

The population theorem assumes an external bound  $|\gamma| \leq \beta$  on a hidden disagreement-conditional calibration residual. The empirical radius  $\varepsilon$  instead describes prediction error for a measured cell target under a calibration protocol. The two quantities live in different layers. Replacing  $\beta$  by a fitted development residual would need a justified bridge from source-like development cells to the target disagreement fibre.

That route was tested and failed. The development-based estimate was between 1.4 and  $50\times$  too small on real evaluation cells. Across the reported configurations, 24–73 percent of cells fell outside the declared class induced by that estimate, and commit error ranged from 0.4 to 16.6 percent. In five of ten configurations the route returned no commitments on all 405 ImageNet-C cells. The estimation route is therefore withdrawn;  $\beta$  must be declared from domain knowledge or another explicit structural assumption instead.

This negative result has a provenance limit of its own. The maintained manuscript records the exact result text, but the named raw beta-sweep artifact is not present in the canonical release bundle. The result is historically documented rather than currently regenerable from the canonical panel. That limitation must accompany the numbers. It would be incorrect either to hide the failed test or to present its historical record as a newly sealed experiment. KGA does not estimate  $\beta$  and does not apply the population inequality numerically.

### 21.2 Search history is not a multiplicity family

One superseded repository census reported 1,387 determinations and 326 positives, or 23.5 percent. A broader archived scan reported 1,427 determinations and 345 positives, or 24.2 percent. These counts overlap, and neither froze a reproducible family definition, exact file list, and repository state before the outcomes were examined. They are historical search summaries, not two independent experiments and not a basis for calculating a release-wide adjusted  $p$ -value.

The current six-contrast CIFAR family is narrower and separately defined. CCT-20 has its own two-comparison inference family. Archived comparator-port values have another historical scope. Pooling these families after the fact would obscure rather than solve selection. A future confirmatory inventory must enumerate candidate-track arms, incomplete runs, exclusions, calibration choices, and both fixed-policy contrasts before target outcomes are opened.

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**Non-evidentiary provenance note: historical POEM/AETTA ports.** The archived CIFAR-10-C Tent-stream comparison used local protocol-matched, non-official ports and an earlier KGA policy that was not recomputed under the current three-way cross-fit. Its historical regrets were 0.00157361 for that earlier KGA policy, 0.00880463 for the POEM-style port, 0.00732986 for the AETTA-style port, 0.00792338 for always-adapt, and 0.12409792 for always-freeze. The stored KGA-minus-port differences were  $-0.00723102$  with unadjusted paired bootstrap interval  $[-0.00878038, -0.00570806]$  against the POEM-style port, and  $-0.00575625$  with interval  $[-0.00714260, -0.00439859]$  against the AETTA-style port. Holm was applied only to the archived paired  $p$ -values; the stored adjusted value was 0.00039996 for each comparison.

These numbers document what the old artifact contains. They do not support the current policy, do not constitute an official reproduction of either comparator, and do not establish superiority. The current claim ledger marks native official reproduction as pending. A usable comparison must run all methods on identical current-policy records and freeze its multiplicity family before scoring. No argument elsewhere in this report relies on the historical ordering.

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## 22 From an Artifact to a Defensible Sentence

The same numerical row can support several sentences of very different strength. The following workflow makes the choice explicit. It is deliberately conservative because the cost of an ambiguous claim is larger than the cost of one additional qualifier.

### 22.1 First identify the direction and target

Write down the sign convention before comparing values. In this report, measured benefit is adapted score minus frozen score. Regret is a nonnegative shortfall from the cell oracle. Paired policy gaps are fixed-policy regret minus KGA regret, so a positive gap favors KGA. Changing the sign convention between a table and its interval reverses the interpretation even when every number is copied correctly.

Then name the target. A cell-benefit outcome, a mean family gap, a location-cluster contrast, and population risk reduction are not synonyms. The interval must cover the same target that appears in the action proposition. If the target is  $B_i = \Delta_i^{\text{cell}}$ , the resulting statement is about the cell. Moving to  $\Delta$  needs an additional sampling and metric bridge.

## 22.2 Then expose the branch that generated the utility

Report  $A/F/U$  before describing regret. A policy can match always-freeze because it correctly rejects many harmful candidates, because every interval is unresolved, or because its action path is unavailable. Those mechanisms have different deployment meaning even when the served predictions are identical. Similarly, low regret on an all-helpful panel can come from adapting on every cell without demonstrating harm detection.

For each wrong-direction numerator, print the all-cell denominator and the action denominator. When a denominator is zero, use “undefined,” not zero. Preserve ties in the error definition: strict directional semantics count ADAPT with  $B_i \leq 0$  and FREEZE with  $B_i \geq 0$  as unsupported directions, even though a tie has zero utility regret for either served model.

## 22.3 Finally state design, dependence, and multiplicity

Label access is not a binary adjective for an entire project. A study can use labeled development cells, keep target cell outcomes unopened until scoring, and still have inspected aggregate target metadata during dataset selection. State each boundary. Likewise, distinguish one checkpoint with five RNG seeds from five independent checkpoints, and distinguish ten candidate backbones from ten environments for one router.

Every interval needs its resampling or calibration unit and its status. A retrospective family bootstrap over six corruptions is not an independently validated confidence interval. A sign-flip enumeration is exact relative to its reference distribution but does not verify the sign-symmetry assumption. Holm adjustment acts on the named collection of  $p$ -values; it neither adjusts ordinary confidence intervals nor repairs an invalid marginal test.

# 23 Interpretation Checklist

Before accepting an empirical sentence from this report, verify every item below. A failed item does not always invalidate the underlying artifact, but it narrows the admissible wording.

1. **Authority.** Is the number expanded from a current macro or copied from the exact generated surface linked to the canonical, CCT-20, or So2Sat authority? Is its digest recorded?
2. **Status.** Is the study current diagnostic, governed target execution, development stop, withheld row, invalid route, or historical provenance? Does the sentence use that status rather than a more favorable one?
3. **Candidate identity.** Are checkpoint, adapter port, learning rate, update schedule, adapted parameters, BatchNorm mode, and state-reset semantics the ones actually evaluated?
4. **Evaluation unit.** Is one action attached to a condition cell, checkpoint–location pair, city–checkpoint pair, or another declared unit? Are images within a cell being mistaken for independent gate decisions?
5. **Metric.** Is the score accuracy, set-membership accuracy, balanced accuracy, or macro-F1? Does the metric match the authority and the population claim, if any?
6. **Action exposure.** Are ADAPT, FREEZE, and ABSTAIN counts visible? Does a reported zero error come with nonzero exposure, or is it mechanical because the branch

never ran?

7. **Error denominator.** Is a false-action numerator divided by all cells or only by the corresponding action count? Is an absent conditional denominator marked undefined?
8. **Outcome flow.** Could the scored  $B_i$  change its own prediction, radius, or action? For the controlled grid, does the whole-pipeline perturbation test still pass under the current implementation?
9. **Dependence.** Do folds share a checkpoint, corpus, environment, or overlapping complement? Are RNG seeds being presented accurately rather than relabeled as independent model replications?
10. **Interval premise.** What target, level, rank rule, and resampling unit does the interval use? Is exchangeability, sign symmetry, or transfer assumed rather than verified?
11. **Multiplicity.** Which contrasts form the family, when was that family frozen, and what exactly was adjusted? Are historical search counts or separate studies being inappropriately pooled?
12. **Utility scope.** Does a regret or accuracy contrast describe only the recorded candidate and panel? Is it kept secondary to directional correctness, exposure, and the protocol that made the action possible?
13. **Population boundary.** Has a measured-cell statement been promoted to population benefit without a sampling-error bridge? Has the empirical radius  $\varepsilon$  been confused with the externally declared population residual budget  $\beta$ ?
14. **Negative evidence.** Are stopped studies, adverse architecture patterns, failed estimators, withheld metrics, and absent official comparisons retained rather than omitted from the claim family?
15. **Reproduction boundary.** Do the available compact sources suffice to recompute the displayed result? If raw data or checkpoints are external or historical identity is incomplete, does the sentence say so?

Passing this sentence-level checklist does not certify a deployment. It shows only that the claim is traceable, action exposure is explicit, and remaining assumptions are visible. That is the reporting standard for directional action under uncertainty; accuracy and regret remain scoped consequences.

## Part IV

# Formal Verification and Reproducibility

## 24 Formal Proof and Reproducibility Atlas

This chapter explains what is proved, what is checked mechanically, what remains a scientific assumption, and how a reader can reproduce the maintained publication surfaces without confusing those four categories. The distinction is central to K-Bound. A theorem may be correct under its encoded assumptions while a benchmark fails those assumptions. A data

validator may establish that a table agrees with a sealed artifact while saying nothing about the theorem. A successful PDF build establishes neither. Treating all three as a single notion of “verification” would erase the boundary that the paper is designed to expose.

The current integrated formal receipt registers 315 selected Lean declarations: 65 retained core declarations and 250 additional registered results, including supporting lemmas. This is a registry count, not the total number of authored lemmas, and it is not a score of novelty or scientific importance. Supporting declarations compile as dependencies, while some declarations that are displayed by the theorem map are intentionally not registry capstones. The precise claim is therefore narrow: the dated strict-core build and transitive-axiom audit report success for the 315 registered names under their encoded assumptions. The source-bound summary `formal/full_report_bridges_verification_20260908.portable.json` links the 8 September combined build and independent reviews. The build uses pinned cached dependencies; it is not a cold dependency rebuild. The older `audits/formal_foundations_2026_08_31.json` records 142 declarations, and the later 150-declaration audit-floor, 161-declaration probability-interface and 221-declaration paper-clause evidence retain their dated scopes. Those historical records are preserved. None reports that every sentence in every earlier theory draft has been formalized. The new 60-name addition covers arbitrary-joint-law disagreement reductions, actual calibration and cell/population probability witnesses, dependent and independent sign-flip clauses, and decision/regret algebra. A subsequent 17-theorem addition proves the literal measurable conformal ceiling/order-statistic/infinity absolute-residual construction with ties and marginal coverage under explicit joint score exchangeability. Eight further results derive finite iid input-batch evidence laws and fixed Markov-policy or independent-common-seed action laws and abstention bounds. Separate marginal equalities do not supply a common observation/seed joint law. Eight further proofs identify the actual conditional kernel’s correctness integral with joint event probability and derive the bounded-score residual, benefit and sign identities for every binary joint target law. Positive disagreement mass is required for conditional interpretation; no Standard Borel hypothesis is added to the input space. Ten further proofs establish the full actual score-defined joint-law class’s strict frontiers, clipped interval, nonemptiness, closed-band/zero witnesses and pointwise rule semantics. Eight further results assemble actual interior/zero witnesses, the fixed augmented-evidence-law strict-direction iff and randomized abstention under a fixed marginal-based observation experiment, with its concrete finite iid joint-sample realization proved. Fourteen further results supply the restricted-class interface using a legal interior opposite-pair construction and boundary-only zero membership. They derive actual residuals and benefits, strict-direction existence iff, maximal commitment and restricted randomized abstention. Individual directional iff/uniqueness explicitly requires a realized nonempty subclass. These minimal membership conditions follow from paper construction closure, without asserting equivalence to every construction. The main text now states the fixed observation experiment and nonempty-subclass condition explicitly. Observed-rank evidence and exhaustive semantic coverage remain separate obligations; arbitrary batch or seed couplings are not inferred from one-input marginals. Three further proofs compose helpful-only scores and regret over the same finite nonnegative weights, with positive total for means. Five further proofs separate risk alignment from strict identification on a fixed fibre, using actual zero/positive joint laws at  $M=\beta>0$ . Empirical helpfulness and risk alignment remain premises. Five further proofs encode the exact feature-coordinate



order and relations, establish dimension9 of that real-linear schema and bound the span of satisfying rows by9. Observed exact rank9, numerical tolerance, probabilistic independence and physical feature realizability are separate. Four further proofs assemble the actual full-class supremum radius and fixed augmented-law fibre radius from an actual extremal target. The full fixed-input-marginal class, positive disagreement and  $0 \leq \beta \leq \frac{1}{2}$  remain explicit. The class supremum needs no sampling premise; the augmented-law fibre equality uses the fixed observation model. These mathematical premises are now explicit in the main text; whether an empirical protocol satisfies them requires separate evidence. The 272, 286, 294, 299 and 303 stages are preserved separately. Twelve additional results close five full-report implication groups: null-disagreement risk equality, joint-observation fibre refinement and radius monotonicity, invariance under fixed measurable recoding, conditional error given positive exposure and a marginal bound, and integrable-proxy decomposition with sufficient signs. The general-loss and multiclass-proxy paragraphs now state the necessary integrability conditions; no general-loss exact frontier is claimed. The separate semantic census is `formal/full_report_bridges_semantic_20260908.portable.json`.

## 24.1 Four independent verification layers

The repository separates four layers that are often collapsed in machine-learning papers.

**Mathematical statement.** A paper theorem specifies objects, assumptions, and a conclusion. The decisive questions are whether the assumptions define a nonempty class, whether every symbol has one type and one unit, and whether the conclusion follows for every member of that class. For the population frontier, the key objects are the target law, the frozen and adapted predictors, the positive-mass disagreement region, and the externally declared residual budget. For the empirical coverage-to-action implication, the target is instead an observed cell-level benefit and the conclusion is an event containment conditional on interval coverage.

**Proof implementation.** Lean encodes selected statements and checks proof terms against a pinned kernel and library. A compiled proof establishes the encoded statement, not a nearby prose statement. Small differences matter. A fixed finite-real radius is not an arbitrary measurable extended-real random radius. A full measurable correctness-field class is not an arbitrary restricted deployment class. Arithmetic showing that two opposite worlds force abstention is not, by itself, the measure-law construction showing that the worlds have identical observable evidence.

**Empirical authority.** Generated tables are checked against current result artifacts and the claim ledger. These checks can detect stale counts, a mismatched hash, or an unsupported table cell. They do not transform a retrospective diagnostic into prospective confirmation, and they do not prove exchangeability or population transfer. The result validators guard factual consistency; the paper’s interpretation must still state the design and denominator.

Table 13: Verification layers and their nonimplications. The rows are deliberately separate: no single green check subsumes the others.

| Layer               | A successful check establishes                                                  | It does not establish                                                                                        |
|---------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Paper mathematics   | A stated argument is valid under its written assumptions                        | That a benchmark satisfies those assumptions, or that the assumptions are learnable from label-free evidence |
| Lean kernel         | The encoded declaration has a checked proof with the audited axiom dependencies | Equivalence between the encoded declaration and every prose generalization                                   |
| Evidence validation | Current displayed values agree with the named current artifacts                 | Prospective confirmation, external validity, or population-risk protection                                   |
| PDF validation      | The maintained sources compile and render legibly                               | Mathematical truth, data provenance, or deployment readiness                                                 |

**Publication rendering.** The TeX build resolves references, lays out tables, and emits the PDFs. Rendering every page can detect clipping, blank pages, malformed glyphs, or an unreadable table. That is a necessary publication check, but it has no authority over the science. In particular, a clean log cannot rehabilitate a stale numerical claim or an invalid theorem module.

Table 13 records the useful negative rule for each layer: what a pass does *not* authorize.

## 24.2 Clause-level proof coverage

The maintained theorem map is `formal/KBound/TheoremMap.lean`. It is an executable index: its `#check` commands ensure that named declarations remain available with types accepted by Lean. The map also documents the intended correspondence between paper clauses and formal objects. That correspondence is more informative than reporting one aggregate proof count.

**Benefit reduction.** Lemma 4.6 has an algebraic finite core and an arbitrary-joint-law binary realization. The actual conditional-kernel correctness integral equals the joint correctness event probability, yielding the score-residual, population-benefit and sign identities for every binary joint target law. Positive disagreement mass is explicit; the input space need not be standard Borel. Separate measurable kernel constructions supply extremal target-law witnesses. These results do not extend the scalar binary residual geometry to unrestricted multiclass kernels or a different loss.

**Matched-evidence nonidentifiability.** Actual legal opposite-tilt witnesses have equal augmented-evidence laws under one fixed marginal-based observation experiment. Finite iid sampling is a proved realization; arbitrary world-dependent batch couplings are not inferred from equal one-input marginals. Full-class and construction-closed restricted-class directional error bounds compose with one fixed Markov policy to give abstention at least  $1 - 2\alpha$ . A

restricted class must contain the required witnesses; its richness is not automatically inferred. The general action-pushforward interface and the finite `Impossibility.lean` arithmetic remain supporting layers. Equal random-seed marginals alone do not establish the required common observation/seed joint law.

**Closed band and strict frontier.** The full actual scoreResidual-defined joint-law class has the clipped identified interval, strict-direction iff, nonemptiness and pointwise maximal rule. The restricted-class interface requires a legal opposite pair in the interior and zero membership only on the boundary; no interior zero is required. Individual directional iff and uniqueness explicitly need a nonempty restricted class. These minimal membership conditions follow from the paper’s construction closure, not conversely, and do not establish richness of a practitioner-selected class.

**Coverage-to-action certificate.** The deterministic certificate checks pointwise containment: a false ADAPT or false FREEZE action can occur only if the named interval misses the named scalar target. A measure-level file proves the corresponding probability inequality for a fixed finite-real radius. A conformal file proves a specialized random-threshold result under exchangeable measurable residual scores, including ties. The `ExtendedRadiusCertificate.lean` interface now proves the more general event-containment and probability bound for a measurable nonnegative radius that can take finite and infinite values on the same probability space. Infinity abstains; finite nonnegative real radii agree with the original certificate. Marginal coverage bounds the measurable union of both strict directional errors by  $\alpha$ . Coverage for the declared target and availability before evaluation-label access remain external premises. Population use requires coverage for population benefit under the declared observation/model scope; exact fibre-wide risk alignment is not an additional premise of this event-containment implication. The separate identification frontier concerns uniformly strict directions over compatible laws. The specialized conformal formalization does not prove that heterogeneous benchmark cells are exchangeable.

**Probability foundations outside the headline.** The strict-core also includes general testing, concentration, optional-stopping, and measurable swap results. They strengthen the library available for future work, but they do not enlarge the headline theorem by proximity. The general Le Cam layer treats arbitrary probability measures and randomized measurable tests. The concentration layer treats bounded independent sums and adapted martingale differences under explicit assumptions. The filtered Ville layer treats nonnegative supermartingales and predictable betting products. The measurable swap layer proves invariance and opposite-benefit constructions for a channel, and the channel counterexample shows why selecting one representative per swap orbit does not orient an entire evidence fibre. None of these facts proves the unfinished historical one-bit, ratio, or rate characterization.

### 24.3 The strict-core gate and the disclosed incomplete layer

The recommended command invokes the pinned Lean build and then the strict-core audit. The audit checks that every registered name exists, scans the maintained formal source

Table 14: Clause-level formal status of the maintained headline. “Exact” always means exact for the encoded assumptions, not for an unstated deployment class.

| Paper clause                  | Status                                                                              | Boundary                                                                              |
|-------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Binary benefit/sign reduction | Exact for the encoded binary zero-one class                                         | No automatic multiclass or arbitrary-loss lift                                        |
| Evidence-fibre witnesses      | Actual full class and construction-closed restricted classes                        | Richness of a practitioner-declared subclass must be justified externally             |
| Matched-abstention corollary  | Actual full/restricted witness and probability composition                          | Fixed observation/policy and world-specific error bounds remain explicit              |
| Closed-band/strict frontier   | Actual full-class and construction-closed subclass interfaces                       | Equality gives no strict direction; individual subclass uniqueness needs nonemptiness |
| Random-radius certificate     | Measurable finite/infinite radius; coverage implies a joint directional-error bound | Declared-target coverage and label custody remain external                            |
| Exchangeable random threshold | Exact specialized conformal result                                                  | Does not establish empirical exchangeability or transfer for the benchmark            |

for forbidden proof holes, and asks Lean to print transitive axiom dependencies. The allowed kernel dependencies are `propext`, `Classical.choice`, and `Quot.sound`. A missing declaration, failed build, proof hole, or extra axiom fails the gate. Running the audit without the build flag is only a static inventory check and is not reported as kernel verification.

The command `--full-foundations` deliberately asks a stronger question. It remains a failure because the sixth, one-bit/channel characterization is incomplete: a verified counterexample refutes the earlier orbit-selection sufficiency idea, and the associated H/ratio and rate claims have not been repaired. This failure is informative rather than cosmetic. It prevents a local success on the five maintained probability layers from being advertised as a proof of every proposed extension. The full report uses the strict-core receipt and preserves the stronger failure as a boundary.

Three counts should therefore never be conflated:

1. the number of registered capstones audited transitively;
2. the number of declarations displayed by the theorem map; and
3. the total number of theorem and lemma declarations compiled as source or dependency.

The first count is 315 in the current integrated receipt. The second is 326 across all namespaces, including eleven entries outside the registry; the third is not claimed by either number. Reporting this distinction avoids the incorrect phrase “315 total proofs.”

## 24.4 Authority graph for numerical claims

The numerical publication path is a directed graph, not a collection of manually copied table cells. Raw or reconciled result artifacts feed bounded generator scripts. Generated TeX macros feed shared manuscript sources. The claim ledger records allowed wording, supporting artifacts, and prohibited interpretations. A storage manifest binds mutable current authorities by byte count and SHA-256, while sealed evidence entries retain their own immutable checksums. The manuscript validator then compares these surfaces. Finally, the PDF builder validates before it regenerates presentation files.

The order matters. If a current claim-ledger byte count changes, refreshing only the PDF cannot repair the broken storage binding. If a table macro changes without its current source artifact, the validator should fail rather than accept the visually plausible number. If a historical seal does not match, the bounded mutable-authority refresh must not rewrite that seal. Each repair must occur at the layer that owns the discrepancy.

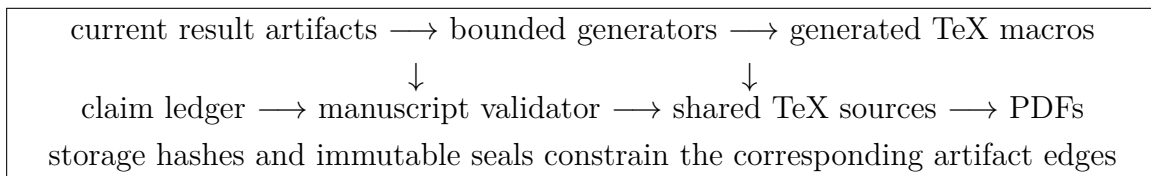


Figure 3: Conceptual authority graph. Arrows denote a checked derivation or binding, not an assumption that two files agree.

This graph also explains why the canonical panel hash appears in the primary result presentation. It is an identity for the scored panel, not a decorative reproducibility token. A reviewer can use it to distinguish the current three-way cross-fit outcome from an older panel with similar method names. CCT-20 and So2Sat have separate receipt-linked authorities and are not silently folded into that panel identity.

## 24.5 Reproduction modes and mutation boundaries

Reproduction is safer when read-only verification is separated from regeneration. The current workflow has three practical modes.

**Inspection.** A reader may inspect the claim ledger, current manifests, generated macros, and theorem map without changing any artifact. This is the right starting point for peer review. It can answer questions such as: Which artifact supports this count? Is a result current evidence or non-evidentiary provenance? Which formal file covers this theorem clause? Inspection cannot establish that the local checkout is complete or that every external dataset is mounted.

**Bounded manuscript verification.** The manuscript-claim validator checks the maintained TeX closure, canonical panel, release consistency surfaces, direct storage hashes, and sealed evidence hashes. The PDF build runs this check before table and figure regeneration. New output is compiled in a fresh temporary directory so an old auxiliary file cannot supply a

missing label or citation. Only allowlisted PDF and log names can be published back to the manuscript directory. These controls make it harder for an old successful build to mask a broken current source.

**Full release.** A publication release is stronger than a manuscript build. It requires a reviewed clean source freeze, pinned Python and external-tool content, the complete declared test inventory, the Lean strict-core build, archive and privacy verification, source sealing, rendered-page inspection, and a final outer checksum inventory. The repository’s older end-to-end release wrapper, page-render helper, and checksum verifier must be checked against this publication family’s actual inventory. A bounded manuscript build must be accompanied by all-page rendering inspection and a newly emitted outer SHA-256 inventory before delivery; neither is implied by successful compilation. Those records establish a handoff receipt, not a clean source seal. External publication actions—pushing a tag, uploading an archive, creating a DOI, or submitting to a venue—remain author actions and are not implied by a local pass.

The separation also defines what not to do. A presentation build must not rerun training. A manifest refresh must not reseal historical evidence. A checksum file must be emitted only after the final PDFs are fixed. A dirty-tree checksum is useful for local handoff but is not a clean source seal. A missing cloud-backed Git object cannot be repaired by describing a component test as a release pass.

## 24.6 A tested implementation invariant

The empirical action for cell  $i$  is intended to be outcome-disjoint: its own scored outcome  $B_i$  cannot influence its prediction, radius, or action. The recorded regression test checks that property for index 7 in two synthetic controlled-grid instances. Each instance holds the fold assignment, features, and hyperparameters fixed, perturbs only  $B_7$ , and reruns the recorded cross-fit procedure. For the focal cell it requires equality of the prediction, radius, and action. Changes for other cells are allowed because their fitted models may legitimately use the perturbed cell.

This is a tested implementation invariant, not a software theorem or a population theorem. Its scope is the two fixed synthetic instances; it does not exercise every possible upstream feature, preprocessing, or hyperparameter-selection path. It also does not establish coverage, exchangeability, or generalization. Conversely, a mathematical coverage proposition cannot compensate for an implementation that leaks the scored outcome. The formal and software claims are complementary because they guard different arrows in the authority graph.

The most useful failure diagnostics localize the broken arrow:

- a changed prediction implicates fit, selection, or preprocessing leakage;
- an unchanged prediction but changed radius implicates residual or calibration leakage;
- unchanged prediction and radius but changed action implicates action-rule state or a hidden tie-breaking path;
- a stable focal cell with changes elsewhere can be correct under cross-fitting and should not be rejected automatically.

## 24.7 Reviewer trace procedure

A compact way to audit any important sentence is to trace it backwards through six questions.

1. **Claim type.** Is the sentence a definition, theorem, finite-panel observation, diagnostic interpretation, or future protocol?
2. **Target.** Does it concern population benefit  $\Delta$ , observed cell benefit  $\Delta^{\text{cell}}$ , a fitted prediction  $\hat{\Delta}$ , or an action consequence?
3. **Scope.** Which candidate, dataset family, action exposure, loss, and deployment class are named?
4. **Authority.** Which current artifact or proof clause supports it, and is that identity bound by a current hash or theorem map?
5. **Negative boundary.** Which nearby stronger reading is expressly not supported?
6. **Reproduction.** Which read-only validator or build command would fail if the supporting surface changed?

For example, a sentence about a false ADAPT count is incomplete until it names the ADAPT exposure and the all-cell denominator. A sentence about an interval is incomplete until it names the scalar target and whether the interval is nominal, retrospective, or prospectively covered. A sentence about the population frontier is incomplete until it names the declared class and the external residual budget. A sentence about formalization is incomplete until it distinguishes the paper clause from the encoded declaration.

## 24.8 Reproducibility checklist for this report

The following checklist is intentionally shorter than the repository release runbook. It states the evidence needed to reproduce the publication family itself.

1. Verify the storage manifest and immutable evidence bindings before generating tables.
2. Run the manuscript-claim validator and the focused claim-consistency tests.
3. Run the scored-cell outcome-invariance regression and publication-driver boundary tests.
4. Build the strict Lean core with the pinned toolchain; retain the stronger full-foundations failure rather than relabeling it as a pass.
5. Build each PDF from a fresh auxiliary directory with its explicit artifact flag.
6. Reject undefined references or citations, multiply defined labels, missing files, and serious overfull boxes.
7. Confirm the short main-paper boundary, the standalone supplement boundary, and the minimum full-report page count from the physical PDFs rather than source estimates.
8. Extract PDF text and scan for stale counts, prohibited current-superiority wording, and a missing population-versus-empirical distinction.
9. Render every page of every latest PDF. Inspect contact sheets and full-resolution pages for tables, equations, hash wrapping, and appendix transitions.
10. Only after all artifacts are final, emit and independently verify their SHA-256 inventory.

The checklist cannot guarantee a favorable review. Its purpose is narrower and more useful: it makes contradictions, stale evidence, invalid source boundaries, and rendering defects observable before submission. The scientific conclusion remains the one supported by the theory and current diagnostics: strict action can be justified only inside an explicit scope; abstention is the correct output when that scope does not orient the compatible outcomes.

## Part V

# Current Evidence Appendices

## A Conditional Bridge from Cell Coverage to Population Coverage

The interval proposition covers its declared target. Proposition 6.3 now states the conditional bridge explicitly. This appendix records what its additional premise requires and what it does not establish for the reported experiments.

**Additional condition for population transfer.** For cell outcomes using the same loss as the population risk, such as accuracy and 0/1 loss, suppose that, on the same probability space,

$$\begin{aligned}\mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon) &\geq 1 - \alpha_{\text{cell}}, \\ \mathbb{P}(|\Delta^{\text{cell}} - \Delta| \leq b) &\geq 1 - \delta,\end{aligned}$$

where  $b \geq 0$  is available without the new evaluation labels, then

$$\mathbb{P}(|\hat{\Delta} - \Delta| \leq \varepsilon + b) \geq 1 - \alpha_{\text{cell}} - \delta.$$

As in the main-paper proposition, the triangle inequality applies on the intersection of the two coverage events, and the union bound controls its complement; independence is not needed. A population interval-supported action would use radius  $\varepsilon + b$  and budget  $\alpha_{\text{cell}} + \delta \leq \alpha_{\text{target}}$ . No such sampling-error bound is established for the reported panels, and their gates do not apply this enlarged radius. This conditional implication therefore adds no new population-safety claim to the empirical results.

For paired accuracy on fresh IID evaluation observations, a direct fixed-pair IID concentration argument requires the model pair to be fixed relative to that evaluation sample, under a stated sampling model. Merely hiding evaluation labels while adapting on the evaluation inputs does not establish this premise. Likewise, disjoint windows by themselves do not imply IID sampling or independence across environments. The reported transductive panels are not retroactively assigned a sampling radius.

The maintained Lean namespace `KBound.PopulationTransfer` verifies the triangle, union-bound, and strict-direction implications under their declared premises. Formal verification does not verify that the empirical calibration or sampling premises hold for a dataset.

For a nonlinear score such as macro-F1, an analogous argument must first define the population functional and a suitable sampling-error bound. Neither follows from the binary disagreement identity. Appendix L gives a counterexample with perfect cell-outcome coverage and negative population benefit.

## B Additional Results and Non-Promoted Studies

This section preserves evidence outside the two primary protocols. Ordinary accuracy and balanced accuracy remain separate, and no result is pooled across datasets, candidates, or



protocols. The input inventory is bound by source-manifest SHA-256 03b1d2b1e9e5ed1cf835126f871ed83eb8497a1ac81727b3955af0d69eb89742. Historical, incomplete, and invalid studies retain their original status.

## B.1 Accuracy-Based Diagnostic Replays

**Status: retrospective diagnostic.**

Table 15 collects the remaining ordinary-accuracy replays; the dataset-specific interpretations follow below.

Table 15: Appendix ordinary-accuracy diagnostics. Each row is weighted over the evaluation unit named in the table; lower regret is better. Commitment counts ADAPT or FREEZE and is not interval coverage. The rows are retrospective or historical, not confirmatory. Office-Home’s primary and stream-seed rows differ from its invalid five-checkpoint audit. ImageNet-C is candidate-specific. For PACS,  $n = 12$  denotes equal-weight domain-seed regret units, whereas the archive contains 216 repeated action decisions.

| Protocol                 | Status                  | $n$ | KGA    | Adapt  | Freeze | FA <sub>u</sub> | Commitment rate |
|--------------------------|-------------------------|-----|--------|--------|--------|-----------------|-----------------|
| Office-Home primary      | retrospective retention | 35  | 0.0158 | 0.0468 | 0.0158 | 0.0000          | 0.3143          |
| Office-Home stream seeds | descriptive replication | 54  | 0.0215 | 0.0458 | 0.0217 | 0.0000          | 0.2778          |
| ImageNet-C Tent          | candidate-specific      | 135 | 0.0154 | 0.0191 | 0.0145 | 0.0222          | 0.0370          |
| ImageNet-C EATA          | candidate-specific      | 135 | 0.0024 | 0.0001 | 0.0342 | 0.0074          | 0.8074          |
| ImageNet-C SAR           | candidate-specific      | 135 | 0.0227 | 0.0529 | 0.0319 | 0.0000          | 0.2741          |
| PACS (aggregate)         | partial diagnostic      | 12  | 0.0431 | 0.0176 | 0.0446 | 0.0093          | 0.8148          |
| CIFAR-10.1               | locked diagnostic       | 48  | 0.0017 | 0.0190 | 0.0017 | 0.0000          | 0.4583          |

## B.2 Balanced-Accuracy Diagnostic Replays

**Status: retrospective diagnostic on a separate metric scale.**

Table 16 reports the balanced-accuracy replays on their own metric scale, with each recorded candidate and evaluation unit kept distinct.

Table 16: Appendix balanced-accuracy diagnostics; lower regret is better. Each row uses the evaluation-cell weighting recorded by its protocol. The archived contracts name balanced accuracy for Camelyon17, ImageNet-R, and RxRx1; equality with ordinary accuracy can occur only under recorded class balance. ImageNet-R averages candidate backbones rather than one deployed gate. Camelyon17 OOD was opened, its B-v2 rows are separate within-seed diagnostics, and the displayed RxRx1 row is model seed 0. No interval or confirmatory claim is attached to this table.

| Protocol             | Status               | $n$ | KGA    | Adapt  | Freeze | FA <sub>u</sub> | Commitment rate |
|----------------------|----------------------|-----|--------|--------|--------|-----------------|-----------------|
| ImageNet-R backbones | architecture panel   | 480 | 0.0137 | 0.0064 | 0.0325 | 0.0125          | 0.4437          |
| Camelyon17 OOD       | opened one-sided     | 18  | 0.0000 | 0.0000 | 0.1381 | 0.0000          | 1.0000          |
| Camelyon17 B-v2 Tent | opened diagnostic    | 108 | 0.0446 | 0.0097 | 0.0820 | 0.0000          | 0.2500          |
| Camelyon17 B-v2 EATA | opened diagnostic    | 108 | 0.0260 | 0.0040 | 0.0911 | 0.0278          | 0.4722          |
| Camelyon17 B-v2 SAR  | opened diagnostic    | 108 | 0.0289 | 0.0016 | 0.1001 | 0.0093          | 0.4815          |
| RxRx1 model seed 0   | retention diagnostic | 60  | 0.0000 | 0.2531 | 0.0000 | 0.0000          | 1.0000          |

### B.3 Large-Scale Corruption: ImageNet-C

**Status: retrospective candidate-specific diagnostic.**

ImageNet-C is candidate-dependent. At the regenerated cross-fit authority, SAR has KGA/adapt/freeze regrets 0.0227/0.0529/0.0319 and 28/9/98 A/F/U decisions. It has a point advantage over both fixed policies and no false adaptation among 135 cells, but no seed-robust inferential support. Tent has regrets 0.0154/0.0191/0.0145 and 3/2/130 decisions; all three ADAPT decisions are false adaptations. EATA has regrets 0.0024/0.0001/0.0342 and 108/1/26 decisions, including one false adaptation. These rows support candidate-level diagnosis, not a dataset-wide success label. The main paper discloses the nonstandard SAR operating point and the absence of an official-setting control.

### B.4 Office-Home

**Status: retrospective diagnostic; the five-checkpoint route is invalid.**

#### B.4.1 Valid Descriptive Records

The primary Office-Home row (Venkateswara et al., 2017) is conservative. KGA regret is 0.0158, tying always-freeze (0.0158) and below always-adapt (0.0468), with 0 adapt decisions. This is a descriptive point-estimate match that evaluates conservative retention rather than selective routing. The archived protocol does not supply a predeclared uniform stability guarantee for the full-fit versus out-of-fold benefit predictor.

The 54-cell Office-Home stream-seed replication in Table 15 has KGA regret 0.0215 versus 0.0458 and 0.0217, a small point advantage relative to each fixed policy. The three stream seeds reuse the archived source checkpoint. The descriptive seed-bootstrap interval for improvement over always-freeze includes zero; the entire point gain comes from one ADAPT decision among 54 cells. This is not an interval-supported beats-both claim.

Table 17: Historical Office-Home diagnostics with equal weight per recorded condition; lower accuracy regret is better. The rows are separate experiments. The stream-seed replay uses a fixed SAR candidate. The five-checkpoint route is invalid, so only its post-hoc fixed-SAR opportunity regret (.0027) and freeze regret (.0230) remain; its actions have no routing or interval-support meaning. A/F/U denotes ADAPT/FREEZE/ABSTAIN. No interval-supported claim is made.

| Protocol                        | $n$ | Router / Adapt / Freeze | A/F/U   |
|---------------------------------|-----|-------------------------|---------|
| Archived stream-seed replay     | 54  | .0215 / .0458 / .0217   | 1/14/39 |
| Five-checkpoint candidate audit | 15  | invalid / .0027 / .0230 | n/a     |

#### B.4.2 Invalid Five-Checkpoint Route

The post-hoc five-checkpoint Office-Home study is a candidate opportunity audit, not a valid KGA result. It uses five independently trained source checkpoints, the three target-test domains, one fixed IID/tiny condition, and Tent, EATA, and SAR. Checkpoint identity

passed, but routing and serialization did not. The identifying agreement relation is binary-only and does not apply to the 65-class task; the old spectral estimate was sign-indeterminate and unbounded. Thirteen of 15 anchors were negative, 9 of 60 stored  $\hat{b}$  values fell outside  $[0, 1]$ , and EATA and SAR produced exactly the same predictions in 10 of 15 cells. The historical leave-one-out audit has only three cells per checkpoint, whereas a 90% empirical order-statistic single-candidate radius requires at least 10 total cells. All five result files fail strict JSON and contain 57 literal **Infinity** values in total.

The archived exploratory implementation printed ABSTAIN in all 15 cells, but those outputs have no safety, routing, or interval-support meaning. The valid descriptive quantities are checkpoint accuracies and candidate opportunity: freeze is 0.5711, SAR is 0.5914, their paired difference is 0.0203 (nominal post-hoc 95%  $t$ -interval  $[0.0123, 0.0283]$ ), and the per-cell oracle is 0.5941. Oracle minus SAR is 0.0027 (nominal post-hoc 95% interval  $[-0.0024, 0.0077]$ ). The best adapted candidate improves over freeze in 13 cells, ties in one, and harms in one. Because the target was already opened and SAR was chosen after comparing the audit means, these statistics are post-hoc opportunity evidence only.

## B.5 Camelyon17

**Status: opened, one-sided retrospective diagnostic.**

The archived Camelyon17 OOD evaluation (Bandi et al., 2019) gives zero KGA and always-adapt regret, versus 0.1381 for freeze. All 18 cells are helpful. This opened panel is therefore a one-sided descriptive diagnostic, not prospective evidence of selective routing. The B-v2 within-seed diagnostics are separate. Their regenerated Tent/EATA/SAR KGA regrets are 0.0446/0.0260/0.0289, versus always-adapt regrets 0.0097/0.0040/0.0016 and always-freeze regrets 0.0820/0.0911/0.1001. None has lower regret than both fixed policies. The three run seeds do not identify independently trained checkpoints, and the target was already open. These rows remain stress diagnostics, not untouched hospital-domain evidence.

## B.6 RxRx1

**Status: retrospective retention diagnostic.**

The displayed RxRx1 row (Sypetkowski et al., 2023) is model seed 0: KGA and freeze regret are zero against 0.2531 for always-adapt, with no adapt decisions. A separate audit across three independently trained model checkpoints repeats the same all-freeze tie. It strengthens model-variation reproducibility but adds no adapt exposure. RxRx1 therefore evaluates reproducible conservative retention rather than selective routing.

## B.7 iWildCam

**Status: invalid current metric contract; no canonical numerical row.**

iWildCam (Koh et al., 2021) is not assigned a canonical numerical row. The archived scorer used sklearn macro-F1, which averages prediction-only classes, rather than the official WILDS metric, which averages labels present in the target truth. The archived runtime, sample identities, and target population were not sealed under the current contract, so its diagnostic scores and routing actions remain audit-only. A pinned rerun with the official

metric, sealed sample identifiers, and a population hash is required before promotion. The older binary dominance flag and the later replay under the same invalid metric contract remain hash-locked only as superseded audit history.

## B.8 Boundary and Transfer Diagnostics

**Status:** retrospective transfer diagnostics.

### B.8.1 PACS

PACS (Li et al., 2017) KGA regret is 0.0431, compared with 0.0176 for always-adapt. The regret summary has 12 equal-weight domain-seed units. Its 216 action records are repeated decisions within those units, not 216 independent regret observations. The archive lacks the residual detail needed to replay the current gate, so PACS remains an aggregate performance diagnostic.

### B.8.2 ImageNet-R Backbone Breakdown

ImageNet-R (Hendrycks et al., 2021) KGA regret is 0.0137, compared with 0.0064 for always-adapt and 0.0325 for always-freeze, across the ten-backbone architecture panel. No backbone has lower KGA point regret than each fixed comparator. KGA regret is higher than always-adapt regret on 8 of 10 backbones; four of those eight have a 0% harmful base rate, so routing has no harm to avoid. On `efficientnet_b0` (100% harmful) and `swin_t` (60.4% harmful), KGA improves over always-adapt but exactly matches always-freeze and makes no adapt decisions. The pattern is consistent with the regime map: routing can add value only where candidate harm is present, while these two rows evaluate conservative retention rather than selective routing.

### B.8.3 CIFAR-10.1

CIFAR-10.1 (Recht et al., 2018) matches always-freeze at 0.0017 and is retained as a low-margin cross-seed transfer diagnostic. This pattern alone does not prove structural non-identifiability; weak evidence, low margin, estimator inadequacy, or calibration transfer can produce the same behavior.

## B.9 Historical Protocol-Matched POEM and AETTA Ports

**Status:** historical local-port audit; not a synchronized official comparison. The repository retains one historical head-to-head artifact on the CIFAR-10-C Tent stream. Its KGA, POEM-style, and AETTA-style regrets are 0.00157361, 0.00880463, and 0.00732986; the associated always-adapt and always-freeze regrets are 0.00792338 and 0.12409792. Historical KGA-minus-port paired differences and unadjusted 95% paired-bootstrap intervals are

| Port        | Difference  | 95% interval                 |
|-------------|-------------|------------------------------|
| POEM-style  | -0.00723102 | $[-0.00878038, -0.00570806]$ |
| AETTA-style | -0.00575625 | $[-0.00714260, -0.00439859]$ |

Holm adjustment applies only to the archived  $p$ -values; the adjusted value for each of these two comparisons is 0.00039996.

Those estimates are protocol-matched diagnostics from local ports, not results from untouched official implementations. They also use an earlier KGA policy and are not synchronized with the current three-way cross-fit. The numerical ordering is therefore historical audit evidence only: it does not establish current-policy or official superiority. The source artifact is `experiments/kbound/results/mixed_headtohead_v1/HEADTOHEAD_RESULTS_cifar10c_tent_primary.json`; its stored WIN and `kga_beats` labels are withdrawn from the current claim set. A defensible new comparison would replay all methods on identical current-policy records and freeze the multiplicity family before scoring.

## B.10 So2Sat Development and Action-Unit Audit

**Status: development stop before target access.**

The locked So2Sat-LCZ42 v1 study screened Tent and SAR in a development-only, multi-city, multi-checkpoint candidate-feasibility stage. It was not a target evaluation. Tent showed both helpful and harmful development cases, but neither candidate passed every predeclared eligibility check. The locked status is therefore *no feasible candidate; stop before gate calibration*. The detailed development authority and its binding receipt remain outside the default public build closure; the public release records only the protocol shape, stopping decision, and access boundary.

No held-out gate-calibration rows or target inputs, pixels, or labels were accessed. The study therefore contains no target natural-shift score. The gate-fit outcomes are open, so retuning on them cannot create fresh confirmatory evidence. V1 also declares one target action per city while its target runner creates city-checkpoint actions. This mismatch did not affect the stop because no target action was created. Prospective v2 resolves the action unit as city-checkpoint and supplies a source-complete runner. It remains a no-go: no v2 pre-calibration seal or target authorization exists, and gate calibration and target evaluation remain unopened. V2 does not retroactively change the negative v1 record.

## B.11 DomainNet Painting Development and Fixed Diagnostics

These additional results are descriptive on already-opened painting development. The five-checkpoint cohort uses 24 shared check cells per checkpoint: its 120 seed-cell records are not 120 independent environments. A separate reference-source ResNet-50 checkpoint is used by two different recipes. The local entropy probe updates BatchNorm affine parameters using one unlabeled pass, Adam at  $10^{-3}$ , and batches of at most 32; it restores source BatchNorm running buffers before evaluation. It is neither online Tent nor AdaContrast. The AdaContrast-derived single-process bridge uses batch size 16, 15 epochs, square-resize preprocessing and direct-logit inference; it is not an exact reproduction of the published pipeline. Each reference-source check panel contains 24 cells and 3,073 image evaluations. Their frozen/candidate correct totals are 1387/1366 for R50 entropy and 1384/1131 for the bridge. The differing frozen totals mean this is not a matched head-to-head comparison.

Table 18: Opened DomainNet painting development, kept separate from the canonical primary panel. Regret and radii are in percentage points. Repeated seed/cell records and the fixed diagnostic cells do not provide independent prospective trials.

| Saved evaluation           | Units and measured effects                             | Gate or diagnostic result                                                                                                                |
|----------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Five-checkpoint recipe     | 120 seed-cell records; 12 helpful, 15 harmful, 93 tied | 120 ABSTAIN; equal-seed regret 0.1106770833 for gate/freeze, 0.1235465116 for adapt                                                      |
| R50 local entropy          | 24 check cells; 7 helpful, 14 harmful, 3 tied          | 24 ABSTAIN; radius 2.6099386731; inclusion 21/24 (87.5%); regret 0.390625 gate/freeze, 1.07421875 adapt                                  |
| AdaContrast-derived bridge | 24 check cells; 1 helpful, 23 harmful, 0 tied          | 24 ABSTAIN; radius 10.8115435382; inclusion 24/24; regret 0.0651041667 gate/freeze, 8.2995195413 adapt                                   |
| Fixed queue diagnostic     | First 4 DEV_fit cells; 512 paired image evaluations    | Correct totals: capacity 16384, 187; capacity 64, 181; freeze, 243. Smaller queue loses 6/512 (1.171875 points); 3 cells worse, 1 better |
| Fixed BN decomposition     | Same 4 DEV_fit cells and 512 evaluations               | Correct totals: freeze 243; stats-only 107; other-state-only 133; actual adaptation 187. Every altered condition harms all 4 cells       |

All three check evaluations have zero ADAPT and zero FREEZE exposure. Their conditional false-ADAPT and false-FREEZE rates are undefined; zero false commitments under universal abstention supply no directional exposure. The inclusion rates are descriptive, not validation of nominal coverage on unseen groups. These development results do not establish a selective-routing gain over freeze.

The queue intervention changed capacity, negative-pool size, memory age, and the inherited class-aware masking interaction; it does not isolate random initialization. The BN experiment crossed original/adapted running buffers with original/adapted remaining state after identical baseline training; other state includes BN affine parameters. Its accuracy-scale interaction is +37.109375 percentage points. These non-additive effects do not identify a BN-only cause of harm; restoring source statistics alone does not rescue the adapted model. Neither follow-up selects a new adapter, radius, or decision rule. Any retrospective radius or predictor checks remain exploratory.

## B.12 Office-Home mechanism check

**Office-Home: archival selection and changed-recipe smoke.** On the opened 54-cell stream-seed archive, the post-hoc rule “serve SAR iff its recorded affine update norm equals zero exactly” selects 35 candidates and freezes 19 cells. Its accuracy regret is 0.00035613, versus 0.04579772 for always-adapt and 0.02172365 for always-freeze. This rule is not the calibrated KGA gate, and choosing it after inspecting the archive does not constitute a new held-out result. A zero affine norm also does not imply frozen predictions: candidate inference can replace source BatchNorm statistics with current-batch statistics.

A separate MPS execution smoke used one class-balanced validation episode in each of Art, Clipart and Product: 65 evaluation images and 16 disjoint adaptation images per

domain. All three arms used the same 65-image evaluation batch. They were frozen source statistics, batch statistics without parameter updates, and the local SAR helper with one pass over two batches of eight, learning rate  $10^{-3}$ ,  $\rho = 0.05$ , entropy threshold  $0.4 \log(65)$  and reset threshold 0.2. This deliberately smaller recipe differs from the archived aggressive SAR recipe. The checkpoint is identified by observed local bytes and metadata; its published training recipe was not authenticated.

Table 19: Three-arm Office-Home execution smoke: correct predictions out of 65 per domain. These are three development episodes, not independent confirmatory trials.

| Domain  | Frozen | Batch stats | SAR | SAR steps |
|---------|--------|-------------|-----|-----------|
| Art     | 37     | 34          | 34  | 0         |
| Clipart | 26     | 30          | 30  | 0         |
| Product | 41     | 43          | 43  | 1         |

Batch-statistics-only and SAR predictions agree on all 195 evaluation images, including Product, where one optimizer step was accepted and the affine update norm was nonzero. Their totals are both 107/195, versus 104/195 for frozen inference. The archival exact-zero rule applied descriptively to this smoke yields 105/195, below always-SAR. The smoke therefore does not reproduce the archival selector’s advantage. Source tensors and buffers were preserved, and all prediction receipts were written before scoring. No KGA predictor or radius was fitted, calibrated or evaluated. Passing the smoke establishes execution of the three arms and one SAR update branch; it does not establish statistical efficacy or the cause of historical gains.

The runner-recorded interval was 20.43 seconds, beginning after argument parsing and output-directory creation and excluding process startup and module imports. It includes subsequent setup and integrity work as well as execution of all three arms. Per-arm latency, tail latency, sustained throughput and peak memory were not measured; the saved before/after allocator readings are snapshots, not a peak-memory estimate.

### B.13 Withdrawn Development Proxy for the Population Budget

**Status: negative retrospective diagnostic; estimation route withdrawn.**

Without additional structural information, label-free evidence cannot uniformly validate a residual bound below the unresolved radius of its compatible fibre. This does not exclude a valid constant bound at or above that radius, and  $\beta = 0$  is not a safe default. The tested source-like development proxy was  $1.4\text{--}50\times$  too small on real evaluation cells: 24–73% fell outside  $\mathcal{C}_\beta$ , commit error was 0.4–16.6%, and five of ten configurations returned zero commitments on all 405 ImageNet-C cells. The proxy did not observe the theorem’s disagreement-conditional residual, and its named raw artifact is not regenerable from the canonical bundle. We therefore withdraw estimation and require domain knowledge, a declared coupling, or historical labeled deployments under a transfer assumption.

The population layer targets  $\Delta$ , whereas empirical calibration targets  $\Delta^{\text{cell}}$ . Appendix A states the additional sampling argument needed to bridge them. Neither layer establishes risk alignment, estimator transfer, or residual coverage in a new domain.

## C Compact Assumption and Claim Ledger

Table 20 separates formal claims from empirical findings and records the conditions and scope limits for each.

Table 20: Formal half of the claim-to-support ledger. Each row separates its target and assumptions from what is and is not supported.

| Statement and target                         | Required assumption or protocol                                                                                               | Supported conclusion                                    | Not supported                                           |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|
| Disagreement reduction; population $\Delta$  | binary labels, 0/1 loss, fixed pair, $\mu_T(D) > 0$                                                                           | $\Delta = 2\mu_T(D)(M + \gamma)$                        | calibration transfer or observability of $\gamma$       |
| Interior ambiguity; population $\Delta$      | Thm. 4.7; $\beta > 0$ , $ M  < \beta$ , fibre richness                                                                        | evidence-identical laws with opposite nonzero benefit   | ambiguity outside the declared class                    |
| Closed band; population $\Delta$             | Prop. 4.9; strict semantics and richness                                                                                      | no uniformly sound strict action on $ M  \leq \beta$    | unequal task risk at every boundary law                 |
| Strict frontier; population $\Delta$         | Thm. 4.10; fixed evidence and declared $\mathcal{C}_\beta$                                                                    | pointwise maximal sound three-action rule               | learning $\beta$ from the unlabeled batch               |
| Audit floor; residual radius                 | nonempty fibre, bounded attainable residuals, measurable audit, common joint evidence/randomization law, and uniform validity | $\hat{\beta} \geq \Gamma_z$ with the stated probability | a finite-batch estimator or data-learned $\beta$        |
| Coverage-to-action; declared cell target $B$ | Prop. 6.1 and Eq. (9)                                                                                                         | marginal false ADAPT and false FREEZE control for $B$   | population-risk, conditional, group, or anytime control |



Table 21: Empirical and future-evidence half of the claim-to-support ledger. Evaluation units and statuses remain protocol-specific; no row upgrades retrospective evidence.

| Statement and target                     | Required assumption or protocol                                    | Supported conclusion                                                                   | Not supported                                                                  |
|------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| CIFAR-10-C; measured cells               | opened one-checkpoint grid and outcome-disjoint cross-fit          | candidate-specific actions, errors, and point regret; Tent nominal intervals favorable | prospective confirmation; six-contrast Holm rejection                          |
| CCT-20; measured cells                   | locked 45-cell protocol and declared utility endpoint              | all-frozen utility preservation with 0/44/1 actions                                    | selective routing, improvement over always-freeze, or population safety        |
| Other natural replays; measured cells    | Tables 15–16; fixed splits and candidates                          | protocol-specific retention or adverse behavior                                        | one pooled success claim                                                       |
| So2Sat; development eligibility          | Sec. 8.4; locked feasibility rules                                 | no candidate passed the pre-target screen                                              | gate calibration or a target score                                             |
| DomainNet; opened development            | Sec. B.11; shared cells, separate candidates and fixed diagnostics | universal abstention and adverse diagnostic outcomes                                   | independent confirmation, pooled-trial inference, or selective-routing success |
| ImageNet-C; measured cells               | Table 15; candidates and operating points kept separate            | candidate-specific diagnostic evidence                                                 | dataset-wide or published-default conclusion                                   |
| Other replay scope; recorded units       | Tables 15–16                                                       | observed protocol behavior                                                             | structural unknowability or universal transfer                                 |
| Prospective confirmation; future targets | Sec. K; diverse unopened environments                              | a frozen evaluation design                                                             | a current empirical result                                                     |
| Physical camera; future sessions         | locked session split and held-out recordings                       | protocol infrastructure only                                                           | a completed camera result                                                      |

## D Calibration and Evaluation Contracts

### D.1 Controlled-Grid Contract

The CIFAR-10-C grid is an opened, dependent, constructed six-family diagnostic. Its fitting unit is the condition cell. Stable identifiers assign deterministic score folds. For each fold, its complement is divided into disjoint estimator-fit and residual-calibration subsets. The fitted predictor and exact-rank radius are then applied only to the score fold. Consequently, the scored outcome  $B_i = \Delta_i^{\text{cell}}$  enters neither the fit nor the calibration set that determines its own prediction, radius, or action. This is a three-way cell-outcome-disjoint cross-fit, not leave-one-cell-out calibration.

The CIFAR-10-C grid uses one ResNet-18 checkpoint and five RNG stream/run seeds. Within each seed it crosses six corruption families, severities 1 and 5, batch sizes 200/16/8, IID/imbalanced/single-class composition, two update schedules (10 steps at  $10^{-3}$  and 50 steps at  $2.5 \times 10^{-3}$ ), and two repeats: 432 cells per seed and 2,160 per candidate. Point summaries weight cells equally. The family analysis first averages seeds and cells within a family and then weights the six family means equally. These are opened, dependent, constructed cells

coupled through one corpus and checkpoint; the cross-fit does not make them exchangeable or validate coverage on unseen environments.

Tent, EATA, and SAR are local protocol-matched ports. Candidate preview runs in training mode with BatchNorm running statistics disabled, so the measured delta combines evaluation-batch statistic replacement with gradient updates to normalization affine parameters. The two recorded update schedules are part of the grid. They are not untouched reproductions of each method’s published default operating point; the main text gives the additional SAR-specific departure.

## D.2 Natural-Shift Contract

Development conditions fit the benefit model; a disjoint calibration partition supplies residuals; choices are frozen; and the target partition is scored once where the archived protocol records all three stages. When the saved evidence does not support that complete chain, the row is marked descriptive or diagnostic. Replaying a table does not strengthen its original design.

The candidate TTA step itself is transductive. Episodic candidates update on each unlabeled evaluation batch before predicting that batch. Online candidates update on a separate auxiliary stream, but prediction still uses evaluation-batch BatchNorm statistics. Thus “disjoint” above refers to development, residual-calibration, and target partitions (and to the recorded auxiliary stream versus evaluation sample identities); it does not mean that candidate inference is inductive. No target labels enter adaptation, prediction, evidence, or the action.

## D.3 Action Contract

Every action is computed from  $Z$ ,  $\hat{\Delta}$ , and  $\varepsilon$  without target labels. Offline labels determine  $\Delta^{\text{cell}}$ , the measured oracle action, empirical regret, and false-commit indicators only after the action is fixed. Existing artifact fields named `delta` store this cell outcome; their schema and numerical values are unchanged. Abstention resolves to the frozen prediction for measured task-performance accounting, while its log retains the distinction from an interval-supported FREEZE decision.

The whole-pipeline invariant is tested by perturbing one scored  $B_i$  and rerunning all fits. Its  $\hat{\Delta}_i$ ,  $\varepsilon_i$ , and action must remain unchanged. The maintained test `tests/test_controlled_grid_crossfit.py` checks this for the implementation and also exhibits the old radius leak as a negative control. This is a software non-leakage result for the scored cell; it does not prove exchangeability, estimator validity, or future-domain coverage.

## D.4 Configuration Contract

A configuration identity binds the checkpoint, adapter and update settings, evidence schema, benefit estimator, residual construction,  $\alpha$ , and action unit. Section H records that identity once. Any mismatch requires a declared transfer argument or recalibration; otherwise the controller does not commit.

## E Decision Exposure and Repetition Audit

Point regret alone can conceal whether a conservative match exercised the ADAPT branch. Table 22 therefore reconstructs action counts from the same canonical panel used in the main-paper primary CIFAR table 5 and Tables 15 and 16. The three additional DomainNet rows use the separate saved development summaries in Sec. B.11; they do not change that canonical panel or form a pooled set of independent trials. A/F/U denote adapt, freeze, and abstain. PACS is omitted from the action table because the released aggregate cross-checks regret but does not contain the per-cell benefit estimates and residuals required for a complete gate replay. We omit iWildCam because its archived actions were scored under the wrong macro-F1 contract.

Table 22: Action exposure in replayable rows. A/F/U denotes ADAPT/FREEZE/ABSTAIN. Counts are unweighted evaluation cells, not independent experiments; regret tables use the weighting declared by each protocol. False A/A and false F/F use the corresponding action denominator. A dash means that the direction was unexposed or was not reconstructed by the maintained authority; it does not mean zero error. Controlled-grid rows use the current three-way cross-fit; the remaining rows retain their stated diagnostic protocol. The canonical numerical panel is bound to SHA-256 35d4c165843de1ece3cb35ffb4ac50dbfcbfa33b646754e2b6d133fbdfa78c6e; its input inventory is bound separately by the source manifest. DomainNet rows have separate development authorities and are not part of that canonical panel.

| Row                           | $n$  | A    | F   | U   | false A/A | false F/F | repetition / scope                                 |
|-------------------------------|------|------|-----|-----|-----------|-----------|----------------------------------------------------|
| Office–Home primary           | 35   | 0    | 11  | 24  | –         | –         | 2 test-stream seeds; ties freeze                   |
| Office–Home stream-seed repl. | 54   | 1    | 14  | 39  | 0/1       | –         | 3 test-stream seeds; point-only edge               |
| ImageNet–C SAR                | 135  | 28   | 9   | 98  | 0/28      | –         | 5 seeds $\times$ 27 cells                          |
| ImageNet–C Tent               | 135  | 3    | 2   | 130 | 3/3       | –         | 5 seeds $\times$ 27 cells                          |
| ImageNet–C EATA               | 135  | 108  | 1   | 26  | 1/108     | –         | 5 seeds $\times$ 27 cells                          |
| ImageNet–R                    | 480  | 179  | 34  | 267 | 6/179     | –         | 4 seeds $\times$ 10 backbones $\times$ 12 cells    |
| CIFAR–10–C Tent               | 2160 | 1091 | 337 | 732 | 2/1091    | 1/337     | 5 seeds $\times$ 432 cells                         |
| CIFAR–10–C EATA               | 2160 | 1207 | 105 | 848 | 0/1207    | 0/105     | 5 seeds $\times$ 432 cells                         |
| CIFAR–10–C SAR                | 2160 | 1414 | 0   | 746 | 0/1414    | –         | no FREEZE exposure                                 |
| CCT–20                        | 45   | 0    | 44  | 1   | –         | 1/44      | 5 checkpoints $\times$ 9 locations                 |
| DomainNet five-checkpoint     | 120  | 0    | 0   | 120 | –         | –         | 5 checkpoints $\times$ 24 shared development cells |
| DomainNet R50 entropy         | 24   | 0    | 0   | 24  | –         | –         | separate opened development evaluation             |
| DomainNet AdaContrast-derived | 24   | 0    | 0   | 24  | –         | –         | separate opened development evaluation             |
| Camelyon17 OOD                | 18   | 18   | 0   | 0   | 0/18      | –         | opened helpful-only OOD subset                     |
| Camelyon17 B–v2 Tent          | 108  | 27   | 0   | 81  | 0/27      | –         | 3 within-seed grids; diagnostic                    |
| Camelyon17 B–v2 EATA          | 108  | 51   | 0   | 57  | 3/51      | –         | 3 within-seed grids; diagnostic                    |
| Camelyon17 B–v2 SAR           | 108  | 52   | 0   | 56  | 1/52      | –         | 3 within-seed grids; diagnostic                    |
| RxRx1 J                       | 60   | 0    | 60  | 0   | –         | –         | model seed 0; 3-checkpoint audit also all-freeze   |
| CIFAR–10.1 K                  | 48   | 0    | 22  | 26  | –         | –         | locked cross-seed replay                           |

The largest adapt exposure occurs on CIFAR-10-C, making it the main diagnostic of the adapt branch. Tent records two false adaptations among 1,091 ADAPT decisions and one false FREEZE among 337 FREEZE decisions; EATA and SAR record no false adaptations,

although SAR never exposes FREEZE. The CCT-20 row supplies the complementary all-frozen case: one false FREEZE among 44 FREEZE decisions and no ADAPT exposure. Primary Office-Home and RxRx1 have zero adapt exposure, while the 54-cell Office-Home replication has one adapt decision. Those rows chiefly test conservative retention. The invalid five-checkpoint Office-Home route and the invalid-metric iWildCam archive are excluded. Camelyon17 OOD has the opposite limitation: every archived cell is helpful and every action adapts. On ImageNet-C, Tent makes three ADAPT decisions and all are false adaptations, whereas EATA makes 108 ADAPT decisions with one false adaptation and SAR makes 28 with none. Similar regret values can therefore hide different operating behavior.

The repetition column also prevents a common inflation of evidence. Five seeds of one checkpoint and grid protocol are not five independently designed studies. Ten ImageNet-R backbones are a panel of candidate policies, not ten target domains for one frozen gate. Three Camelyon17 within-seed grids permit a stability diagnostic but do not create an untouched hospital-domain replication. RxRx1 is stronger on model variation because the source checkpoint is independently trained three times, but its all-freeze behavior still provides no adapt exposure.

## F Inference Discipline

### F.1 Point Estimates, Paired Gaps, and Clusters

For a baseline  $b \in \{\text{always-adapt}, \text{always-freeze}\}$ , the paired routing gap at condition  $j$  is

$$G_{bj} = \text{regret}_{b,j} - \text{regret}_{\text{KGA},j}.$$

A positive mean gap means lower regret for KGA. All paired routing-gap intervals reported in this paper use this baseline-minus-KGA sign convention. The point-estimate two-comparator criterion requires both means to be positive. Interval support additionally requires the lower confidence limit for both paired means to exceed zero at the declared resampling unit. These definitions are intentionally separate: Table 15 can establish a point ordering, but not an inference claim without the associated unit, interval artifact, and justified coverage assumptions. A positive computed lower endpoint is numerical interval support under the resampling model, not verified coverage. Multiplicity adjustments do not repair invalid marginal intervals or tests.

Condition-level resampling treats cells as exchangeable and can understate uncertainty when severities and compositions share a corruption family. Corruption-family resampling preserves observed within-family dependence, but few clusters limit reliability; conservatism is not guaranteed. Seed resampling measures variation across repeated training or test streams only when the seeds represent the intended deployment randomness. The compact paper therefore does not use one generic “bootstrap CI” label. An inference claim must state whether the unit is cell, corruption-severity pair, corruption family, seed, backbone, domain, or physical session.

For a vector of cluster gaps  $G = (G_1, \dots, G_m)$ , the exact sign-flip reference calculation uses the joint null condition

$$G \stackrel{d}{=} s \odot G \quad \text{for every } s \in \{-1, +1\}^m. \quad (67)$$

Independent zero-symmetric cluster gaps are sufficient. Exchangeability, marginal symmetry, or one global sign symmetry alone is not sufficient. Enumeration removes Monte Carlo error but does not establish this null condition.

## F.2 Current-Policy Corruption-Family Sensitivity

The retrospective family contains all six candidate-by-baseline contrasts. The replay policy, corruption-family unit, sign-flip reference test, and Holm adjustment were selected after the underlying CIFAR outcomes were open. The current-policy artifact averages run seeds and condition cells within each of the six families. Tent’s nominal intervals favor KGA against always-adapt ( $[0.002087, 0.009537]$ ) and always-freeze ( $[0.062822, 0.177573]$ ). EATA’s adapt-side interval crosses zero ( $[-0.001348, 0.004382]$ ), although its freeze-side interval is positive. SAR is worse than always-adapt. These nominal bootstrap intervals are distinct from the retrospective sign-flip/Holm analysis. The within-Tent two-contrast Holm calculation is itself post hoc and is not the six-test family reported in Table 23.

Table 23: Retrospective sensitivity over six opened CIFAR-10-C corruption families. Each candidate uses 2,160 cells from one checkpoint; seeds and conditions are averaged within family, and the six family means receive equal weight. Entries are baseline regret minus KGA regret with nominal, unadjusted 95% family-bootstrap intervals; positive values favor KGA. Holm values adjust the six retrospective sign-flip reference tests. For Tent, the post-hoc within-candidate Holm values are 0.046875 (adapt) and 0.03125 (freeze), while the six-contrast values are 0.140625 and 0.09375. The replay, unit, tests, and adjustment were not preregistered. The table supplies no verified coverage, independent-checkpoint inference, prospective confirmation, natural-shift evidence, or official POEM/AETTA comparison.

| Candidate | Adapt gap [95% CI]            | Freeze gap [95% CI]        | Six-way Holm $p$ (A/F) |
|-----------|-------------------------------|----------------------------|------------------------|
| Tent      | 0.00606 [0.00209, 0.00954]    | 0.12202 [0.06282, 0.17757] | 0.14062/0.09375        |
| EATA      | 0.00164 [-0.00135, 0.00438]   | 0.12970 [0.06893, 0.18746] | 0.31250/0.09375        |
| SAR       | -0.00145 [-0.00294, -0.00027] | 0.13869 [0.07972, 0.19683] | 0.90625/0.09375        |

## F.3 Retrospective Interval Inclusion by Family

This audit summarizes the stored per-cell authority produced by the three-way cross-fit. For every scored cell, its measured outcome was absent from both estimator fitting and residual calibration. The audit performs no tuning, new target access, or training; it verifies the stored predictions, radii, actions, and input/code/output digests.

The observed inclusion counts are 1,976/2,160 for Tent, 1,950/2,160 for EATA, and 1,949/2,160 for SAR. Tent also has one false FREEZE among 337 FREEZE decisions. These are in-sample cross-fit diagnostics on opened, dependent cells, not independent estimates for new environments. Family rows localize misses and widths but do not supply group-conditional guarantees. The nominal 90% setting is therefore not a validated coverage promise.

Table 24: Overall retrospective interval diagnostics for 2,160 equally weighted cells per CIFAR-10-C candidate. Inclusion concerns measured cell benefit, not population benefit. Marginal directional errors divide by all cells; conditional errors divide by the corresponding action count and are undefined when that action is absent.

| Candidate | $n$  | Nominal | Inclusion | Mean width | Median width | Commitment | $FA_u/FA_c$   | $FF_u/FF_c$   |
|-----------|------|---------|-----------|------------|--------------|------------|---------------|---------------|
| TENT      | 2160 | 0.9000  | 0.9148    | 0.0478     | 0.0457       | 0.6611     | 0.0009/0.0018 | 0.0005/0.0030 |
| EATA      | 2160 | 0.9000  | 0.9028    | 0.0397     | 0.0389       | 0.6074     | 0.0000/0.0000 | 0.0000/0.0000 |
| SAR       | 2160 | 0.9000  | 0.9023    | 0.0286     | 0.0290       | 0.6546     | 0.0000/0.0000 | 0.0000/-      |

Table 25: Current-policy interval and decision diagnostics by corruption family. Each row aggregates equal-weight opened cells after the three-way cross-fit; the scored cell outcome enters neither its fit nor calibration set. Width is  $2\epsilon$ , and A/F/U means ADAPT/FREEZE/ABSTAIN. Conditional errors use the corresponding action count as denominator and are undefined when that action is absent. The table is retrospective and is not held-out group-coverage validation.

| Candidate | Family           | $n$ | Inclusion | Mean width | Commitment | False A/A; $FA_c$ | False F/F; $FF_c$ |
|-----------|------------------|-----|-----------|------------|------------|-------------------|-------------------|
| TENT      | contrast         | 360 | 0.8361    | 0.0478     | 0.6917     | 0/180; 0.0000     | 0/69; 0.0000      |
| TENT      | defocus_blur     | 360 | 0.9417    | 0.0480     | 0.7083     | 0/180; 0.0000     | 0/75; 0.0000      |
| TENT      | fog              | 360 | 0.9472    | 0.0480     | 0.6917     | 0/180; 0.0000     | 0/69; 0.0000      |
| TENT      | gaussian_noise   | 360 | 0.8667    | 0.0480     | 0.8556     | 0/305; 0.0000     | 1/3; 0.3333       |
| TENT      | jpeg_compression | 360 | 0.9694    | 0.0478     | 0.3472     | 2/66; 0.0303      | 0/59; 0.0000      |
| TENT      | pixelate         | 360 | 0.9278    | 0.0475     | 0.6722     | 0/180; 0.0000     | 0/62; 0.0000      |
| EATA      | contrast         | 360 | 0.8417    | 0.0396     | 0.5694     | 0/180; 0.0000     | 0/25; 0.0000      |
| EATA      | defocus_blur     | 360 | 0.9278    | 0.0398     | 0.5694     | 0/180; 0.0000     | 0/25; 0.0000      |
| EATA      | fog              | 360 | 0.9139    | 0.0399     | 0.5750     | 0/180; 0.0000     | 0/27; 0.0000      |
| EATA      | gaussian_noise   | 360 | 0.8528    | 0.0399     | 0.9722     | 0/350; 0.0000     | 0/0; -            |
| EATA      | jpeg_compression | 360 | 0.9694    | 0.0396     | 0.4056     | 0/137; 0.0000     | 0/9; 0.0000       |
| EATA      | pixelate         | 360 | 0.9111    | 0.0393     | 0.5528     | 0/180; 0.0000     | 0/19; 0.0000      |
| SAR       | contrast         | 360 | 0.8694    | 0.0287     | 0.5000     | 0/180; 0.0000     | 0/0; -            |
| SAR       | defocus_blur     | 360 | 0.9306    | 0.0287     | 0.5000     | 0/180; 0.0000     | 0/0; -            |
| SAR       | fog              | 360 | 0.8944    | 0.0283     | 0.5000     | 0/180; 0.0000     | 0/0; -            |
| SAR       | gaussian_noise   | 360 | 0.8417    | 0.0286     | 1.0000     | 0/360; 0.0000     | 0/0; -            |
| SAR       | jpeg_compression | 360 | 0.9667    | 0.0283     | 0.8944     | 0/322; 0.0000     | 0/0; -            |
| SAR       | pixelate         | 360 | 0.9111    | 0.0288     | 0.5333     | 0/192; 0.0000     | 0/0; -            |

## F.4 False-Adapt and False-Freeze Accounting

The event in Eq. (4) is unconditional and concerns measured cell benefit. Its reported frequency divides ADAPT decisions with  $\Delta_j^{\text{cell}} \leq 0$  by all evaluated units, including freezes and abstentions. This need not estimate the population-harm event  $\{g = \text{ADAPT}, \Delta \leq 0\}$ . Conditional false-adapt divides by adapt decisions and is undefined when no adaptation occurs. A zero observed count should be accompanied by the number of adapt decisions; otherwise a controller that never adapts appears perfect. Table 22 exposes that denominator.

False FREEZE is the symmetric event: the policy chooses FREEZE when measured cell benefit is nonnegative. Its marginal rate divides by all evaluated units, whereas its conditional rate divides by FREEZE exposure. The latter is undefined when the policy never freezes, as in the current SAR grid. Every directional-error statement therefore appears with its action denominator.

The three-way split removes the direct path from  $B_i$  to its own prediction, radius, and

action. It does not make the overlapping fitted models or the constructed cells independent. Pooled cross-fit inclusion therefore remains a descriptive reuse of the opened panel and supplies neither exchangeability nor a finite-sample guarantee for the next corruption family. A genuinely held-out family or environment remains a different protocol.

## F.5 Multiplicity and Search History

The repository contains many exploratory configurations. One superseded census reported 1,387 determinations and 326 positives (23.5%); a broader archived scan reported 1,427 determinations and 345 positives (24.2%). Neither census froze one family definition, file list, and commit, so neither is the release multiplicity family. The reconciled panel is also post hoc, although some component protocols were individually locked. Confirmatory work must declare every candidate-track test, calibration and target unit, seed structure, inference unit, and two-comparator criterion before labels are opened, while retaining failures and incomplete runs.

## G Track-Level Provenance and CCT Ledger

Separate archival records preserve two additional scope checks outside the canonical panel. FMoW Protocol L reached held-out scoring and matched always-freeze, while always-adapt had lower regret; its status remains not-cleared. PovertyMap stopped at its preregistered development screen because the harm-detection AUC was 0.637, below the 0.65 threshold, so held-out validation and test scoring did not run. These records remain in the claim inventory and are not pooled with the canonical rows.

The provenance table is part of the result, not administrative metadata. A reviewer can trace a canonical row to compact records and their original archive hashes. If the original large artifact is not shipped, the manifest still records its byte count and digest. This guards against silent replacement of a result while allowing the release to omit raw datasets and oversized transient checkpoints.

Table 26: Release-status inventory, not a performance comparison. “Replayable” means stored quantities suffice to recompute the displayed gate output under the named scorer; it does not imply fresh data, independent units, or confirmatory status. Units and weighting remain protocol-specific.

| Track           | Released status                                                      | Present evidence                                                                                                   | Remaining limitation                                                                                                                                                                                              |
|-----------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CIFAR-10-C      | replayable diagnostic                                                | per-cell $\Delta^{\text{cell}}$ , $\hat{\Delta}$ , radius, action; 5 RNG seeds; 3 candidates                       | opened, dependent, constructed grid from one checkpoint; no untouched family                                                                                                                                      |
| ImageNet-C      | replayable diagnostic                                                | per-condition records for 5 seeds and Tent/EATA/SAR                                                                | opened 27-cell controlled configuration; candidate-specific, not one policy                                                                                                                                       |
| Office-Home     | replayable                                                           | canonical primary and 54-cell stream-seed records; later five-checkpoint audit reported separately                 | full-fit-versus-OOF transfer-stability premise not predeclared                                                                                                                                                    |
| iWildCam        | invalid archived diagnostic                                          | opened test-split records scored with the wrong macro-F1 definition                                                | no completed current-protocol or official-metric result; population and overlap were not sealed                                                                                                                   |
| Camelyon17      | replayable diagnostic                                                | archived opened OOD row plus separate B-v2 diagnostics                                                             | OOD row is one-sided and non-prospective; B-v2 is not untouched-domain evidence                                                                                                                                   |
| RxRx1           | replayable                                                           | displayed model-seed-0 row plus three model-checkpoint records and source hashes                                   | all three checkpoints freeze throughout; no helpful test conditions                                                                                                                                               |
| CCT-20          | release-complete; cell-outcome-unopened, internally sealed execution | 5 independent checkpoints, 9 target locations, actions, predictions, score, manifest, and receipts                 | aggregate target-label metadata had previously been inspected; not globally label-unopened or publicly preregistered; only two fit locations; no ADAPT exposure; trans-target extrapolation remains an assumption |
| So2Sat-LCZ42    | sealed development stop                                              | Tent and SAR candidate bundles; multi-city, multi-checkpoint gate-fit screen; no gate-calibration or target access | no feasible candidate; gate-fit outcomes are open; no gate calibration or target result                                                                                                                           |
| PACS            | partial                                                              | three-seed aggregate and source cross-checks                                                                       | missing per-cell $\hat{\Delta}$ and residuals prevent full gate replay                                                                                                                                            |
| ImageNet-R      | replayable                                                           | 4 seeds, 10 candidate backbones, 12 conditions each                                                                | architecture aggregate is diagnostic, not one deployable router                                                                                                                                                   |
| CIFAR-10.1      | replayable                                                           | locked cross-seed condition records                                                                                | low-margin conservative match; selective routing not exercised                                                                                                                                                    |
| Physical camera | protocol only                                                        | dashboard, logging schema, and planned session split                                                               | no completed fresh recording, held-out session, or result table                                                                                                                                                   |

## G.1 CCT-20 Inference, Location, and Release Ledger

**Provenance ledger.** This was a cell-outcome-unopened, internally sealed execution: individual cell outcomes remained unopened until one-shot scoring. Aggregate target-label metadata had previously been inspected, so the study was not globally label-unopened or publicly preregistered. Item-level target labels were deserialized once, after the one-shot scoring marker was spent. The release chain also records the numeric-v2 repair in a sealed addendum created before outcome access. That repair replaced an unstable divergence calculation with a stable Gaussian KL computation; it did not change the model, data split, or scoring rule. We treat it as protocol provenance, not result-driven tuning. All primary outcomes and action/effect counts come from `paper/generated/cct20_release_manifest.json`. Its sidecar `paper/generated/cct20_release_manifest.json.receipt.json` binds the file-level digest; the generated macros instead record canonicalized inference-document SHA-256 5d1ba74d9a8b16ef24746ca3d95969e635d34fd4112806f6bf91b4b45a6cf907.



**Locked thresholds.** Strong success required all six conditions: all 45 cells complete; both nominal 97.5% Bonferroni intervals strictly above zero; both exact nine-location sign-flip tests surviving Holm at 0.05; ADAPT rate at least 0.10; FREEZE rate at least 0.10; and total commitment at least 0.30. Completion, freeze exposure, and commitment passed. The interval and test requirements failed against always-freeze, and ADAPT exposure was zero. The release therefore fails the strong-success criterion.

The nominal pointwise 95% intervals in the CCT utility-preservation table 6 belong to the locked utility-preservation endpoint. The nominal 97.5% Bonferroni intervals in the strong-success audit in Table 27 belong to a separate, stricter two-comparison criterion; they are not alternate versions of the same interval.

Table 27: Locked CCT-20 primary contrasts over 45 equal-weight checkpoint–location cells from five independent checkpoints and nine location clusters. The estimand is baseline regret minus KGA regret, so positive values favor KGA. A paired two-way product bootstrap resamples checkpoint rows and location columns. Two nominal 97.5% intervals target a nominal 95% Bonferroni family level; actual family coverage still requires valid marginals. Reference  $p$ -values enumerate one-sided sign flips of nine location means, with Holm adjustment. This cell-outcome-unopened execution reproduces the locked calculation but does not establish a population guarantee.

| Comparator    | Baseline regret – KGA regret | Nominal 97.5% CI         | Sign-flip $p$ | Holm $p$   | Holm flag (.05) |
|---------------|------------------------------|--------------------------|---------------|------------|-----------------|
| Always adapt  | 0.18190227                   | [0.08501071, 0.31066374] | 0.00195312    | 0.00390625 | yes             |
| Always freeze | 0                            | [0, 0]                   | 1             | 1          | no              |

Finite-sample sign-flip validity requires the null joint law of the nine location gaps to satisfy Eq. (67). Independent zero-symmetric location gaps conditional on the evaluated checkpoints would suffice, but that assumption is unverified here. Exchangeability alone does not suffice. The sign-flip analysis conditions on the evaluated checkpoints, whereas the two-way bootstrap also resamples checkpoint levels. These are different uncertainty models, and neither establishes distribution-free safety for unseen locations.

**Primary result.** KGA exactly matched always-freeze. Its contrast against always-freeze was 0 with nominal Bonferroni-adjusted 97.5% bootstrap interval  $[0, 0]$ , enumerated location sign-flip  $p = 1$ , and Holm  $p = 1$ . Against always-adapt, the contrast was 0.18190227 with interval  $[0.08501071, 0.31066374]$ , sign-flip reference  $p = 0.00195312$ , and Holm  $p = 0.00390625$ . Measured cell benefit is adapted accuracy minus frozen accuracy: 1 cell was helpful, 0 were exactly zero, and 44 were harmful. These counts explain why freezing was useful here. They do not show that the gate learned when to adapt. KGA made 0/44/1 ADAPT/FREEZE/ABSTAIN decisions. All 45 served predictions used the frozen model, but the sole helpful cell received an interval-supported FREEZE, yielding one false FREEZE among 44 FREEZE actions (1/45 overall). This is all-frozen fail-closed service. It passes only the locked utility-preservation rule and does not demonstrate selective routing or strong success.

**Limits and next step.** All 55 development cells showed lower performance after Tent than under the frozen model, so the gate had no helpful ADAPT example. The ridge fit used 10 checkpoint–location rows from only two camera units for 11 recorded features. Calibration used 45 rows collapsed to nine calibration camera locations, while the locked targets came from held-out target locations. Coverage therefore needs an unverified cross-split exchangeability assumption. Target Mahalanobis diagnostics ranged from 7.12 to 485.67 (median 28.73), showing strong extrapolation; this diagnostic never changed an action and is not a formal coverage guarantee. One natural target study also cannot establish transfer of selective routing across natural shifts. Cell-level 16-indicator macro-F1 is archived as descriptive secondary evidence; no post-hoc aggregate or inference claim is made.

Table 28 reports all nine location clusters. Evaluation  $n$  is the number of evaluation images per checkpoint at that location. Each A/F/U triplet and each helpful/zero/harmful triplet therefore has denominator five, one cell for each checkpoint. These labels describe measured cell outcomes. Accuracy differences use one sign throughout: adapted minus frozen for adaptation benefit, and KGA minus the named fixed policy for the two policy columns.

Table 28: Complete cell-outcome-unopened CCT-20 location ledger. A/F/U denotes ADAPT/FREEZE/ABSTAIN, and  $+/0/-$  counts positive/zero/negative measured-benefit cells. Difference columns are equal-weight means over five checkpoint cells; positive policy differences favor KGA. Values are rounded to four decimals, while the sealed manifest retains full precision. Cells are repeated-model observations, and the inferential cluster unit is the camera location.

| Location | $n/\text{checkpoint}$ | A/F/U | Benefit $+/0/-$ | Adapt $-$ freeze | KGA $-$ adapt | KGA $-$ freeze |
|----------|-----------------------|-------|-----------------|------------------|---------------|----------------|
| 0        | 1302                  | 0/5/0 | 0/0/5           | -0.5418          | 0.5418        | 0.0000         |
| 7        | 1283                  | 0/5/0 | 1/0/4           | -0.0293          | 0.0293        | 0.0000         |
| 28       | 911                   | 0/5/0 | 0/0/5           | -0.0562          | 0.0562        | 0.0000         |
| 40       | 800                   | 0/5/0 | 0/0/5           | -0.1232          | 0.1232        | 0.0000         |
| 46       | 4432                  | 0/4/1 | 0/0/5           | -0.0723          | 0.0723        | 0.0000         |
| 78       | 1633                  | 0/5/0 | 0/0/5           | -0.0839          | 0.0839        | 0.0000         |
| 100      | 3441                  | 0/5/0 | 0/0/5           | -0.2035          | 0.2035        | 0.0000         |
| 105      | 1540                  | 0/5/0 | 0/0/5           | -0.2892          | 0.2892        | 0.0000         |
| 130      | 983                   | 0/5/0 | 0/0/5           | -0.2376          | 0.2376        | 0.0000         |

The release ledger contains all 45 cells: 0 ADAPT, 44 FREEZE, and 1 ABSTAIN actions; 1 helpful, 0 zero, and 44 harmful adaptation effects. The release verdict remains SAFE UTILITY ONLY. The location table is descriptive; sign-flip enumeration uses the nine location means under the reference model in Eq. (67), and no post-hoc aggregate is claimed for the secondary macro-F1 outcome.

## H Extended KGA Implementation Contract

### H.1 State Isolation

The frozen and candidate predictors must not share mutable normalization buffers, optimizer state, or parameter storage during preview. The candidate is created from a checkpoint,

adapted on the unlabeled window, and evaluated without overwriting the active state. Only an adapt decision may replace the active checkpoint. Freeze and abstain discard the temporary state. A proposed deployment test should hash the active parameters before and after each non-adapt decision and require equality.

## H.2 Evidence Determinism

Evidence extraction must define preprocessing, batch order, numerical precision, class ordering, normalization constants, and missing-value behavior. Paired features require frozen and candidate outputs on the same samples in the same order. Source statistics and ATC thresholds are immutable calibration artifacts. Evidence schema versions should be explicit rather than inferred from vector length, because two schemas can share a dimension while changing semantics.

## H.3 Calibration Identity

The calibration identity should hash the source checkpoint, candidate adapter class, adapted parameter names, optimizer and learning rate, update steps, evidence schema, benefit-model hyperparameters, feature scaler, residual construction,  $\alpha$ , and inference-unit definition. The radius belongs to that identity. Reusing it after a candidate or evidence change is not a benign software optimization; it changes the random variable whose residual is being covered.

## H.4 Runtime Accounting

A deployment report should time capture/preprocessing, frozen inference, candidate adaptation, candidate inference, evidence extraction, benefit prediction, interval decision, copy creation, and rollback separately. Mean and tail latency should be reported at the decision-window level. Peak memory should include both model states and optimizer buffers. The compact manuscript leaves these cells unfilled because the canonical cross-dataset manifest does not provide one comparable runtime measurement. Omitting an unsupported number is preferable to quoting controller-only latency as end-to-end cost.

# I Evidence Inventory

## I.1 Protocol-Specific Feature and Model Schemas

The controlled-grid implementation records 11 label-free coordinates. Two exact schema relations imply row rank at most nine. The previously reported effective rank nine remains an unverified empirical assertion here without a matrix-specific receipt and numerical tolerance; it is not inferred from the proved upper bound. Two coordinates are deterministic differences:

$$\begin{aligned}\text{entropy\_drop} &= \text{pre\_entropy} - \text{post\_entropy}, \\ \text{pbal\_drop} &= \text{pre\_pbal} - \text{post\_pbal}.\end{aligned}$$

The panel therefore contains 11 stored values, not 11 independent information sources. Together they describe confidence, class balance, change from the frozen model, and update size.

The exact common coordinate order is

`pre_entropy, pre_conf, pre_pbal, post_entropy, post_conf, post_pbal, pbal_drop, entropy_drop, frac_highconf, marginal_KL, and update_norm.`

The canonical evidence-contract digest for this ordered schema is `76b3951e77c42bbb1359123abd8d937b087c9c42ba0f14e11e61f9aca132f7f7`. The current controlled cross-fit passes these recorded coordinates directly to the GBRT; it applies no separate feature scaler. Nonfinite evidence or outcomes are rejected, and a split that cannot support the exact rank receives an infinite radius and therefore no commitment.

The shared historical builders use natural-log predictive entropy. They normalize class-marginal entropy by  $\log K$ , stabilize logarithms at  $10^{-9}$ , define the high-confidence fraction with a fixed candidate-confidence threshold of 0.9, and retain marginal KL on its raw scale. The Office-Home builder uses the analogous definitions with  $10^{-12}$  log stabilization. These constants are part of the recorded schema: changing one creates a new evidence variable even when the coordinate name and dimension remain unchanged.

Table 29: Evidence-schema identity crosswalk. Digests bind ordered names and dimensions for the archived contracts; the CCT row instead records its explicit trace schema and sealed gate document. Preprocessing and invalid-input behavior are part of the protocol, not interchangeable defaults.

| Authority                 | Dim.    | Schema identity                                                                                                                              | Model-side preprocessing                                         | Invalid or unavailable state                                              |
|---------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------|
| Common<br>trolled/archive | con- 11 | <code>76b3951e77c42bbb1359123abd8d937b087c9c42ba0f14e11e61f9aca132f7f7</code>                                                                | raw coordinates to candidate-specific GBRT                       | controlled cross-fit fails closes malformed or nonfinite matrices         |
| Camelyon17 OOD            | 17      | <code>bf6eb06e5048705c02293c85038c732e0aacc6e6737f6cb481b3bc5df83ca746</code>                                                                | raw coordinates to archived GBRT                                 | record validator rejects wrong width or nonfinite values                  |
| Office-Home               | 18      | <code>a363c05a2475f3170976b8f7efc30763859e29b9cfe1714f9c4f91a4f1e892bc</code>                                                                | raw coordinates to archived GBRT                                 | absent source logits use recorded zero fallbacks for energy shift and ATC |
| RxRx1                     | 11      | <code>names unrecovered; 8444fd58264350b2379bff08f5ccc0259f381265db7a03f9e0b48415e562805e</code>                                             | archived model only; coordinate names are not inferred           | validator rejects wrong width or nonfinite values                         |
| CCT-20                    | 11      | <code>kbound_cct20_label_free_trace_features_v1; sealed gate document 74db83cca3641c505735bcc9db4f76c588148f1ebb4d7d7c2b713bfb6eda0e7</code> | fit-location mean and population-SD standardization before ridge | exact-schema, finiteness, or support failure returns AB-STAIN             |

Natural-shift artifacts do not all use the same schema. The Camelyon17 OOD record uses the common 11 coordinates plus `disagreement_rate`, `entropy_gap`, `energy_shift`, `bn_kl`, `atc_acc_est`, and `conf_drop`; its 17-coordinate digest is `bf6eb06e5048705c02293c85038c732e0aacc6e6737f6cb481b3bc5df83ca746`. The separate Camelyon17 B-v2 diagnostic uses the common 11-coordinate schema. Office-Home adds `disagreement`, `energy_shift`, `bn_kl`, `atc_acc_est`, `conf_drop`, `prior_kl_source`, and `cand_disagree_mean`; its 18-coordinate digest is `a363c05a2475f3170976b8f7efc30763859e29b9cfe1714f9c4f91a4f1e892bc`. The

RxRx1 authority records dimension 11 and digest 8444fd58264350b2379bff08f5ccc0259f381265db7a03f9e0b48415e562805e, but explicitly marks the coordinate names as unrecovered. We therefore do not silently assign it the common names. The iWildCam record is withheld from release-level numerical claims because its archived scorer used scikit-learn macro-F1 rather than the official WILDS label-present macro-F1 contract. Its canonical evidence metadata nevertheless recovers the common ordered 11-coordinate schema and digest above.

A fitted model and its radius remain specific to their recorded evidence schema. Missing-input behavior is protocol-specific. The current controlled cross-fit fail-closes malformed or nonfinite numeric matrices to ABSTAIN, and CCT-20 returns ABSTAIN on an exact-schema or finiteness failure. The maintained natural-shift record validator instead rejects wrong-width or nonfinite records before routing; historical feature builders encode unavailable BatchNorm statistics as zero, and the Office-Home builder also defines zero fallbacks for absent source logits in its energy-shift and ATC coordinates.

The controlled and archived GBRT protocols use scikit-learn’s `GradientBoostingRegressor` (Pedregosa et al., 2011) with squared-error loss, 250 trees, depth 2, learning rate 0.05, subsample 0.8, and random seed 0. The fit is specific to each dataset and adapter. CCT-20 instead uses its sealed ridge model, as described in the main-paper design in Section 7.2. Thus “adapter agnostic” describes the interface: any adapter can propose a candidate that is evaluated and either committed or rolled back. It does not mean that a benefit model fitted for Tent transfers validly to EATA, SAR, or another adapter.

**CCT-20 trace and ridge contract.** CCT-20 uses a separate ordered trace schema:

```
frozen_mean_entropy, tent_mean_entropy, entropy_change, frozen_mean_confidence, tent_mean_confidence, confidence_change, prediction_disagreement, marginal_jensen_shannon_divergence, normalized_predicted_class_effective_count, normalized_tent_update_norm, and batchnorm_batch_source_statistic_divergence.
```

The sealed gate identifies this contract as `kbound_cct20_ridge_gate_v1`; the release manifest binds its canonical document by SHA-256 74db83cca3641c505735bcc9db4f76c588148f1ebb4d7d7c2b713bfbb6eda0e7. Every feature must be present exactly once and finite. Fit-location means and population standard deviations standardize the coordinates; a zero standard deviation is replaced by one. The ridge penalty is 10, the intercept is unpenalized, and calibration at  $\alpha = 0.10$  takes the maximum absolute residual across checkpoints within each location before applying the exact-rank rule over nine locations. A support or schema failure returns ABSTAIN; Mahalanobis distance is diagnostic and never changes the action.

Before ridge standardization, the CCT trace extractor normalizes each entropy by log 16, the Jensen–Shannon divergence by log 2, and predicted-class effective count by 16. It defines the update coordinate as

$$\frac{\|\theta_{\text{after}} - \theta_{\text{before}}\|_2}{\max\{\|\theta_{\text{before}}\|_2, 10^{-12}\}}$$

over the official Tent BatchNorm-affine parameters. The final coordinate is a channel-weighted Gaussian KL divergence from the probe-batch statistics to the stored source BatchNorm statistics, using the layer’s BatchNorm epsilon. Probability values are clipped only at

machine tiny before a logarithm. These normalizations are distinct from the archived GBRT coordinates above.

Table 30: Non-empirical feature-family summary for the maintained implementation. Each row names one feature family, the state required to compute it, and its protocol-specific validation or fallback. All quantities aggregate unlabeled inputs, frozen/candidate outputs, or the proposed update; the table reports no performance result.

| Family                  | Representative coordinates                            | Required state                      | Failure handling                                         |
|-------------------------|-------------------------------------------------------|-------------------------------------|----------------------------------------------------------|
| Predictive uncertainty  | mean entropy, confidence, quantiles                   | frozen/candidate soft-max           | reject NaN or malformed probabilities                    |
| Marginal balance        | normalized class-marginal entropy and change          | predicted-class marginals           | fixed class count and normalization                      |
| Model change            | entropy gap, confidence drop, prediction disagreement | paired frozen/candidate outputs     | require aligned sample order                             |
| Collapse signal         | fraction above adapted confidence 0.9                 | candidate probabilities             | threshold fixed before scoring                           |
| Distribution shift      | free-energy shift, batch/running-statistic divergence | frozen logits and source statistics | protocol-specific validation or documented zero fallback |
| Update magnitude        | parameter-update $\ell_2$ norm                        | shadow candidate delta              | configuration hash must match calibration                |
| Source-calibrated proxy | ATC-style threshold estimate                          | source-fixed threshold              | target labels never used                                 |

These protocol-specific dimensions are part of the fitted estimator contract, not universal sufficient statistics. Feature selection may improve prediction, but it cannot defeat the label-kernel construction in the main-paper Lemma 4.5 over an unrestricted target fibre.

## J Implementation Failure Modes and Conservative Responses

The implementation must distinguish a supported negative interval from an unavailable assessment. The deployment contract must preserve the frozen service on an unavailable assessment. Some interfaces return ABSTAIN for invalid features, while malformed authority documents can fail-stop with an integrity exception; the caller must record that failure and must not relabel it as certified FREEZE. Recovery requires a predeclared rule that restores a validated state. The CCT-20 Mahalanobis distance is diagnostic and is not an action trigger; a declared support test is a separate condition.

Table 31: Non-empirical deployment checklist. Each row maps one detectable operational failure to the conservative response required by the implementation contract; it reports no benchmark outcome.

| Failure                            | Detection                     | Response                                        |
|------------------------------------|-------------------------------|-------------------------------------------------|
| Checkpoint/adaptor mismatch        | stored hash mismatch          | rollback and recalibrate                        |
| Predeclared support-rule violation | declared support test         | withhold commitment                             |
| Missing/invalid feature            | schema and finite-value check | retain $f_0$ ; abstain                          |
| Insufficient calibration size      | rank $k > n$                  | infinite radius                                 |
| Batch too small                    | minimum-size contract         | wait or abstain                                 |
| Residual transfer concern          | group/residual diagnostic     | withhold claim; recalibrate or justify transfer |
| Continual-state drift              | state and stream monitor      | restore the last validated state                |

The audit log should store a timestamp, dataset or stream identifier, checkpoint and adapter hashes, evidence schema version, evidence vector, benefit estimate, radius, interval endpoints, action, rollback outcome, latency components, and configuration hash. Labels, if later revealed, are joined in a separate offline audit table. This separation makes it possible to verify that the live decision was target-label-free.

## K Prospective Natural-Shift Confirmation Protocol

CCT-20 completed a cell-outcome-unopened, internally sealed natural-shift evaluation but produced all-frozen fail-closed service, not selective routing (Section G.1). So2Sat-LCZ42 then stopped at its development-feasibility gate before gate calibration or target access (Section B.10). Its city-level outcome summaries are now open, and its declared city-level target action conflicts with the runner’s checkpoint–city action unit. A new confirmation must therefore be versioned independently, exclude those opened outcomes from tuning, and resolve the action unit before target execution. The design below is future evidence, not a reinterpretation of either completed record. The painting-development evaluations and their queue and BN follow-ups are also already opened and excluded from confirmation.

### K.1 Lock Before Target Labels

The protocol should name a fixed, predeclared set of independently trained source checkpoints, one candidate adapter, one evidence schema, one benefit-model class, one calibration rule, one  $\alpha$ , and one target-domain family. Adapter hyperparameters must be selected on development domains and then frozen. The lock should hash the checkpoints, code revision, configuration, calibration records, and planned scoring script. It must also name the action and aggregation units, the test unit, and the multiplicity family. A later change creates a new protocol version; it does not overwrite an earlier development-stopped or incomplete arm.

Natural heterogeneity should be built through independently defined environments rather than by pooling whichever benchmark conditions produce opposite signs. Examples include hospitals, camera locations, laboratory batches, acquisition days, physical devices, or visual domains. Environment membership may be observed at deployment, but the measured cell-benefit label remains hidden. The target test environments should be absent from estimator fitting, residual calibration, threshold tuning, and candidate selection.

### K.2 Three-Way Data Separation

Development data fit  $h_\theta$  and select the evidence pipeline. A separate calibration set produces residuals for Eq. (14). The held-out target set is opened exactly once after all software and configuration hashes match the lock. Splitting adjacent frames, patches, or samples from the same session across these stages is leakage even when filenames differ. The split must occur at the environment or session level that defines deployment exchangeability.

The target stream is replayed identically for always-freeze, always-adapt, simple gates, KGA without a radius, and the full empirical benefit-interval gate. Model randomness should either be frozen for paired replay or repeated through independently trained checkpoints

declared in advance. Every policy receives the same ordered windows. Labels are stored behind a separate audit interface and are unavailable to the live decision process.

### K.3 Endpoints and Decision Rule

The primary endpoint is paired regret against each fixed policy. Success requires both baseline minus KGA regret gaps to be positive and both predeclared intervals to exclude zero, with coverage justified for the declared estimand and sampling model. If the intended claim concerns population risk rather than measured cells, the protocol must also justify that transfer. The false-adapt endpoint is  $FA_u$  with the adapt count reported beside it. Secondary endpoints are  $FA_c$ , adapt rate, freeze rate, abstain rate, commitment rate, task accuracy, and latency. Conditional false-adapt is descriptive unless a separate selective-risk result is proved.

The primary benefit-interval rule remains unchanged: adapt if  $\hat{\Delta} - \varepsilon > 0$ , freeze if  $\hat{\Delta} + \varepsilon < 0$ , and abstain otherwise. The experiment must not lower  $\varepsilon$ , change  $\alpha$ , or select a different benefit regressor after observing target labels. Exploratory sensitivity analyses may be reported, but they remain outside the primary bar.

Table 32: Protocol requirements for a future confirmatory natural-shift KGA experiment. Each row states what must be frozen before target labels are revealed and the corresponding audit record; the table is a design specification, not a completed-study result.

| Component              | Must be fixed                                                                               | Audit evidence                                        |
|------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Model and candidate    | checkpoint, adapter, optimizer, steps, adapted parameters                                   | hashes and immutable configuration                    |
| Evidence and estimator | schema, normalization, model class, hyperparameters, seed                                   | feature contract and fitted-artifact hash             |
| Calibration            | split unit, residual construction, exact rank, $\alpha$                                     | calibration IDs, residuals, radius, no target overlap |
| Target evaluation      | environments, windowing, order, exclusion rules, transductive/inductive candidate semantics | sealed manifest and one-time scoring log              |
| Primary inference      | two paired regret gaps, cluster unit, interval method                                       | code and random seed fixed before label reveal        |
| Safety                 | $FA_u$ and adapt exposure                                                                   | action log joined to labels only offline              |
| Multiplicity           | all candidate-track arms in family                                                          | time-stamped inventory including failures             |
| Runtime                | end-to-end components and peak memory                                                       | raw timing rows on the target hardware                |

### K.4 Interpreting Outcomes

Four outcomes are scientifically useful. An interval-supported reduction in regret relative to each fixed comparator, with nontrivial ADAPT and FREEZE exposure, supports routing utility under the locked natural-shift class. Matching the better fixed policy with few errors supports conservative commitment but not selective routing. High abstention with low error indicates a conservative interval gate and motivates better evidence or more calibration data. A comparator shortfall fails the predeclared routing-utility criterion; excess false adaptation challenges the corresponding safety or calibration criterion. Both outcomes must remain in the result family. None of these outcomes alone establishes the population band: that conclusion still requires the declared class and its structural assumptions.



## L Theorem Dependencies, Counterexamples, and Non-claims

The compact paper uses three related logical statements. First, the disagreement reduction is an identity under the binary setup. Second, the population frontier is a uniform statement over a rich, declared class. Third, the empirical coverage-to-action implication is marginal and conditional on interval coverage. The population statements concern  $\Delta$ ; the reported empirical calibration concerns  $\Delta^{\text{cell}}$ . Connecting these targets requires an additional validity argument. Neither the assumptions nor the quantifiers are interchangeable.

**Why the multiclass interface does not extend the binary theorem.** Under multiclass 0/1 loss,  $\Delta = \mathbb{P}_T(D)(p_a - p_0)$ , where  $p_a = \mathbb{P}_T(f_a = Y \mid D)$  and  $p_0 = \mathbb{P}_T(f_0 = Y \mid D)$ . Only the binary setting forces  $p_0 = 1 - p_a$ . KGA may predict a multiclass scalar score difference, but that interface does not inherit the binary population identity.

Table 33: Logical dependency map for the paper’s formal and empirical statements. Each row lists a statement’s inputs, conclusion, and excluded inference; it reports no additional empirical result, and a check in one row does not validate premises in another row.

| Statement                      | Inputs                                                         | Conclusion                                             | Not concluded                                                     |
|--------------------------------|----------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------|
| Disagreement reduction         | binary labels, 0/1 loss, $\mu_T(D) > 0$                        | $\Delta = 2\mu_T(D)(M + \gamma)$                       | calibration transfer or observability of $\gamma$                 |
| Interior impossibility         | label-free fibre, $\beta > 0$ , $ M  < \beta$ , class richness | evidence-identical laws with opposite nonzero benefits | every empirical abstention is structural                          |
| Closed-band result             | strict directional semantics and boundary law                  | no uniformly sound strict action on $ M  \leq \beta$   | adapt/freeze have different task risk at $\Delta = 0$             |
| Frontier sufficiency           | declared $ \gamma  \leq \beta$ and fixed $M$                   | direction is fixed when $ M  > \beta$                  | label-free estimation of $\beta$                                  |
| Coverage-to-action proposition | marginal interval coverage for a declared target $B$           | marginal false-adapt/freeze control for $B$            | conditional error control or risk alignment                       |
| KGA benchmark                  | fitted $h_\theta$ , declared residual rule, frozen protocol    | measured regret and actions on saved units             | population-benefit coverage or universal transfer to a new domain |
| Lean build                     | encoded definitions and assumptions                            | kernel-checked formal statements                       | correctness of unencoded data and deployment assumptions          |

**Why the boundary is separate.** For  $|M| < \beta$ , the construction chooses two latent residuals that yield opposite nonzero benefits. At  $|M| = \beta > 0$ , one admissible residual can make the benefit zero while another retains the sign of  $M$ ; opposite nonzero signs need not exist. Both cases block a uniform strict commitment, but for different reasons. At  $M = \beta = 0$ , every law in the zero-residual class has zero benefit under the model. Abstention remains the maximal action only under the paper’s strict-direction convention, not because either task-risk choice is worse.

**Why a calibration residual need not be drift.** Let  $P_T = P_S$ , with equiprobable inputs  $a, b$ , deterministic labels  $Y(a) = 1$ ,  $Y(b) = 0$ , and candidate  $f_a \equiv 1$ . Set  $f_0(a) = 0$ ,  $f_0(b) = 1$ , and  $s \equiv \frac{1}{2}$ . The candidate is correct on half the source inputs, so this constant score is source-calibrated in the ordinary sense. Yet disagreement occurs only at  $a$ , where the candidate is always correct. Thus  $M = 0$ ,  $\gamma = \frac{1}{2}$ , and  $\Delta = \frac{1}{2}$ , although the source and target laws are identical. Conditioning on disagreement, not a distribution change, explains

the nonzero residual. The frontier requires a bound on this residual regardless of its cause. Top-label calibration conditions on the predicted class (Gupta and Ramdas, 2022); neither it nor ordinary score calibration automatically implies calibration conditional on disagreement.

**Why coverage is the premise.** The coverage-to-action proof is elementary once Eq. (9) holds: on the coverage event, an interval wholly above zero cannot contain a nonpositive  $\Delta^{\text{cell}}$ . The scientific burden is therefore to justify coverage for the declared outcome and calibration/deployment pair. Joint exchangeability of the residual scores is one route; known shift correction or another valid construction may be another. Risk alignment can explain why a useful predictor transfers, but it is not an additional logical premise after coverage is given.

**Why cell coverage is not population coverage.** Consider the fixed binary predictors  $f_0 \equiv 0$  and  $f_a \equiv 1$ . Let the target input be  $a$  with probability 0.45 and  $b$  with probability 0.55, with deterministic labels  $Y(a) = 1$  and  $Y(b) = 0$ . Population benefit is  $\Delta = -0.10$ . In a one-item evaluation cell, a fixed predictor that maps  $a$  to  $+1$  and  $b$  to  $-1$  predicts  $\Delta^{\text{cell}}$  perfectly without reading the new label. With radius zero, cell coverage is perfect, yet the gate selects ADAPT with probability 0.45 despite negative benefit under the fixed population law. This constructed example does not contradict the main-paper Proposition 6.1; it shows why its covered target cannot be changed.

**Why exchangeability alone does not validate sign flips.** As a separate null example, let all nine cluster gaps equal the same fair random sign  $S$ . This vector is exchangeable, each gap is symmetric about zero, and a global sign change preserves its law. Nevertheless, when  $S = +1$ , independent-coordinate sign-flip enumeration for the mean gives a reference value of  $1/512$ . That occurs with probability  $1/2$ , not  $1/512$ . The vector fails Eq. (67). This example explains the required assumption; it is not a model fitted to the CCT-20 observations.

**Why a null is not an impossibility result.** A null benchmark row can arise because the true effects are small, the candidate is one-sided, the evidence map is weak, the benefit model underfits, the radius is wide, or calibration does not transfer. The population impossibility requires evidence-identical target laws inside the declared class with incompatible strict directions. The canonical benchmark table does not estimate that full fibre. It therefore reports weak or adverse empirical regimes without calling them proven unknowable worlds.

**What would falsify the practical claim.** The practical claim is conditional and testable. A clean target replay with the locked pipeline can record a high  $\text{FA}_{\text{u}}$ , low inclusion, or regret above both fixed policies. A checkpoint or adapter change, or a family-held-out split, can provide evidence of limited transfer or optimistic cell-level cross-fitting. Such findings can reject a predeclared empirical endpoint or provide evidence against transfer or coverage assumptions; they do not by themselves logically refute a marginal guarantee. Poor utility can also coexist with valid coverage. A claim of coverage or transfer failure requires its own appropriately designed statistical assessment; none of these finite outcomes refutes the population identity.

## M Formalization and Artifact Scope

The Lean project under `docs/research/kbound/formal/` provides a kernel-checked inventory for the theorem names it exports. The compact paper relies on the LaTeX statements above; it does not claim that every sentence has a one-to-one Lean theorem.

In particular, deployment-class richness, risk alignment of a learned evidence map, exchangeability of a benchmark split, and correctness of data preprocessing remain outside the proof assistant unless separately encoded.

### M.1 Proof traceability and limits

The formal evidence is a source-bound, scoped verification record. The 8 September 2026 integration receipt records a successful Lean 4.29.1 strict-core build and transitive-axiom audit for 315 registered declarations: 65 retained core declarations and 250 additional results, including supporting lemmas. The theorem map contains 326 distinct `#check` entries across all namespaces; eleven are outside that registry. Thirteen public-root example files contain 80 examples. These are different inventories, not a percentage of the paper proved.

The source-bound summary `formal/full_report_bridges_verification_20260908.portable.json` links the combined audit, example checks and independent reviews by hash. The build is incremental with cached dependencies, not a cold dependency rebuild. The separate claim census is `formal/full_report_bridges_semantic_20260908.portable.json`. Earlier dated receipts retain their historical scopes. The audited axiom dependencies are `propext`, `Classical.choice` and `Quot.sound`.

**Population reduction and frontier.** `JointKernelScore.joint_score_benefit` derives the score-residual identity from the conditional kernel of an actual binary joint law. `ActualWorldFrontier.actual_identified_interval` identifies the clipped benefit interval over the full fixed-input-marginal residual class. `ActualWorldEvidence.actual_augmented_fibre_strict_direction_iff` connects this class to a fixed marginal-based observation experiment. The restricted-class results in `ActualWorldSubclass` require the legal interior opposite-pair and boundary zero constructions; individual directional uniqueness additionally requires a nonempty class. These results do not establish those membership conditions for an arbitrary deployment subclass or justify conditioning on a model pair adapted on the same observed batch.

**Audit floor.** `measurable_audit_floor` proves the bound at the possibly unattained supremum of a nonempty bounded residual fibre under a common observation law. `ActualFibreRadius.actual_augmented_fibre_radius` supplies the actual full-class radius equality under positive disagreement mass,  $0 \leq \beta \leq \frac{1}{2}$ , and the fixed observation experiment. This is not an estimator of a valid deployment  $\beta$  from unlabeled data.

**Coverage and decision.** `ExtendedRadiusCertificate.directional_error_probability` proves that coverage for the named scalar target bounds the union of false strict directions, allowing measurable finite or infinite radii. `ExactConformal.literal_`

`residual_certificate` constructs the literal ceiling-rank absolute-residual interval, including ties and the infinite-radius case, under joint score exchangeability. Neither result establishes benchmark exchangeability, calibration transfer, or operational label custody. For a population-target application, valid coverage of population benefit is the required statistical premise; exact fibre-wide identification is a separate question. The `PopulationTransfer` module combines cell and sampling coverage by a union bound, but does not supply the reported experiments with a population sampling-error radius.

**Supporting statements.** `WeightedHelpful` assumes per-cell helpfulness and common nonnegative weights, with positive total weight for means. `FeatureRank` proves a nine-dimensional real-linear schema and row rank at most nine under the exact coordinate relations; it does not establish observed numerical rank or probabilistic independence. `RiskAlignment` proves that alignment is weaker than strict identification, including an actual zero/positive-law witness.

**Full-report implications.** `InformationRefinement` covers null-disagreement risk equality and refinement of actual joint-observation fibres, with nonempty/bounded premises for real radii. `ConditionalExposure` derives conditional error bounds from marginal error and positive action exposure. `IntegrableProxy` proves the expectation decomposition and sufficient signs with both terms integrable under the same law; it does not prove general-loss exactness.

The current prose states the fixed observation, model-pair, seed-coupling and nonempty-subclass qualifications. Their empirical applicability remains unproved. Manuscript revisions require their own final source binding and semantic review; unchanged Lean sources do not make every revised sentence kernel-checked. The historical full one-bit/H/ratio-rate extension remains excluded. Formal verification, empirical provenance, implementation tests and publication rendering therefore retain separate scopes.

One historical extension remains outside the verified scope. A measurable label swap preserves label-free evidence and reverses population benefit, but selecting one representative from each swap orbit need not identify the sign: distinct orbits can share the same evidence law and have opposite selected signs. Lean verifies this counterexample and the corresponding set-theoretic fibre-consistency criterion, not a general measurable decoder or the older H/ratio-rate extension. Those stronger claims are excluded from this compact submission. The `-full-foundations` gate therefore remains failing for the sixth layer; the scoped kernel audit checks the registered results and rejects nonstandard proof axioms.

The formal build and empirical build answer different questions. Lean checks that a proposition follows from formal assumptions. The reconciliation scripts check that displayed aggregates match saved records under implemented definitions. Neither checks the truth of an unencoded deployment assumption, and neither substitutes for independent experimental replication.

The authoritative compact artifacts are therefore layered:

1. this source and its compiled PDF for the scoped narrative;
2. shared theorem inputs for the exact statements and proofs;
3. canonical JSON and generated LaTeX for numerical results;

4. the source manifest for record-level provenance;
5. package tests and Lean logs for implementation and formal checks.

Historical drafts, dashboards, and exploratory search folders can remain useful development records, but they are not evidence sources for this compact submission unless referenced by the canonical manifest.

## N Reproducibility and Release Procedures

### N.1 Canonical Numerical Source of Truth

The CIFAR rows and appendix reconciled point estimates share one numerical source: `experiments/kbound/results/reconciled_panels_v1/canonical_panel_results.json`, whose SHA-256 is `35d4c165843de1ece3cb35ffb4ac50dbfcbfa33b646754e2b6d133fbdfa78c6e`. Its input inventory is bound separately by source-manifest SHA-256 `03b1d2b1e9e5ed1cf835126f871ed83eb8497a1ac81727b3955af0d69eb89742`; the canonical JSON also records generator, implementation-dependency, and per-array bindings. Auxiliary artifacts retain narrower roles. The family analysis is the retrospective sensitivity in Table 23; the POEM/AETTA artifact is historical; and the So2Sat record is a locked development stop. The generated result manifest indexes each reported claim and scope. None of these auxiliary records overwrites the canonical point-estimate panel.

### N.2 Reproduction Pipeline

The full gate uses Python 3.12 and `requirements-research.txt`. The smaller paper-only profile does not include the Torch imports used during repository-wide test collection. The model runtime uses PyTorch (Paszke et al., 2019). From the repository root, the release command is:

```
KBOUND_PYTHON=.venv/bin/python bash
docs/research/kbound/runbooks/release_candidate.sh all
```

It runs the read-only preflight, validates and regenerates result authorities, executes software and formal checks, builds the four current role PDFs, renders every PDF page, and emits release checksums. A DOCX may be produced as a nonrelease convenience artifact, but it is not a fifth current role. The gate consumes frozen model outputs and never launches training.

For presentation-only rebuilding from already validated authorities, the command is:

```
SOURCE_SNAPSHOT_COMMIT=<12-hex-source-commit>
BUILD_LONG_TMLR=1 BUILD_SHORT_MAIN=1
BUILD_SHORT_SUPPLEMENT=1 BUILD_FULL_REPORT=1 bash
docs/research/kbound/scripts/build_pdfs.sh
```

This shorter command validates the frozen authorities, regenerates presentation tables and figures, and builds the four maintained role PDFs. It does not replace the full release gate or create new empirical evidence.

### N.3 Calibration Identity and Implementation Contract

The maintained API binds an interval-gate record to the checkpoint, adapter configuration, evidence schema, benefit estimator, residual construction, safety level, and scoring unit. The deployment path must fail closed when any identity differs. `KGA.certify_evidence` requires a frozen estimator, the exact schema, a protocol hash, and a separate residual set. The lower-level `certify` function is an audit interface; it does not construct evidence or validate transfer by itself.

### N.4 Formalization and Lean Scope

Lean artifacts and validators check the encoded theorem cores and selected finite constructions. Formalization of learning-theory results has precedent, including a Rademacher-complexity generalization bound (Sonoda et al., 2025); we cite its pinned 2025 version as context, not as a proof of K-Bound’s assumptions. These checks do not certify dataset correctness, preprocessing, empirical transfer, exchangeability, or every deployment assumption stated in prose. Machine checking strengthens the statements it encodes; it does not validate the benchmark pipeline outside those statements.

### N.5 Release Checklist and Audit Logs

The Phase-1 provenance audit matches all 106 canonical source/compact hash pairs, all 10 recoverable configuration hashes, and all 6 archived checkpoint hashes; it also records a 21-file current-code snapshot. Historical runs that did not serialize full identity remain explicitly partial rather than being backfilled. The release gate runs reconciliation tests, claim-ledger and result-manifest validation, manuscript checks, LaTeX builds, checksum verification, and visual PDF inspection. Build logs, the canonical manifest, the provenance seal, and the final PDFs are archived together.

Table 34: Release-stage audit contract. Each stage has an explicit authority and a fail-stop condition, preventing a mixed-generation or partially verified manuscript set from being published as current.

| Stage                    | Checked authority                                                        | Failure response                                                       |
|--------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|
| Numerical reconciliation | canonical panel, source manifest, CCT manifest and receipt               | stop on hash or claim mismatch                                         |
| Source binding           | reviewed source snapshot and release-role map                            | stop on missing or ambiguous identity                                  |
| Compilation              | four maintained TeX drivers from one source closure                      | stop on compile, reference, or overflow defect                         |
| Structure and anonymity  | page structure, fonts, metadata, links, and anonymous TMLR author fields | reject malformed output or identity leakage                            |
| Visual inspection        | fresh rendering of every maintained PDF page                             | repair clipping, blank regions, orphaned captions, or displaced floats |
| Publication              | exact four-PDF inventory and SHA-256 manifest                            | publish atomically; never mix generations                              |

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